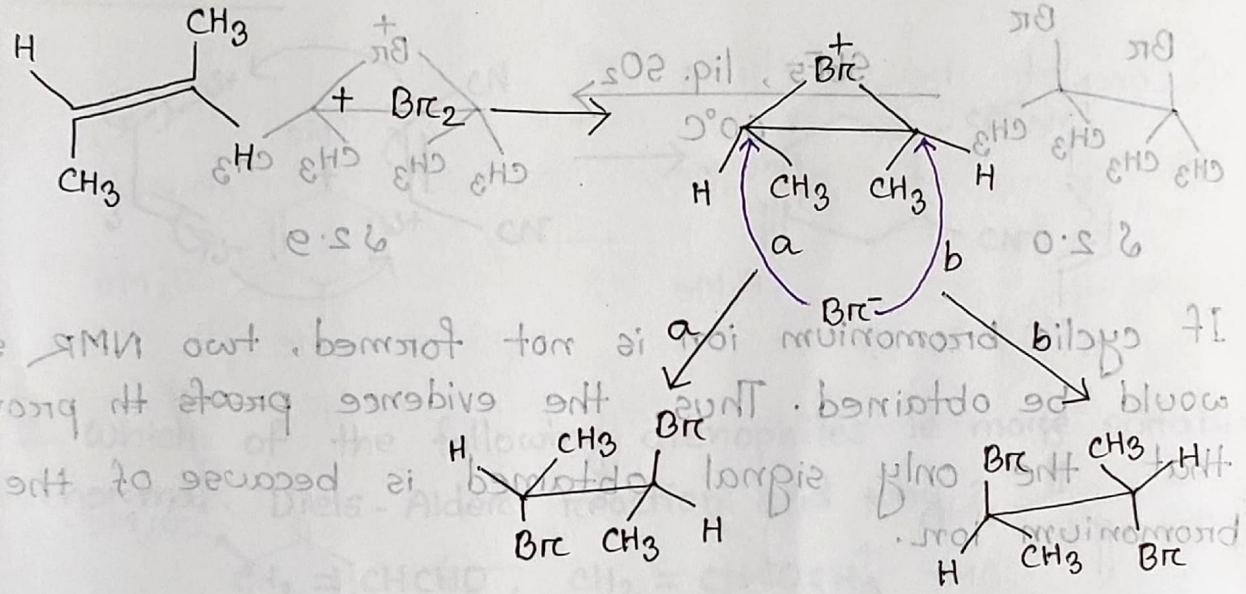
ADDITION REACTIONS

1. Trans-2-butene reacts with bromine to give the dibromo product in highly stereoselective manner. Explain the setereoselectivity with evidence.



[Meso compound]

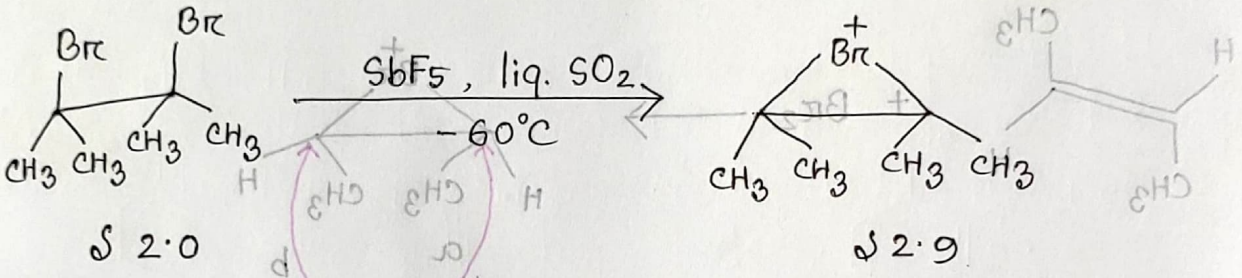
[2,3-dibromobutane]

Anti

In the reaction, Br_2 forms cyclic bromonium ion with trans-2 butene and Br^- ion can attack in both the carbon and then anti addition occurs. 100% anti addition leads to the formation of meso product.

The stereoselectivity of 2,3-dibromobutane can be explained by the formation of bromonium ion. Therefore the evidence of bromonium ion would be enough for explaining the setereoselectivity of the equation.

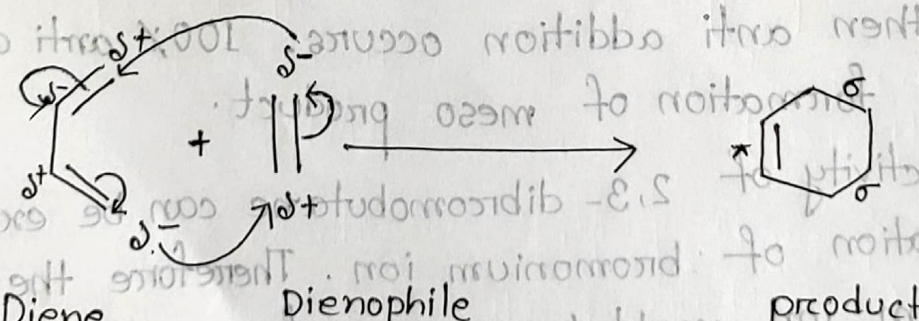
when 2,3- dimethyl-2-butene reacts with SbF_5 in liq. SO_2 at -60°C, it leds to the formation of a compound that doesn't exhibit two NMR spectrum signal expected of. Instead one signal is obserbed indicating that all the twelve protons are in same state.



If cyclic bromonium ion is not formed, two NMR signal would be obtained. Thus, the evidence ~~proofs~~ th proves that the only signal obtained is because of the cyclic bromonium ion.

2. Give a plausible mechanism for Diels Alder reaction between a suitable diene and a dienophile.

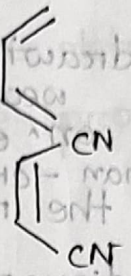
Mechanism:



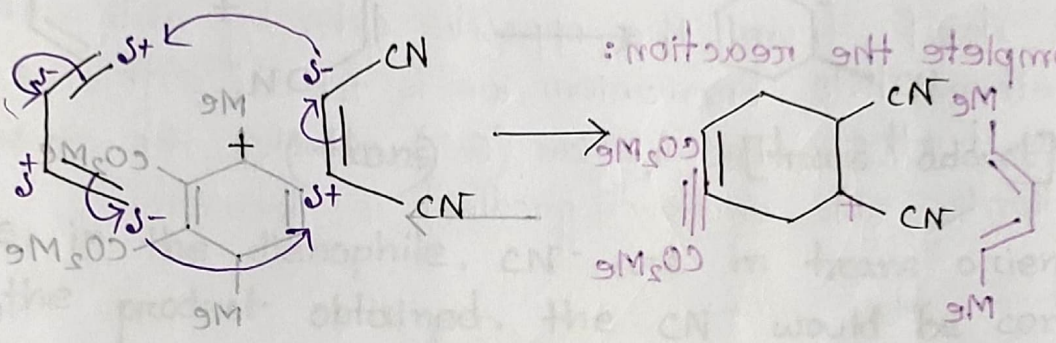
Diene Dienophile product

Let us consider,

A diene → 1,3-butadiene

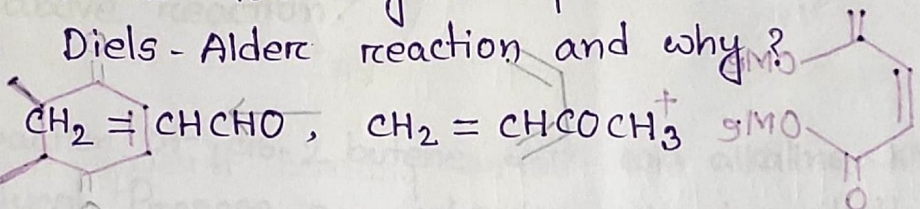
A dienophile → 

Mechanism:



3. Which of the following dienophiles is more suitable in normal Diels-Alder reaction and why?

$CH_2 = CHCHO$, $CH_2 = CHCOCH_3$



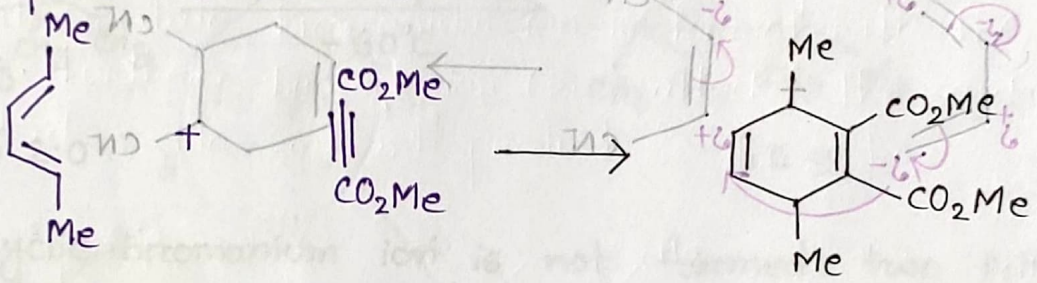
Between the mentioned compounds $CH_2=CHCHO$ is more suitable dienophile for Diels-Alder reaction.

The rate of the Diels-Alder reaction depends upon the dienophile that has substituents. If the substituent is an electron donating group, the rate of reaction decreases and if it is an electron withdrawing group, rate of reaction increases. When an electron withdrawing group withdraws electron from the double bonded carbon, it causes vacancy of electron at that carbon. Thus the carbon pulls the π- cloud more efficiently than before and the partial negative charge produced can attack

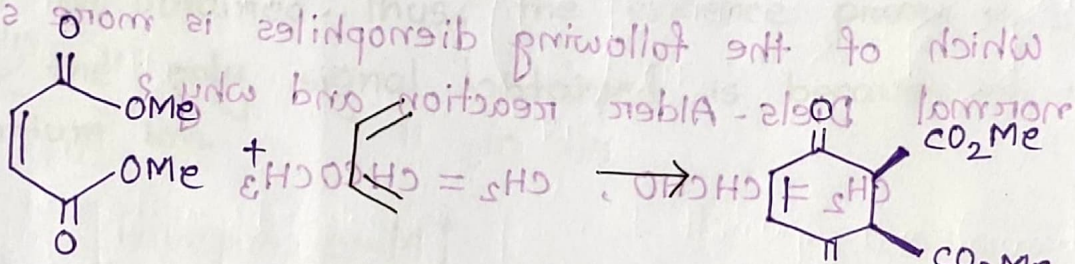
form σ bond with the diene more efficiently. As -CHO is a strong electron withdrawing group, it increases reaction rate. But $-OCH_3$ is a weak electron donating group which perhaps slows down the reaction. Therefore $CH_2=CHCHO$ contribute less than -CHO in is more preferred dienophile than $CH_2=CHCOCH_3$

4. Complete the reaction:

(i)

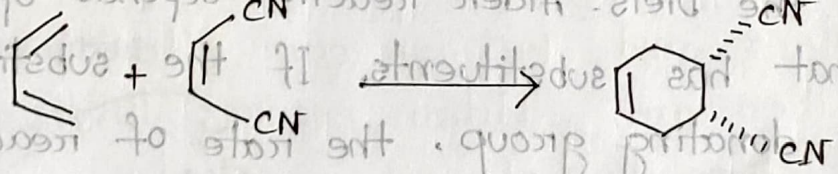


(ii)



5. Complete the reaction and explain their stereo-specific nature.

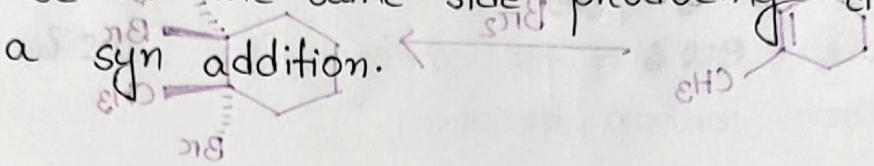
(i)



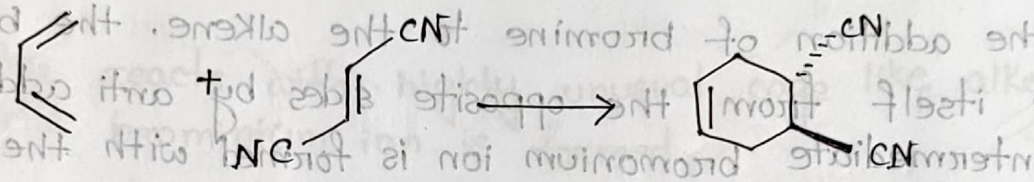
(cis) [cis adduct]

Diels Alder reaction is highly stereospecific. If we start with cis-dienophile, the product will be syn and if we start the reaction with trans-dienophile the product obtained is by anti addition.

As the CN^- present here are cis in orientation, the diels alder product will be obtained where the CN^- ions will be at the same side producing cis adduct i.e. by a syn addition.



(ii)

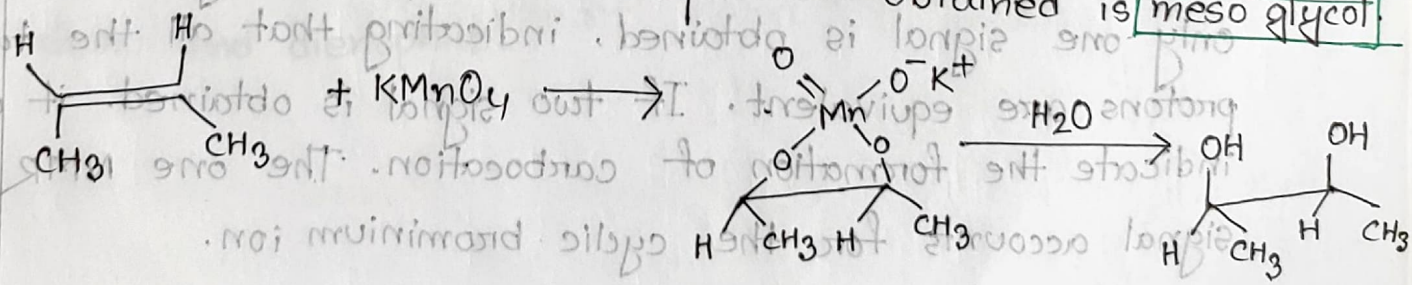


(trans) [trans adduct]

Here, in the dienophile, CN^- are in trans orientation. Thus in the product obtained, the CN^- would be connected from the opposite side. Thus anti addition takes place for the above reaction.

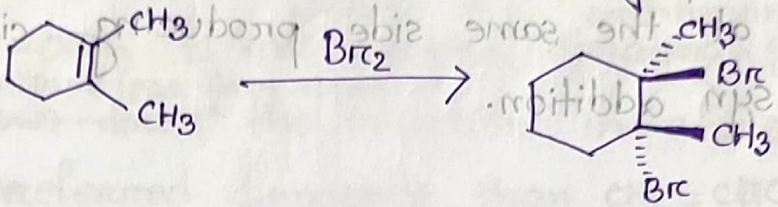
6. The reaction of cis-2 butene with cold alkaline $KMnO_4$ gives meso glycol. Propose a mechanism showing all the intermediates to justify the above statement.

Alkene cis-2-butene reacts with cold $KMnO_4$ and the reaction is syn reaction. In the reaction cyclic permanganate ester is formed. The final product obtained is meso glycol.

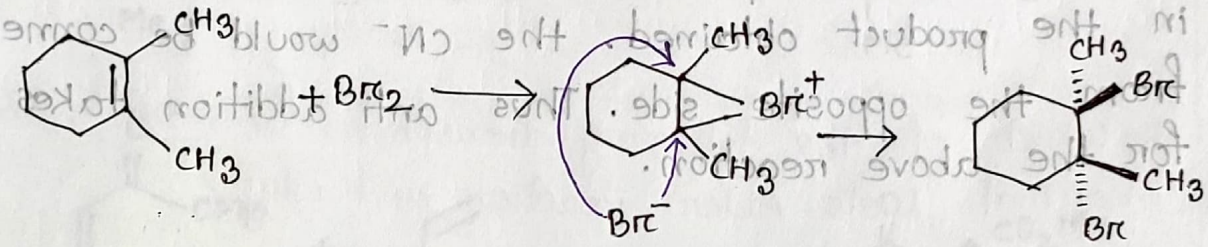


[Meso glycol]
[Syn addition]

7. Suggest a mechanism for the following reaction which explains the observed stereochemistry:



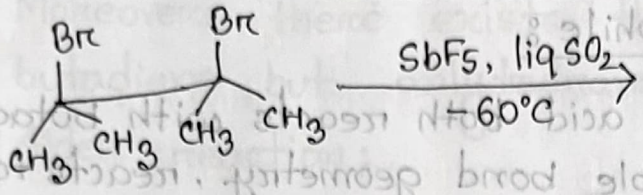
In the addition of bromine to the alkene, the bromine adds itself from the opposite sides by anti addition. An intermediate bromonium ion is formed with the cyclo alkene. Further, bromide ion (Br^-) attacks the carbon from the back side and anti product is formed.



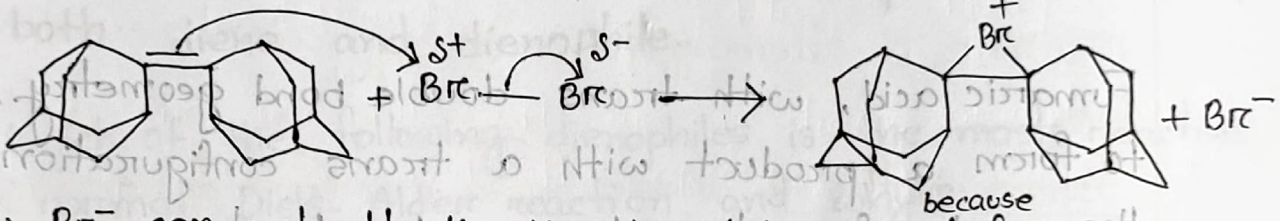
8. Bromination of alkenes proceed through the formation of a cyclic brominium ion intermediate - Explain the statement.

Evidence for brominium ion:

1. when 1,2- dibromide reacts with BrF_2SbF_5 in liq NH_3 at $-60°C$, it leds to the formation of an intermediate that doesn't exhibit two NMR spectra signal. Instead only one signal is obtained, indicating that all the twelve protons are equivalent. If two signal is obtained, it will indicate the formation of carbocation. The one NMR signal accounts for the cyclic brominium ion.



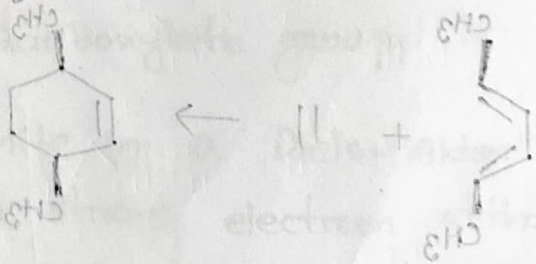
2. Br_2 reacts with highly unusual cage like alkene and cyclic brominium ion is formed.



Here, Br^- can not attack the brominium ion because of cage like structure. But if carbocation is formed, then dibromide is found which is not obtained from the reaction. It is the evidence of the formation of cyclic brominium ion.

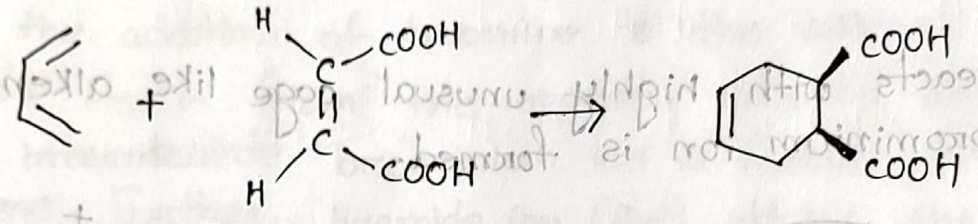
9. The Diels Alder reaction is stereospecific with respect to both the diene and dienophile - explain with suitable example

When conjugated diene reacts with dienophile it gives cyclo compound, then the reaction is called Diels Alder reaction. The Diels-Alder reaction is stereospecific for both the diene and dienophile.

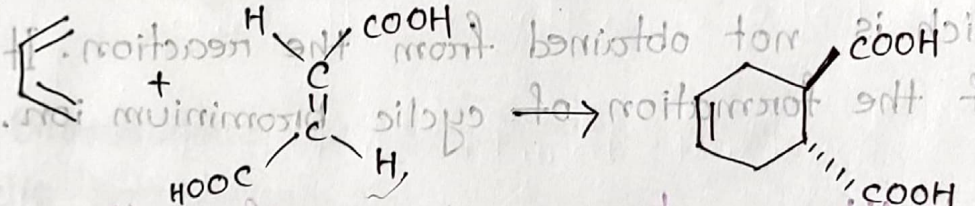


• **Stereospecifity of Dienophile :**

Maleic acid and fumaric acid both reacts with butadiene. Maleic acid, with cis double bond geometry, reacts to form a product with a cis configuration at the newly formed carbon - carbon bond.

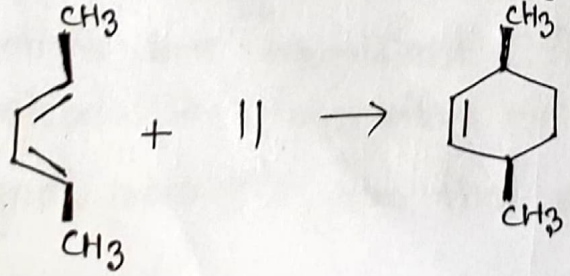


Fumaric acid, with trans double bond geometry, reacts to form a product with a trans configuration at the newly formed carbon - carbon bond.

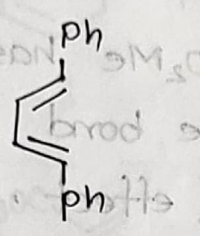


• **Stereospecificity of Diene :**

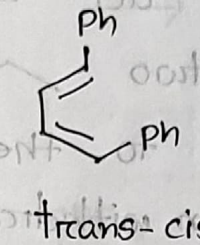
The substituted cis butadiene reacts with a dienophile to form a product with the methyl groups on the same side of the newly formed ring.



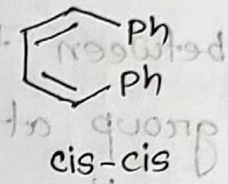
Moreover, there exists three isomerides of 1,4-diphenyl butadiene but only trans trans form will undergo Diels-Alder reaction:



trans-trans



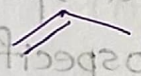
trans-cis

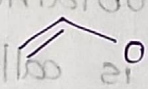


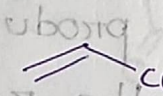
cis-cis

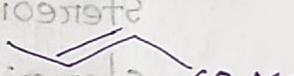
Thus, the Diels Alder reaction is stereospecific with respect to both diene and dienophile.

10. Which of the following dienophiles is the most reactive in normal Diels-Alder reaction and why?





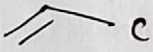




Ans:

$CO > OCOMe > CO_2Me$

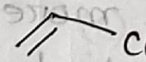
Here,



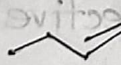
COMe contains a carbonyl group

OCOMe contains an ester group.

Wrong

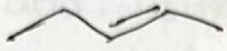
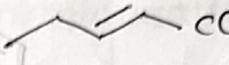
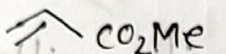


CO_2Me contains a carboxylate group.



CO_2Me contains a carboxylate group.

The most reactive dienophile in a Diels-Alder reaction is typically one that has strong electron withdrawing groups that stabilize the transition state. As.,

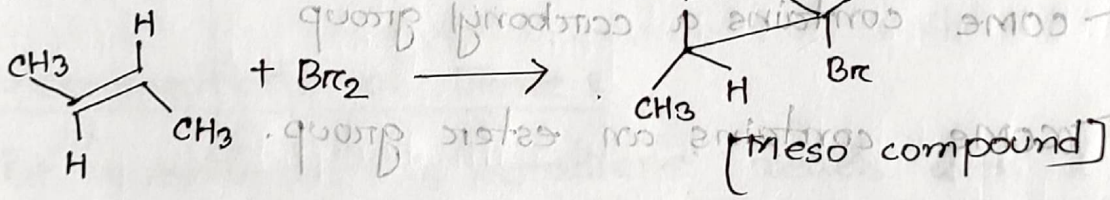
carboxylate group is more electron withdrawing than either of the group present here, and  act as more suitable one. But between these two  has an alkyl group attached to the double bond which may decrease the electron withdrawing effect. Thus  act as the most suitable dienophile.

□ What is stereospecific and regioselective reaction?

Ans:

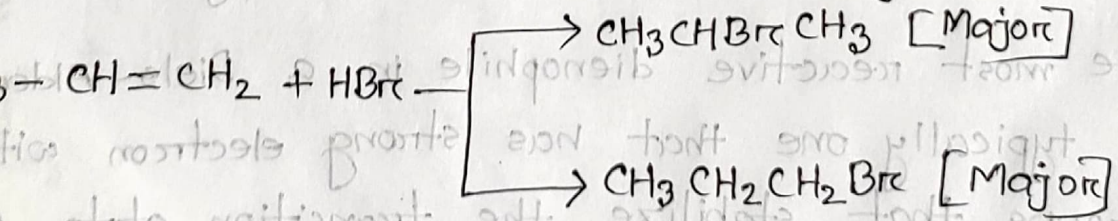
Stereospecific: When in a reaction, a specific stereoisomeric product is obtained from a specific stereoisomer, the reaction is called stereospecific reaction.

Meso compound is obtained when trans-2 butene reacts with $BrCl$.



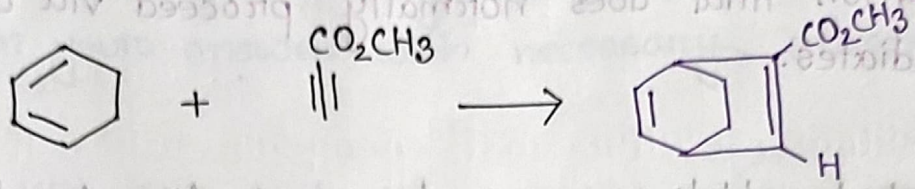
[Meso compound]

Regioselective: When in a reaction, two or more products are obtained but only one is the sole product from all of them, the reaction is called regioselective reaction.

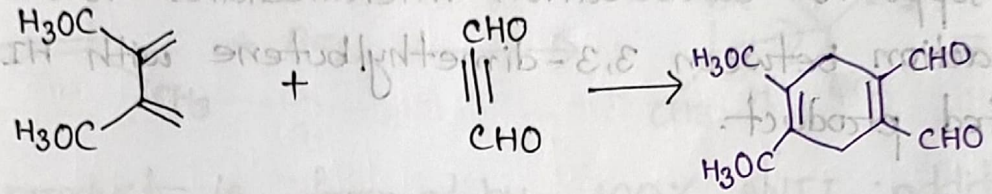


11. Complete the following reaction :

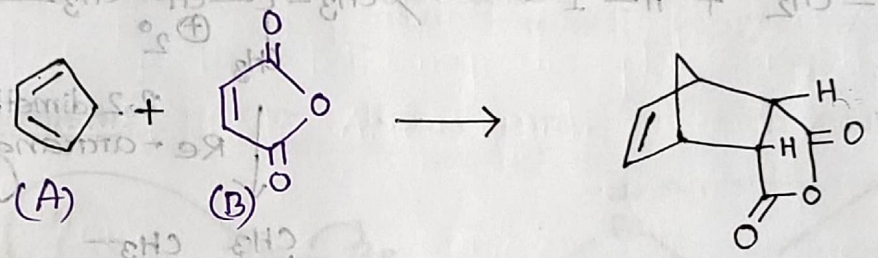
(i)



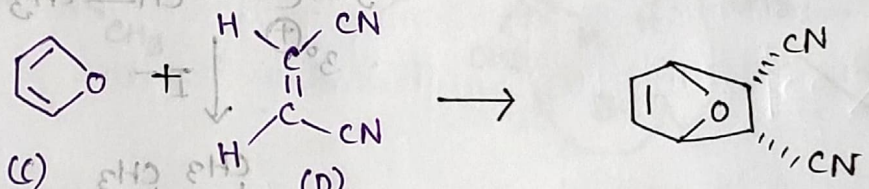
(ii)



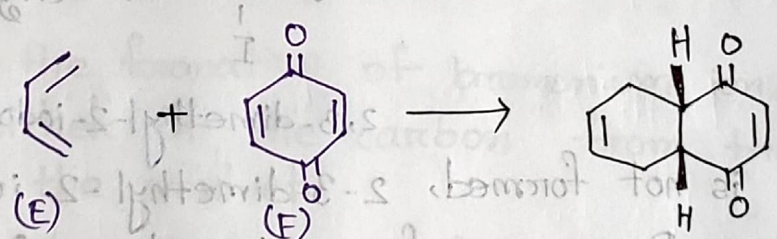
(iii)



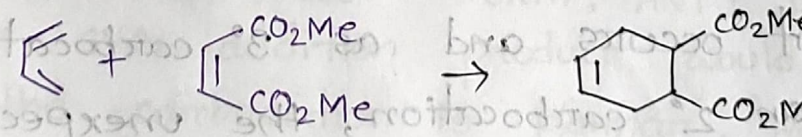
(iv)



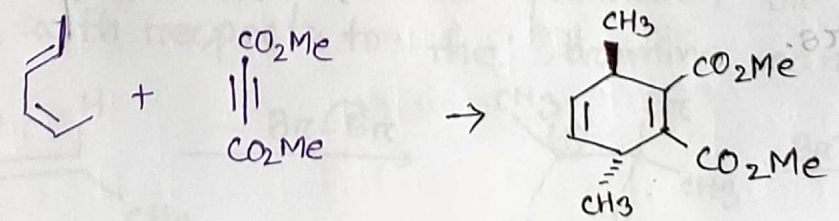
(v)



(vi)



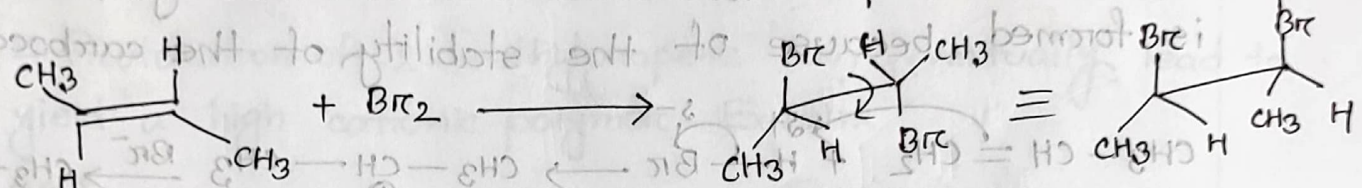
(vii)



□ Bromine addition to trans-2-butene is stereoselective and 100% ANTI. What conclusion you can draw from this statement? Explain your answer with necessary reasoning.

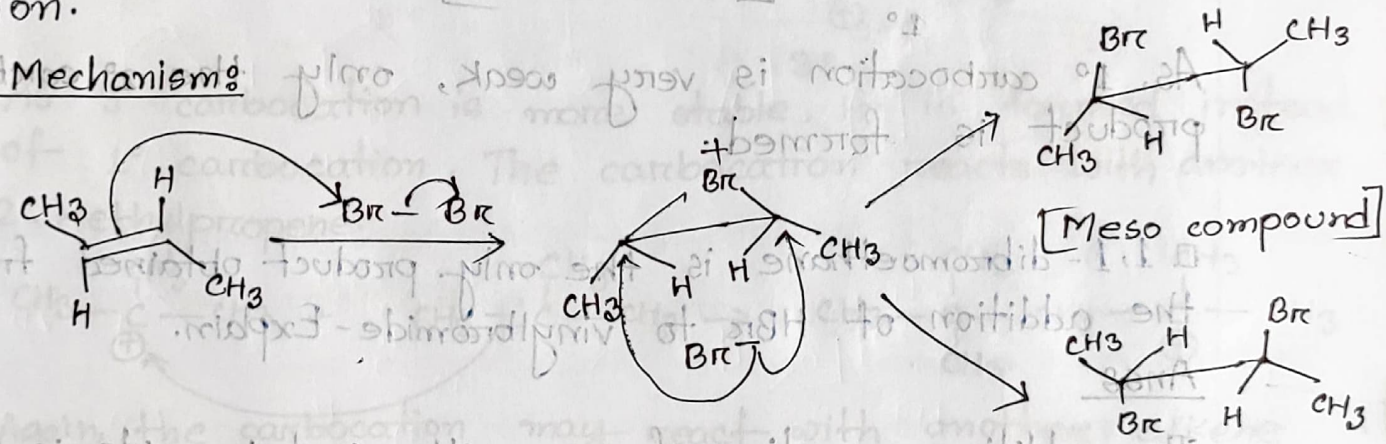
Ans:

Bromine addition to trans-2-butene produces meso compound.



This product is formed by 100% ANTI addition which accounts for the formation of intermediate bromonium ion.

Mechanism:

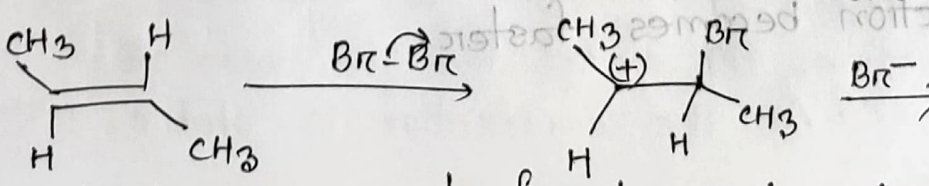


[Meso compound]

[Meso comp]

After the formation of bromonium ion, the bromide attacks at either of the carbon from the opposite side to yield the product.

If carbocation would form, ANTI product along with some SYN addition product would also be formed. As the carbocation is plannar, Br^- ion from either side with respect to the bromine can attack.

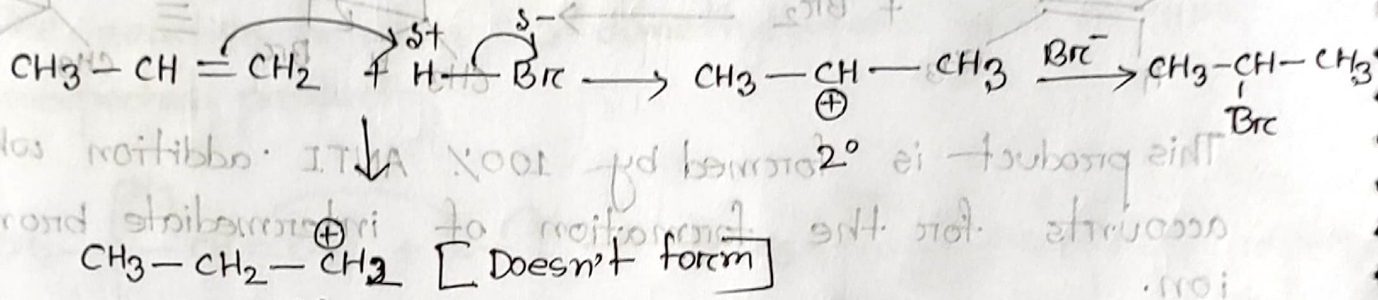


Thus 100% ANTI accounts for bromonium ion intermediate.

□ In the addition of HBrc to propene under polare conditions only product 2-bromopropane is obtained - Explain.

Ans:

The addition of HBrc with propene under polare condition follow Markonikov's rule and only product is formed because of the stability of the carbocation.

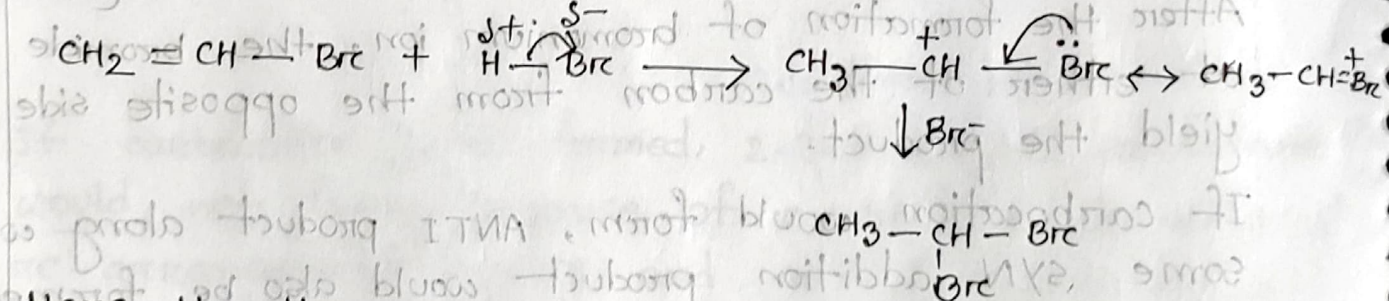


As, 1° carbocation is very weak, only the 2° carbocation product is formed.

□ 1,1-dibromoethane is the only product obtained from the addition of HBrc to vinylbromide - Explain.

Ans:

The addition reaction of HBrc with vinyl bromide is given below:



Here, the the carbocation is stabilized by the lone pair electron of bromine. When the carbocation is stable, reaction becomes faster.

But if carbocation is formed via Anti-Markonikov,

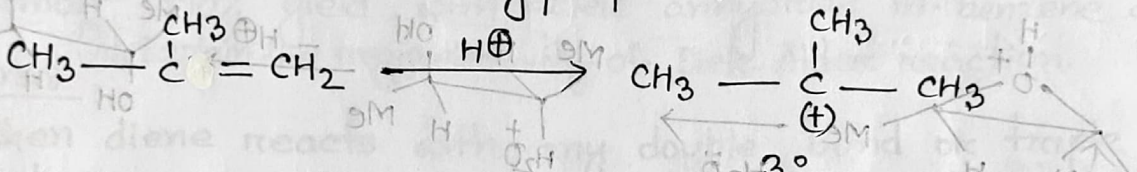
$CH_2=CH-Br + HBr \longrightarrow \overset{(+)}{C}H_2-CH_2-Br$

This carbocation can't be stabilized and no further product occurs. Only 1,1-dibromoethan is formed.

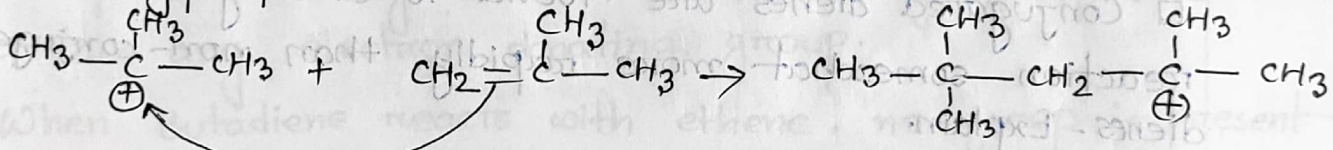
□ Protonation of 2-methylpropene can eventually lead to yield a high cationic polymer - Explain.

Ans:

Protonation of 2-methylpropene,



As 3° carbocation is more stable, it is formed instead of 1° carbocation. The carbocation reacts with another 2-Methylpropene.

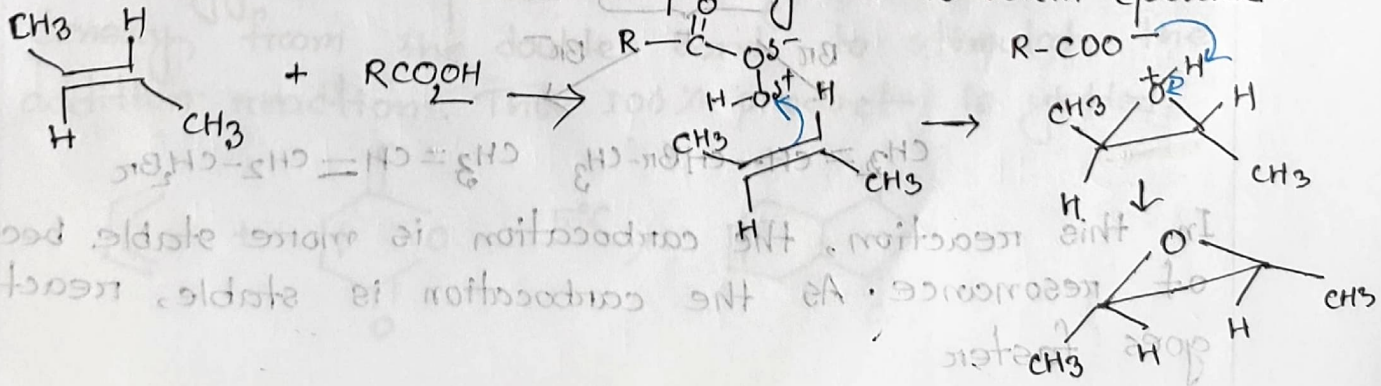


Again, the carbocation may react with another alkene which results in cationic polymer.

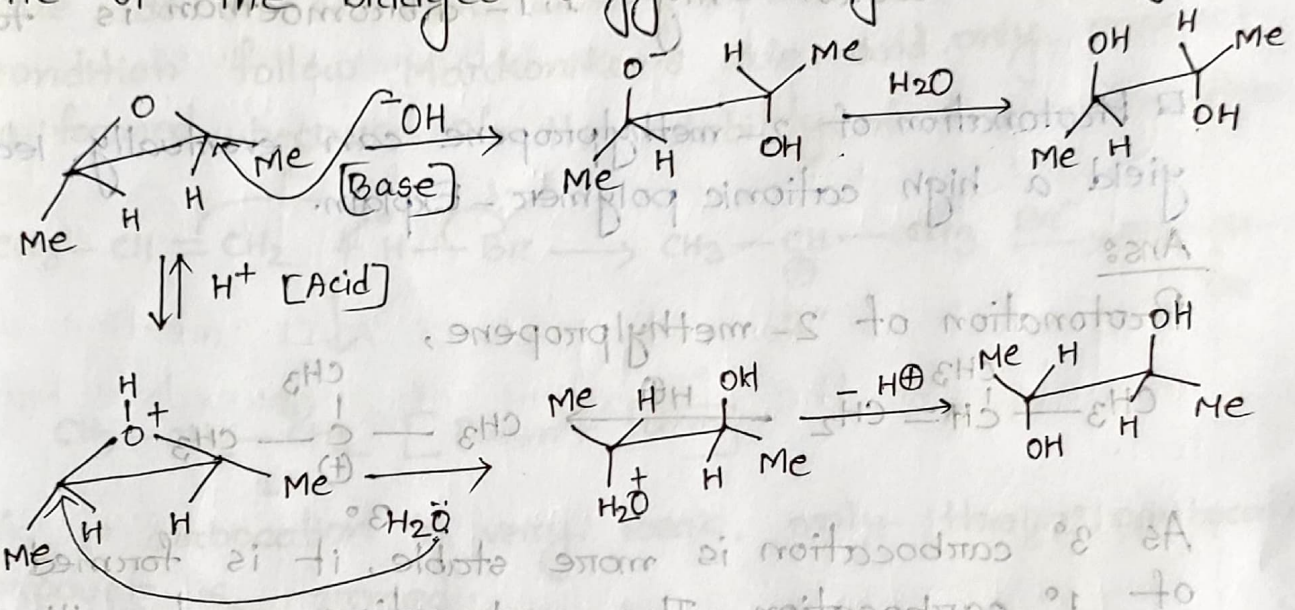
□ An overall ANTI hydroxylation takes place during the reaction of trans-2-butene with a peroxyacid - Explain.

Ans

Trans-2-butene reacts with peroxy acid to form epoxide.



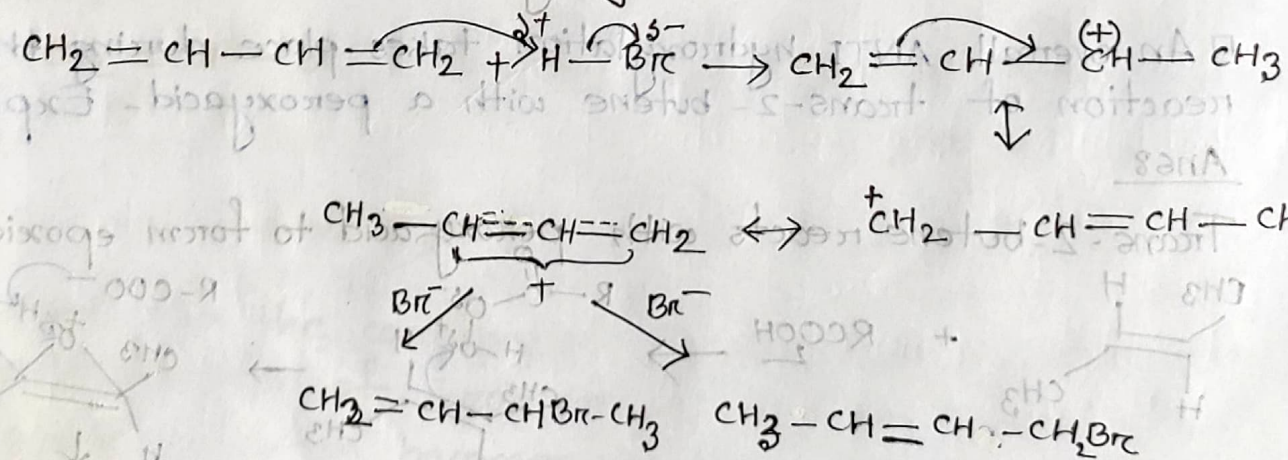
Now, the nucleophile attack under either acid or base catalyzed conditions and produce 1,2-diol. But the nucleophile must attack from the opposite site of the bridged oxygen to yield ANTI hydroxylation.



□ Conjugated dienes are found to undergo addition reaction somewhat more rapidly than non-conjugated dienes - Explain.

Ans:

The reaction of a conjugated diene with a HHal,



In this reaction, the carbocation is more stable because of resonance. As the carbocation is stable, reaction goes faster

But incase of isolated diene no such resonance stabilization occures.

$$CH_2=CH-CH_2-CH=CH_2 + H-Br \longrightarrow CH_3-\overset{\oplus}{C}H-CH_2-CH=CH_2$$

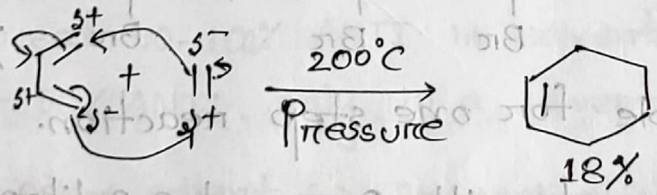
Thus, conjugated dienes undergo rapid addition reaction with nucleophiles than non-conjugated dienes.

□ Butadiene reacts with ethene only at 200°C under pressure and even then to the extent of but 18% compared with almost 100% yield with maleic anhydride in benzene at 15°C. Explain Or, Explain the regioselectivity of Diels-Alder reaction.

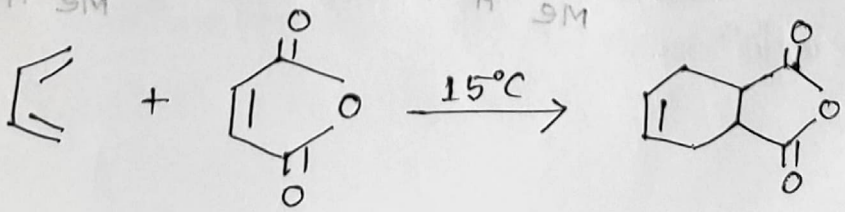
Ans 8

When diene reacts with any double bond or triple bond and give cyclic compound, then the reaction is called Diels Alder reaction. This reaction is influenced when dienophile contain any electron withdrawing group and diene contain any electron donating group.

When Butadiene reacts with ethene, no group is present with either of the reactant which yields a poore product.



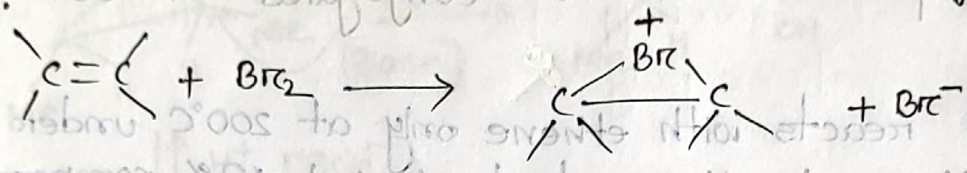
But in case of Malic anhydride, strong electron withdrawing two oxygen are present. They withdraws the electron density from the double bond to stimulate the addition reaction. Thus 100% product is yielded.

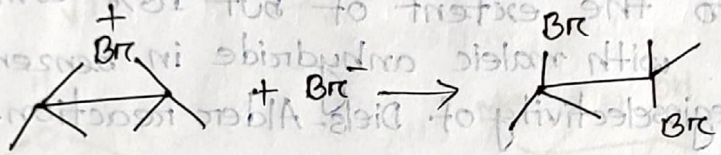


□ Discuss the experimental evidence that refutes the assumption that addition of bromine to carbon-carbon double bond may take place in one step.

Ans:

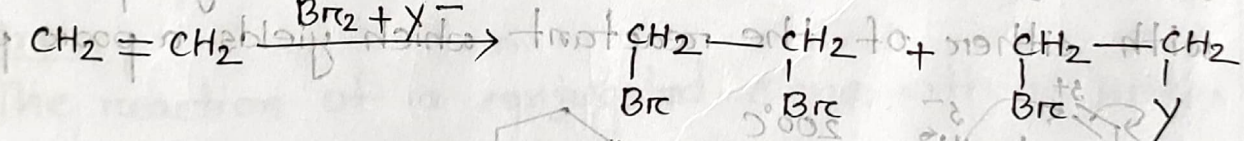
Alkene reacts with Br_2 in the following steps:

1. 

2. 

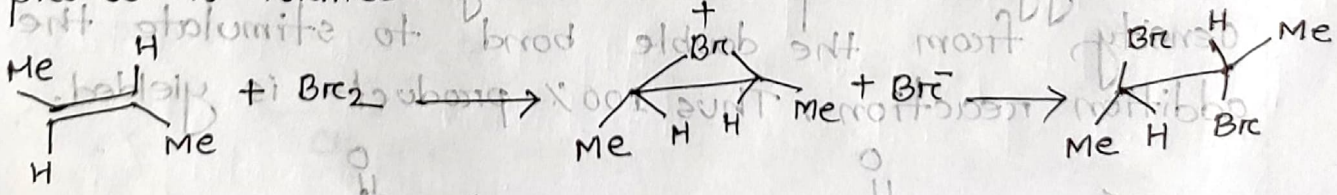
Evidence for two step reaction:

1) If the bromine addition is carried out in presence of another nucleophile Y^-, then two product is obtained: One is expected 1,2-dibromide and another product is one which contains both Br and Y.



This would not possible for one step reaction.

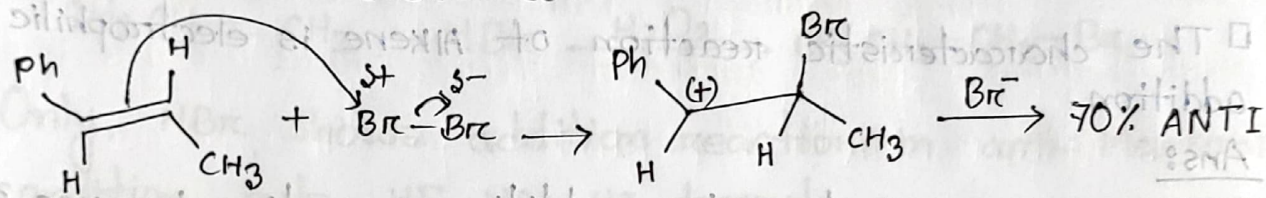
2) Trans 2-butene reacts with Br_2 and 2,3 dibromide is produced and this reaction is 100% Anti and meso product is formed.



□ Why does trans-1-phenyl propene afford only 70% ANTI-addition product when reacted with bromine?

Ans: [Tutorial]

Trans-1-phenyl propene reacts with Br_2 and gives 70% ANTI-addition. This ANTI addition decreases because of the stability of the carbocation. If the stable carbocation is formed because of the stabilization by phenyl group, ANTI addition decreases.



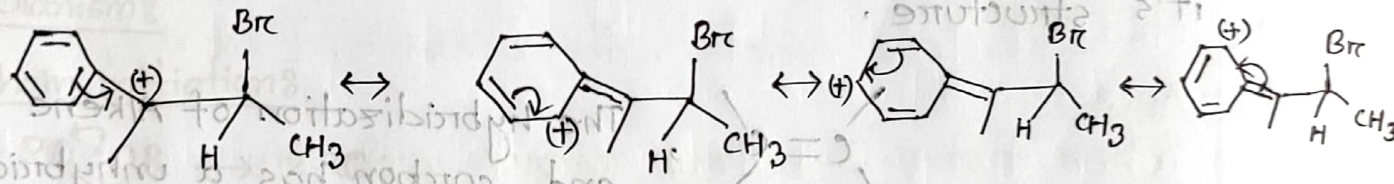


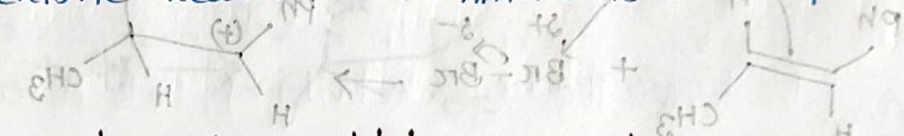
Fig: Resonance stabilization of -ph group.

□ Addition of bromine to 1,2-diphenylethene was found to proceed 90-100% ANTI in solvents of low dielectric constant, but 50% ANTI only in a solvent with $\varepsilon = 35$. Explain.

Non-polar solvent → Bromonium ion → Anti

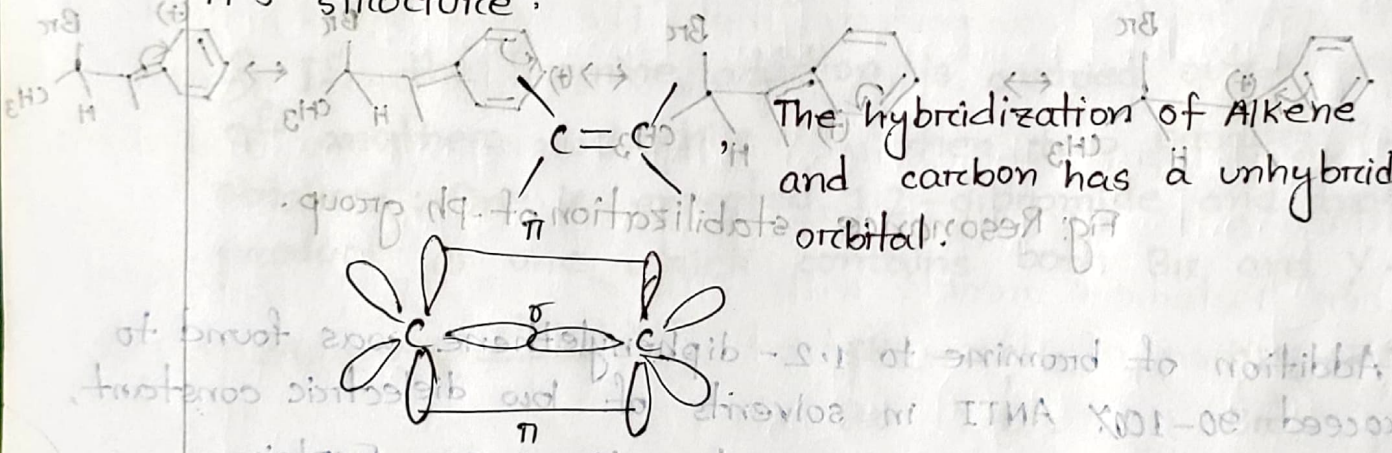
polar solvent → stabilize carbocation → Anti + syn

□ The characteristic reaction of Alkene is electrophilic addition.



Ans:

Alkene shows electrophilic addition reaction because of it's structure.



The hybridization of Alkene is sp^2. and carbon has a unhybridized p orbital.

In alkene, π electron is present which act as a nucleophile. When any electrophile attacks, carbocation is formed. Also π electrons prevent any nucleophile to attack. The reaction is two step reaction of which the addition of the electrophile is the step determining one. Thus, the characteristic reaction of alkene is electrophilic addition.

□ What is anti-Markonicov rule? Explain the fact that HBrc is the only one of the four hydrogen halids which will add readily to alkenes via a radical pathway.

Ans:

Anti Markonikov rule: When alkene reacts with HBrc in presence of peroxides, then hydrogen atom would attach itself to the carbon which has lowerc ~~carbon~~ hydrogen ato

$$CH_3CH=CH_2 + HBrc \xrightarrow{H_2O_2} CH_3CH_2-CH_2-Brc$$

Only HBrc shows addition reaction in anti-Markonikov condition but HF, HCl, HI doesn't.

Mechanism:

Chain initiation:

1. $RO:OR \longrightarrow 2RO^{\cdot}$

2. $RO^{\cdot} + H-Brc \longrightarrow ROH + Brc^{\cdot}$

Chain propagation:

3. $CH_3-CH=CH_2 + Brc^{\cdot} \longrightarrow CH_3-\dot{C}H-CH_2Brc$

4. $CH_3-\dot{C}H-CH_2Brc + H-Brc \rightarrow CH_3-CH_2-CH_2Brc + Brc^{\cdot}$

Chain Termination:

5. $Brc^{\cdot} + Brc^{\cdot} \longrightarrow Brc_2$

In the reaction forc HBrc addition, both the chain propagation steps are exothermic which stimulates the addition process.

$X^{\cdot} + CH_2 = CH_2 \rightarrow CH_2 - \dot{C}H_2 + HX$

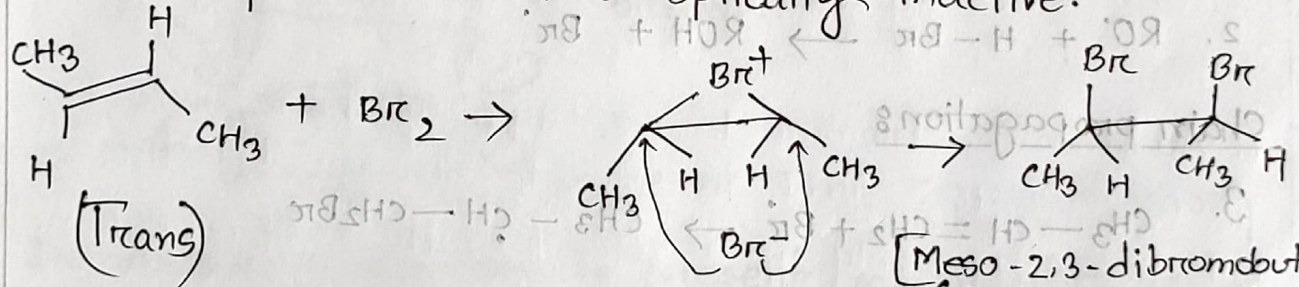
		X	
HF	-188		+155
HCl	-109		+21
HBr	-21		-46
HI	+29		-113

But for other hydrogen halide, either of the step is endothermic. Thus without HBr, none of the hydrogen halide shows anti-Markonikov addition.

□ What products would you get when trans-2 butene and cis-2-butene reacts with bromine? [Tutorial]

Ans:

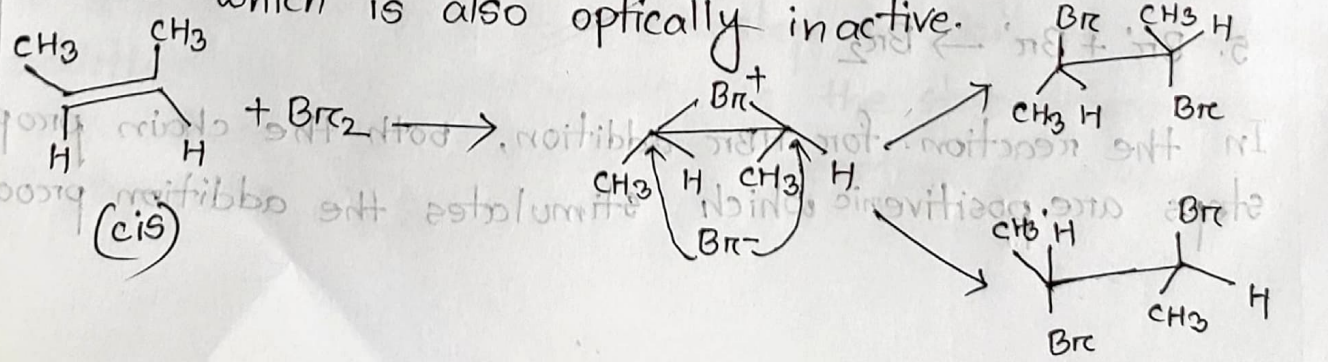
When trans 2-butene reacts with Br_2, it produces Meso compound which is optically inactive.



(Trans)

[Meso-2,3-dibromobutane (Optically inactive)]

When cis-2-butene reacts with Br_2, it produces racemic mixture which is also optically inactive.



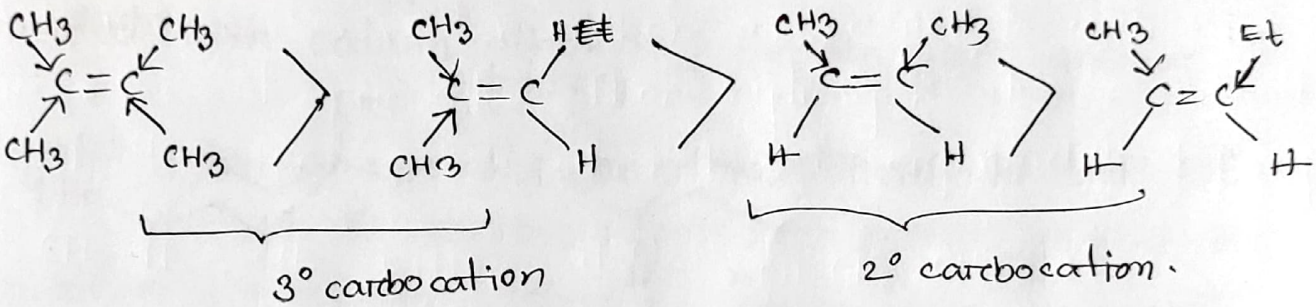
(cis)

Rac-2,3-dibromobutane

□ Arrcongge the following alkenes with respect to their relative rates in electrophilic addition reaction.

$(CH_3)_2C = C(CH_3)_2$, $CH_3CH = CHCH_3$, $EtCH=CHCH_3$, $CH_2 = C(CH_3)_2$

Ans:

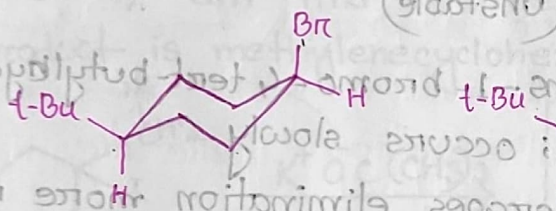


3° carbocation

2° carbocation.

ELEMINATION REACTIONS [Faria-50]

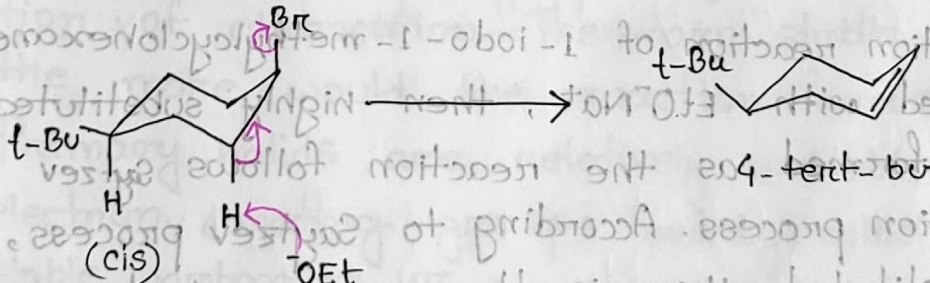
□ Which diastereomer of 1-bromo-4-t butylcyclohexane, the cis or the trans, undergoes elimination more rapidly when treated with sodium ethoxide? Explain your answer.



Ans:

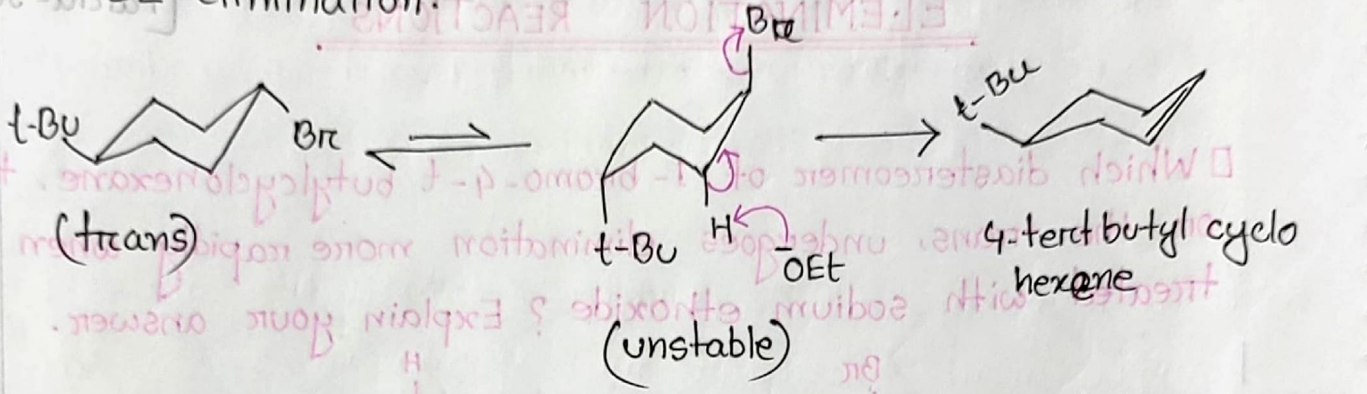
Here, the bulky tert-butyl group remains present at the equitorial position of the chair conformation of the cyclohexane.

In cis-1-bromo-4-tbutylcyclohexane, the removal of H and Br atom is easy as the atoms remain present in an antiperiplanar position to facilitate the E2 elimination with sodium ethoxide in ethanol.



But, trans-1-bromo-4-t-butylcyclohexane doesn't show antiperiplanar orientations. However, it must undergo unfavourable ring flipping to achieve the antiperiplanar orientations of the removal H and Br atoms to produce

an E2 elimination.

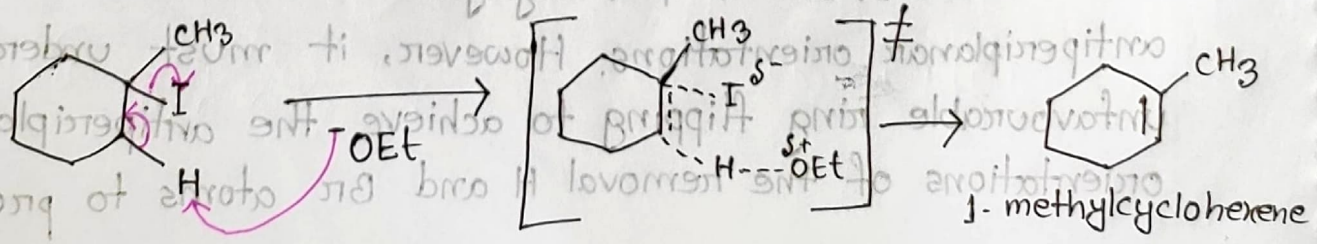


So, the elimination in trans-1-bromo-4 tert-butylcyclohexane is highly energy requiring; occurs slowly.
Therefore, cis isomer undergoes elimination more rapidly.

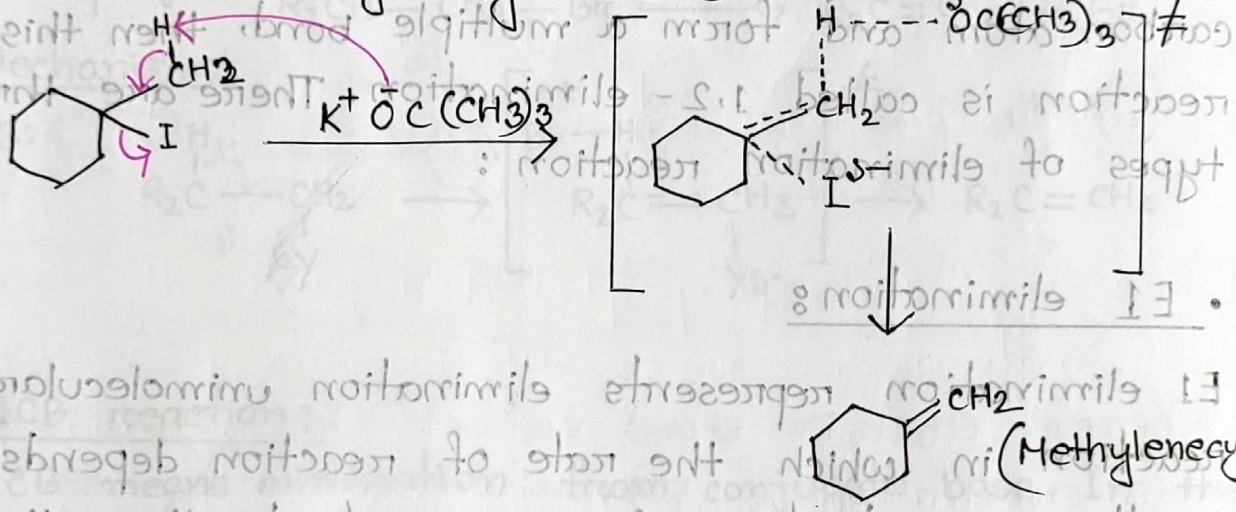
□ When 1-iodo-1 methylcyclohexane is treated with sodium ethoxide the more highly substituted alkene product predominates whereas the same reaction treated with potassium tert-butoxide gives less substituted alkene predominantly. Give the structures of the two products and offer an explanation.

Ans:

The elimination reaction of 1-iodo-1-methylcyclohexane when treated with EtO^-Na^+, then highly substituted product is formed as the reaction follows Saytzev elimination process. According to Saytzev process, highly substituted alkene is the major product. It follows simple E2 mechanism as EtO^-Na^+ has no steric effect.



But, when the compound is treated with potassium-t-butoxide it yields Hofmann product. According to Hofmann rule, less substituted alkene is formed. The base is strong and bulky which is highly sensitive to steric interactions. So, it attacks the least hindered and most accessible β-H which it can find in the methyl ($-CH_3$) group. Thus the product is methylenecyclohexane.



□ Primary halids hardly undergo E1 elimination. Explain.

E1 elimination is basically the process which has two steps of which the rate determining step includes the formation of carbocation. The more stable the carbocation, the more would the reaction favour E1 mechanism. Primary halids are relatively unstable as they lack electron donating groups nearby which can make a stable carbocation. Thus, primary halids generally doesn't undergo E1 elimination.

□ Show the three different schematic representation of mechanism for 1,2A elimination reaction.

Ans:

1,2 or β-elimination: The elimination reaction in which the atoms or groups are eliminated from adjacent carbon atom and form a multiple bond, then this reaction is called 1,2- elimination. There are three types of elimination reaction:

• E1 elimination:

E1 elimination represents elimination unimolecular. The reaction in which the rate of reaction depends only on the concentration of one reactant, then the reaction is called E1 mechanism. eli reaction.

$$\text{Rate} = K_1 [\text{substrate}]$$

Example: $CH_3-CR_2-Y \xrightarrow{EtOH} CH_2=CR_2 + Y^- + EtOH_2^+$

Mechanism:

step-01:

$$H_2C(H)-CR_2-Y \underset{K_1}{\overset{slow}{\rightleftharpoons}} H_2-C(H)-CR_2^+ + Y^-$$

step-02:

$$EtOH + H_2C(H)-CR_2^+ \xrightarrow{fast} CH_2=CR_2 + EtOH_2^+$$

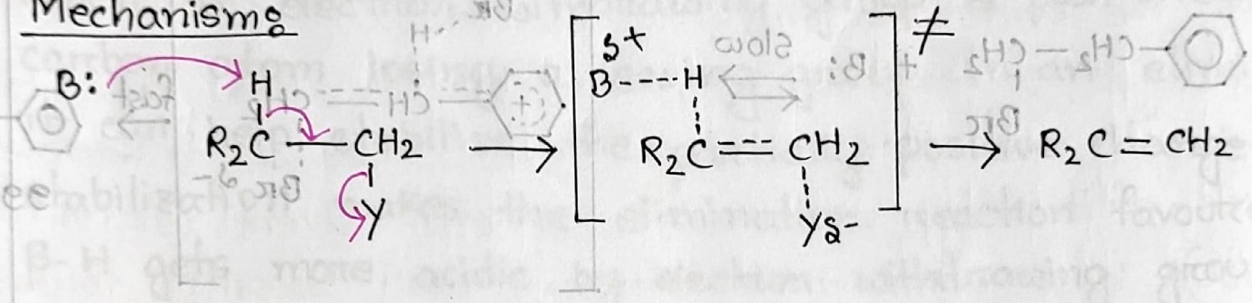
E2 reaction:

E-2 means elimination bi-molecular. The elimination reaction in which rate of the reaction depends on the concentration of both substrate and base; this reaction is called E2 reaction

$$\text{Rate} = K[\text{substrate}][\text{Base}]$$

Example: $R_2CH-CH_2-Br \xrightarrow{\text{Base}} R_2C=CH_2 + Br^-$

Mechanism:



E1CB reaction:

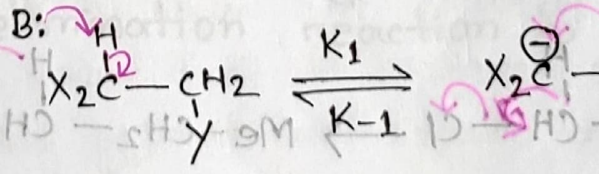
E1CB means elimination from conjugate base. In the reaction base abstract β-H and form carbanion intermediate and finally product is found. The first step is fast and reversible, whereas the 2nd one is slow and the rate determining one where y- group leaves.

$$\text{Rate} = K[\text{substrate}][\text{Base}]$$

Example: $X_2C(H)-CH_2Y \rightleftharpoons X_2C=CH_2 + Y^-$ (Base)

Mechanism:

Step 01:

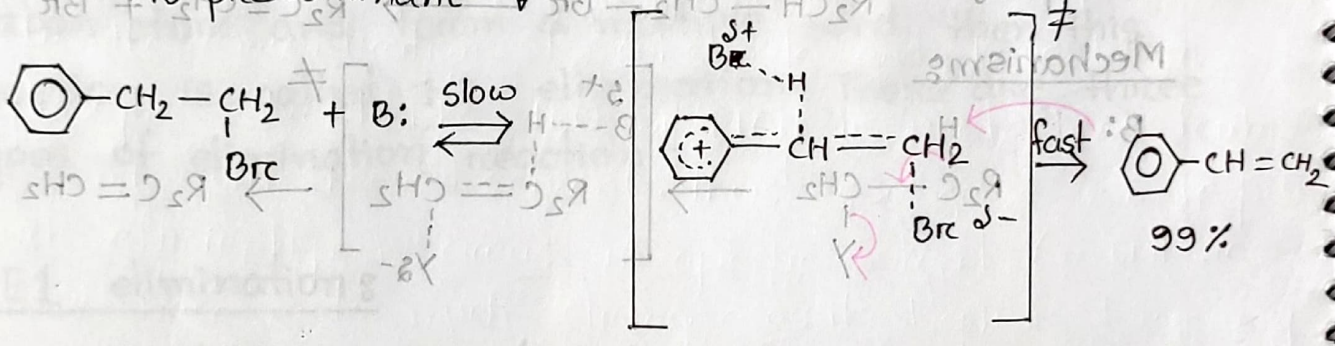


Step 02: $X_2\overset{\ominus}{C}-CH_2Y \longrightarrow X_2C=CH_2 + Y^-$

☐ Under comparable conditions, CH_3CH_2Br yields ≈ 1% alkene while $PhCH_2CH_2Br$ yielded ≈ 99% alkene - explain.

Ans:

$PhCH_2CH_2Br$ gives elimination reaction and 99% alkene is produced because the phenyl group present can stabilize the developing double bond by conjugation. Thus elimination product is predominant.



Whereas, CH_3CH_2Br shows no net stabilization and for this substitution is predominat rather than elimination (1%).

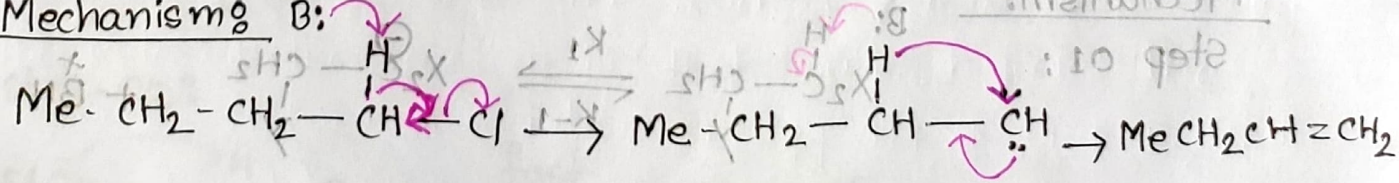
☐ What do you mean by 1,1-elimination reaction. Give example.

Ans:

Reaction in which both H and the leaving group, Y are lost from the same (α) carbon atom is termed as 1,1-elimination reaction. They tend to favoured by powerful electron-withdrawing Y groups and very strong bases.

Example: $MeCH_2CH_2CH_2-Cl \xrightarrow{Base} MeCH_2CH=CH_2$

Mechanism: B:

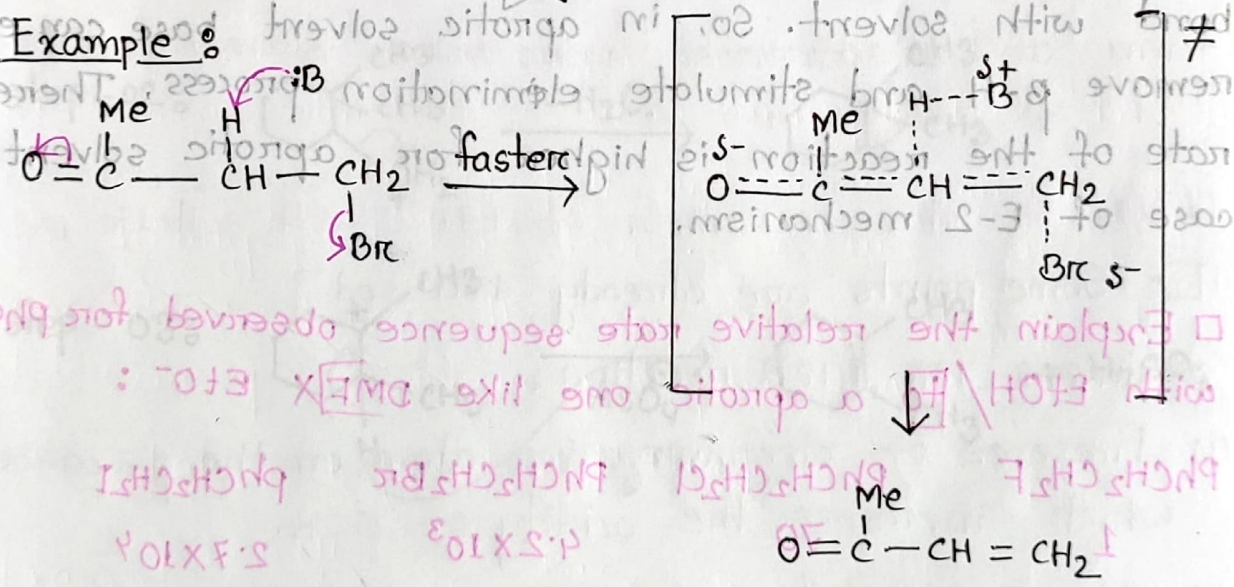


□ Elimination is promoted by electron withdrawing substituents, Explain the statement with appropriate example.

Ans:

Elimination is promoted by electron withdrawing group such as $-CF_3$, $-NO_2$, $-CHO$, $C=O$ etc. Electron withdrawing substituents pull electron density away from the molecule. When an electron withdrawing group is positioned near the carbon atom losing a leaving group in an elimination, it can help stabilize the partially positive charge. This stabilization makes the elimination reaction favourable. Also, β-H gets more acidic by electron withdrawing group.

Example:



Thus, extra stabilization by electron withdrawing group causes the elimination reaction to be faster.

Reason-2:

(i) stabilize partially positive charge

(ii) makes β-H more acidic

□ With a given base, transfer from hydroxylic solvent like H_2O or EtOH to a aprotic one like DMF or DMSO can have a profound effect on the rate of E2 reactions- explain.

Ans:

In E2 mechanism, the rate of the reaction depends on both the concentration of substrate and base. When hydroxylic solvent like H_2O or EtOH is used, base form H-bond with these solvent. Thus base can't remove β-H much efficiently. Whereas in aprotic solvent (ie. DMF or DMSO) base acts much stronger as it doesn't form any bond with solvent. So, in aprotic solvent base can easily remove β-H and stimulate elemination process. Therefore, rate of the reaction is higher for aprotic solvent in case of E-2 mechanism.

□ Explain the relative rate sequence observed for $PhCH_2CH_2Hal$ with EtOH/ to a aprotic one like DMF X EtO^- :

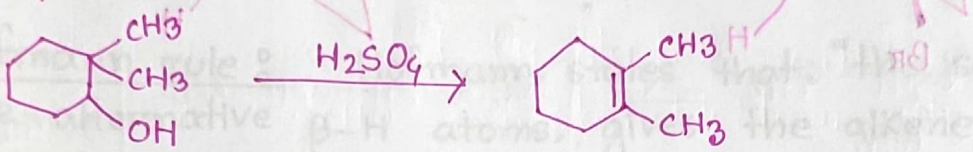
$PhCH_2CH_2F$	$PhCH_2CH_2Cl$	$PhCH_2CH_2Br$	$PhCH_2CH_2I$
1	70	4.2×10^3	2.7×10^4

Ans:

In elimination reaction, the much C-Y bond is weaker, the rate of the reaction is that much higher. The sequence of bond dissociation energy R-F > R-Cl > R-Br > R-I. Thus R-I has the weakest bond and I^- acts as a good leaving group. Thus for elimination reaction breaking breaking bond between carbon and I is easier than

others. Thus moving forward from F to I, bonds get weaker and elimination reaction rate get faster.

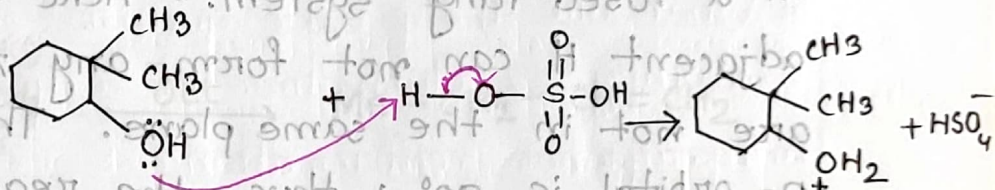
□ Propose a mechanism:



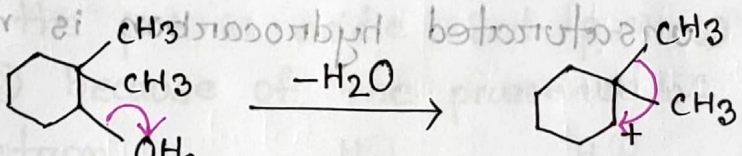
Ans:

The above reaction follows E-1 pathway where the carbocation is formed, then rearrangement occurs.

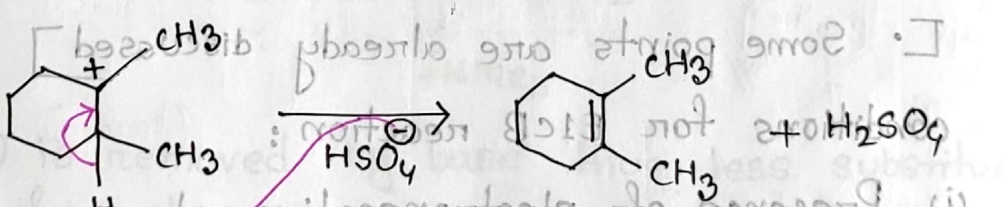
step-01:



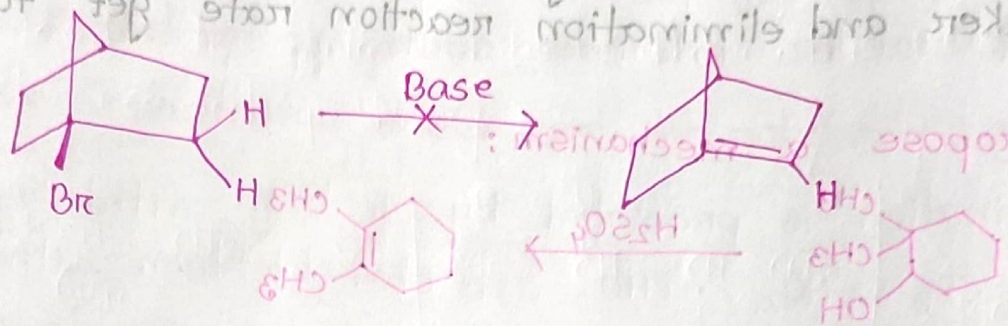
step-02:



step - 03:



□ The following reaction will not take place. explain:



Ans:

According to Bredt's rule, "It is not normally possible to introduce a double bond on a bridged carbon atom in a fused ring system." Here p orbitals of Br and adjacent H can not form any π-bond because they are not in the same plane. The angle between two p-orbital is 90° ; thus the reaction of elimination to form an unsaturated hydrocarbon is not possible.

□ Discuss the mechanism of E1cB with suitable example.

[· Some points are already discussed]

conditions for E1cB reaction :

(i) Presence of electronegative atom on the β-carbon which increase the acidity of β-H.

(ii) Stabilization of carbanion.

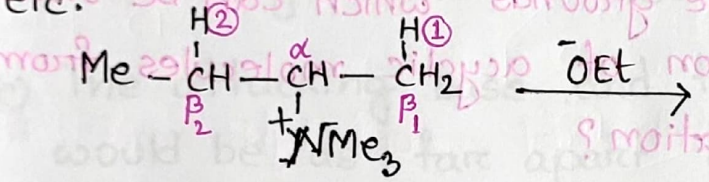
(iii) A poor leaving group.

□ What are the Hofmann and Saytzev rules of elimination? Explain the rules using appropriate examples:

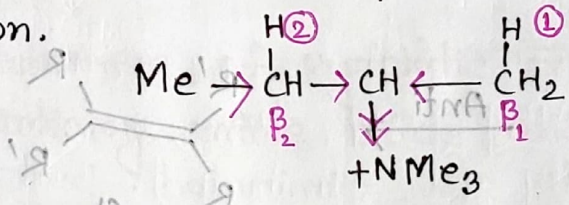
Ans:

• **Hofmann rule:** Hofmann states that, "the substrate which have alternative β-H atoms, gives the alkene as a major product with least alkyl substituents."

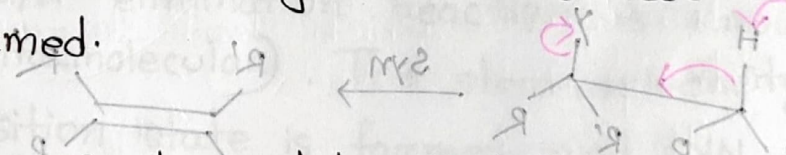
Hofmann elimination occurs when the leaving group is a powerful electron withdrawing group like $\overset{+}{N}Me_3$, $\overset{+}{S}Me_2$ etc.



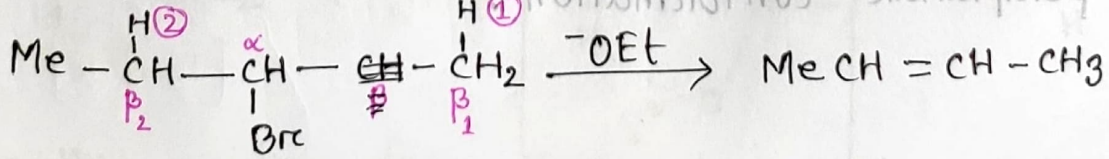
Here both the β-H is acidic, But β-H (1) is more acidic than β-H (2) because of the presence of Me group which donate electron.



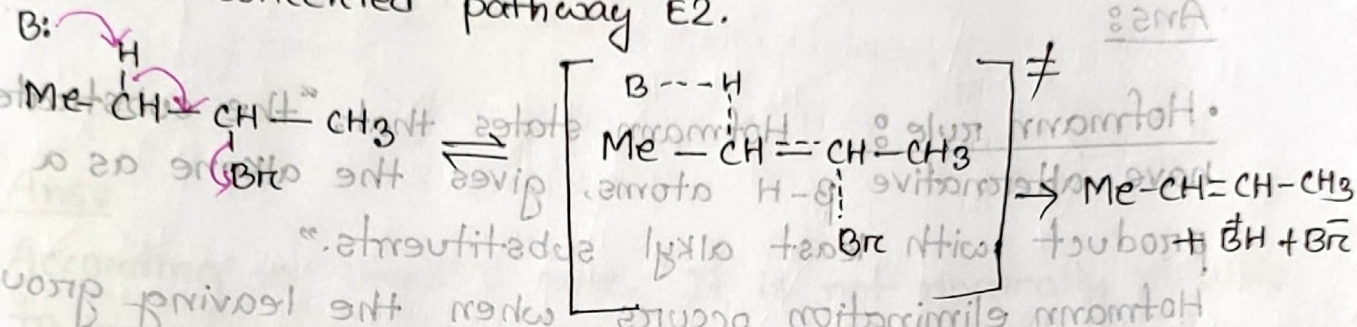
So, β-H (1) is removed by base thus less substituents product is formed.



• **Saytzev rule:** Saytzev states that, " In dehydrohalogenation, the preferred product is the alkene that has the greater number of alkyl groups attached to the doubly bonded carbon.

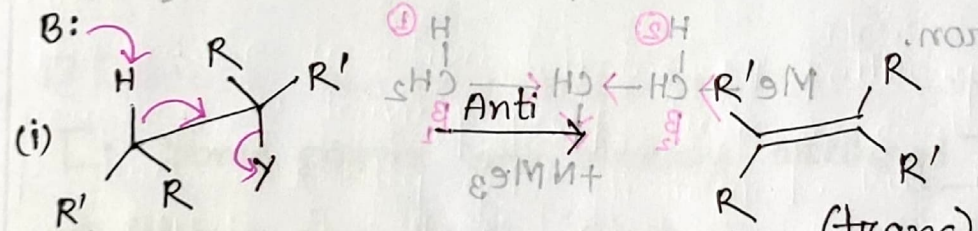


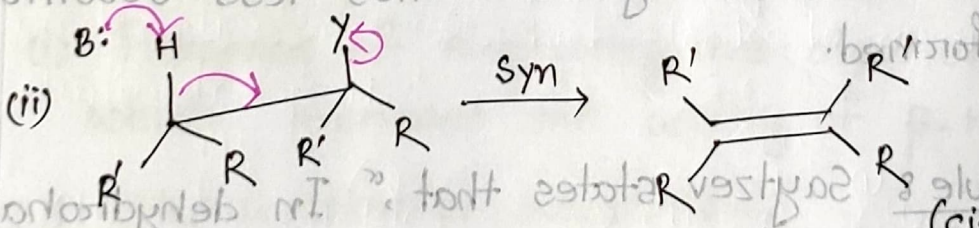
This elimination appears when leaving group is neutral which leads to more stable alkene. This reaction follows concerted pathway E2.



□ What are the possible grounds which can be stated for favouring E2 elimination of acyclic molecules from the antiperiplanar conformation?

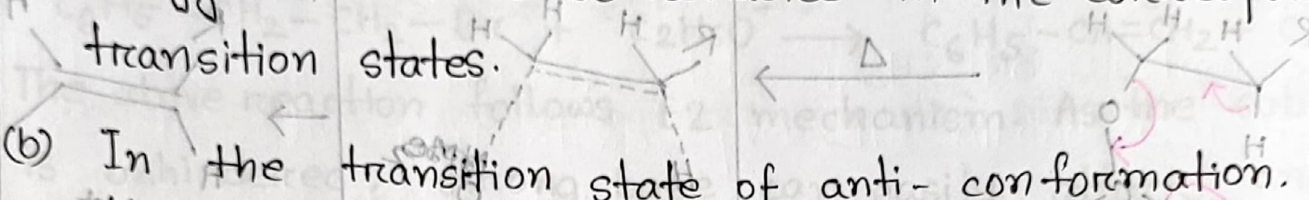
Elimination should have been occurs for both anti and syn periplanar conformation.





But there are three possible grounds which can be stated for favouring E2 elimination of acyclic molecules from anti- periplanar conformation:

(a) In elimination reaction, staggard conformation must be present because of containing lower energy. On the other hand, eclipsed conformation contain larger energy which is to be reflacted in the corresponding transition states.



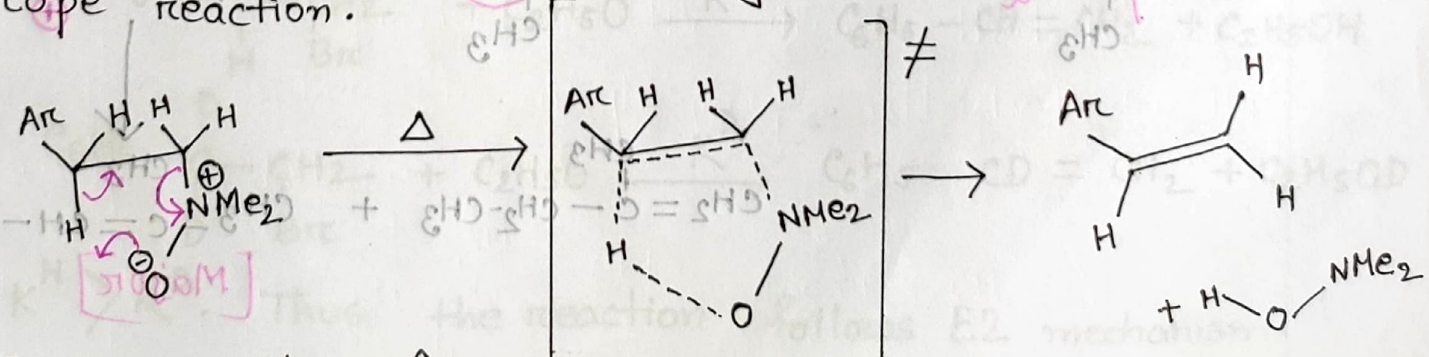
(b) In the transition state of anti- conformation, both the carbon of, H and leaving groups are in the same plane which causes their p orbital to overlap at a larger extent. Thus the elimination is stimulated.

(c) The attacking base and the departing leaving group would be as far apart from each other as possible. Thus, anti periplanar configuration is more favourable.

□ Give the mechanism of pyrolytic SYN elimination reaction considering tertiary amine oxide and xanthate as examples.

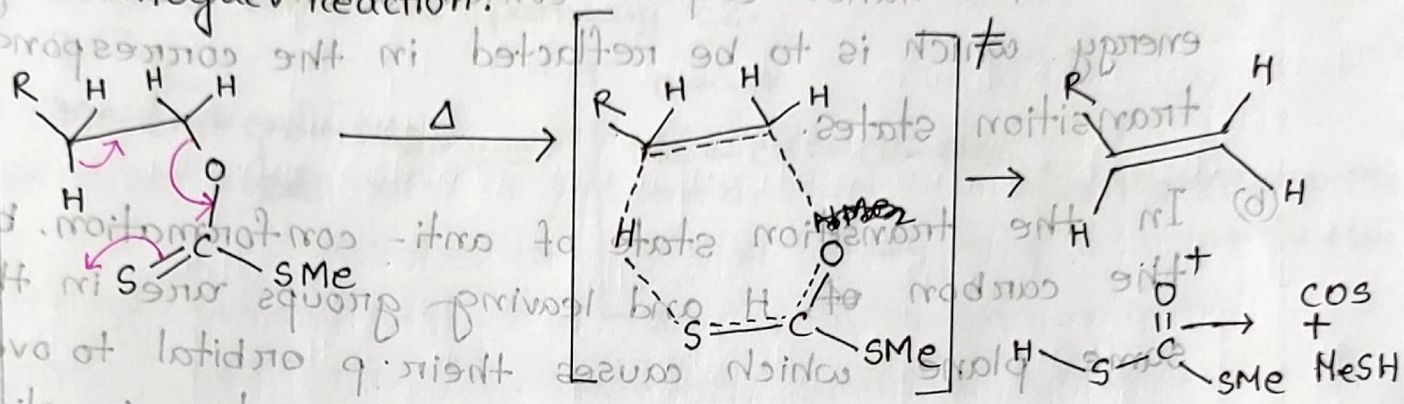
Ans:-

The pyrolytic SYN elimination reaction is known as Ei (Elimination intramolecular). The stereoselectivity is SYN and cyclic transition state is formed and SYN product is found. The reaction of tertiary amine is also known as cope reaction.



In this reaction, five membered transition state is formed.

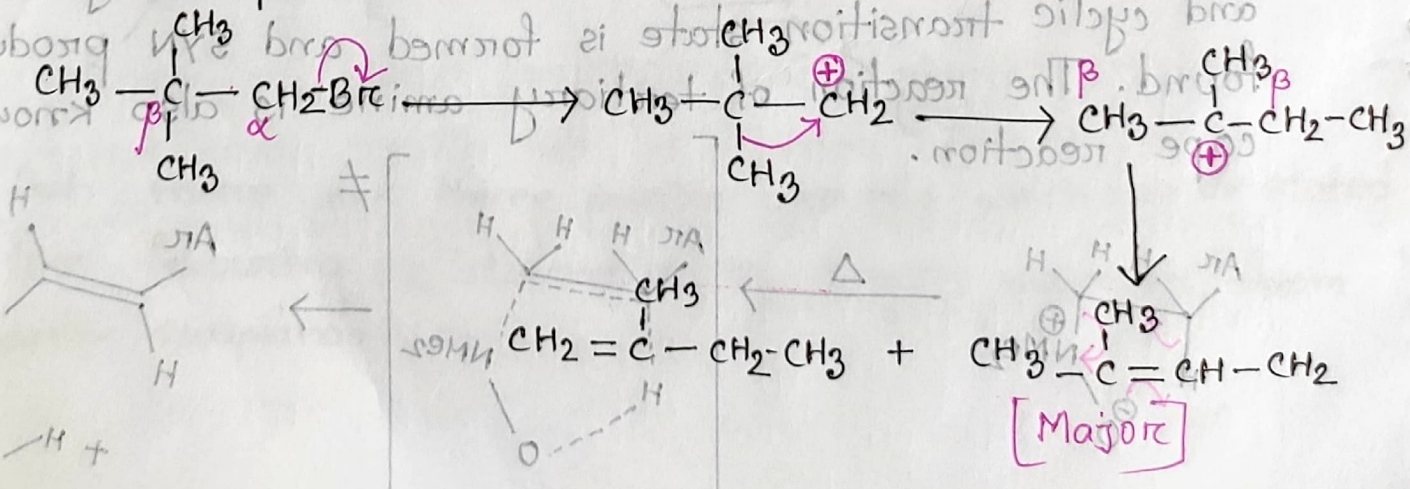
Now, the reaction of xanthates and in the transition state six membered ring is formed. This reaction is known as Chugaev reaction.



□ In the elimination reaction of neopentylbromide, the major product found is $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$. Propose a mechanism for the reaction.

Ans:

The above reaction follows E1 pathway. For 1,2-elimination reaction, a leaving group from α-carbon and a H from β-carbon usually gets eliminated. But in neopentylbromide no β-H is found. By following E1 mechanism at first carbocation occurs which leads to re-arrangement to eliminate β-H.

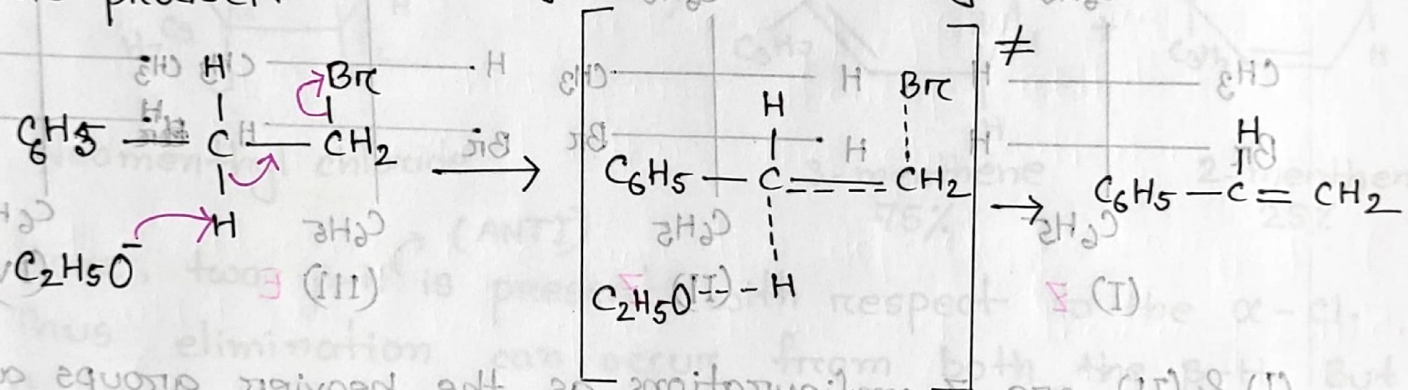


□ What pathway $C_6H_5-CH_2-CH_2-Br$ follows when reacted to $C_2H_5O^-$ to produce unsaturated hydrocarbon? Give evidence

Ans:

$$C_6H_5-CH_2-CH_2-Br + C_2H_5O^- \longrightarrow C_6H_5-CH=CH_2 + C_2H_5OH$$

The above reaction follows E2 mechanism. As the substrate is unhindered, strong base can easily attack and form the product.



The mechanism can be explained by isotopic effect. If we replace the β-H with isotope deuterium (D), it is found that the reaction rate would be higher with β-H than β-D. As D is heavier than H, it takes more time to break the bond between D-C than H-C. It seems that rate determining step must involve the bond breaking of H-C / D-C which causes the rates to be different.

$$C_6H_5-\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}}-CH_2 + C_2H_5O^- \xrightarrow{K^H} C_6H_5-CH=CH_2 + C_2H_5OH$$

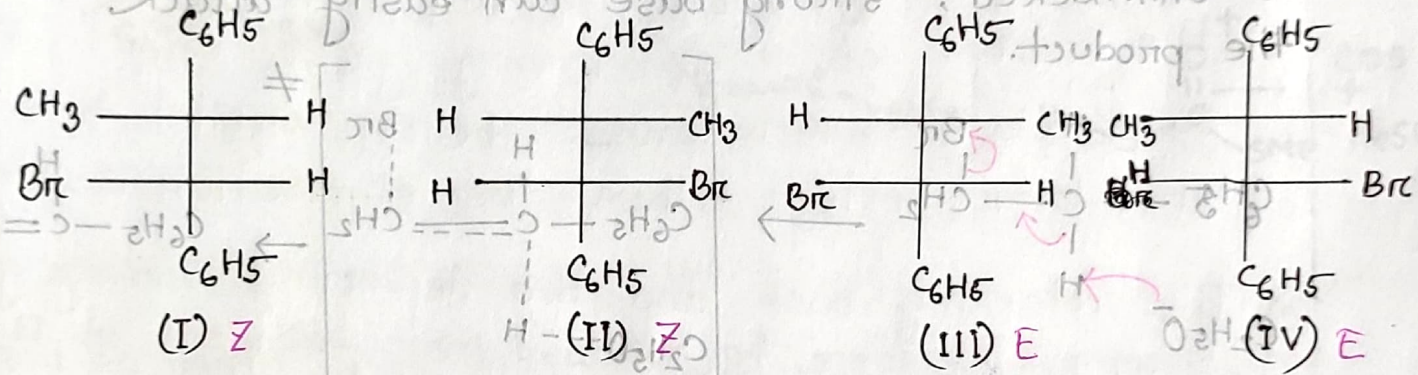
$$C_6H_5-\underset{\underset{D}{|}}{\overset{\overset{D}{|}}{C}}-CH_2 + C_2H_5O^- \xrightarrow{K^D} C_6H_5-CD=CH_2 + C_2H_5OD$$

$K^H > K^D$. Thus the reaction follows E2 mechanism.

□ Which configuration (E/Z) of $C_6H_5-\underset{\underset{Br}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}-C_6H_5$ would produce elimination product?

Ans:

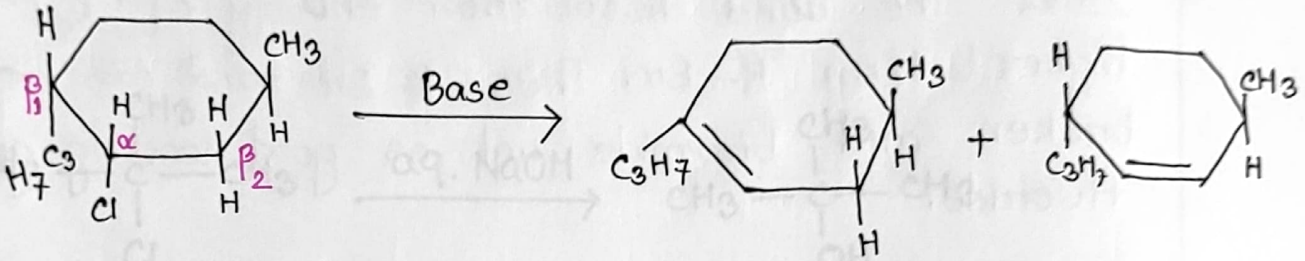
As $C_6H_5-\overset{*}{\underset{\underset{Br}{|}}{CH}}-\overset{*}{\underset{\underset{CH_3}{|}}{CH}}-C_6H_5$ has two chiral centers, total four isomeric (stereo) configuration can be found.



(I) & (II) are Z configurations as the heavier groups at same side and (III) & (IV) are E configurations as the heavier groups at different side. In case of Z configurations, the Br and H are at anti position which may successfully leads to elimination. But at E configuration, Br and H atoms are at syn position which can't favour 1,2- elimination reaction. Thus the product obtained would be from Z isomers.

□ Neomenthyl chloride produces 3-menthene as major product whereas Menthyl chloride produces only 2-menthene in elimination reaction. Explain.

Ans:

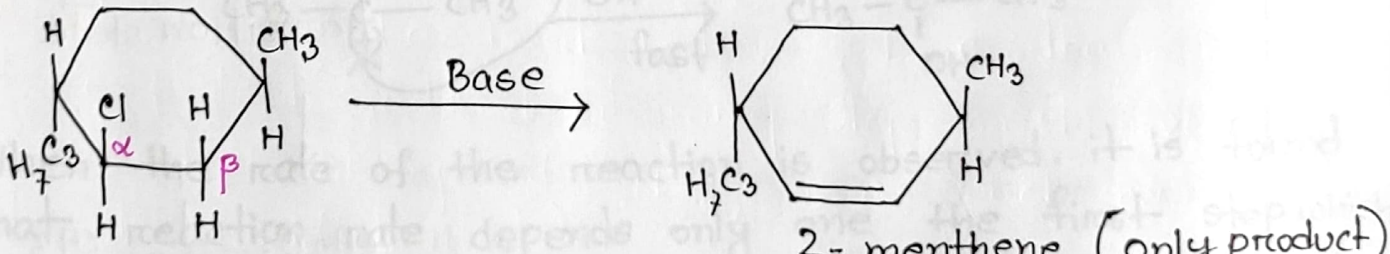


Neomenthyl chloride

3-menthene 75%

2-menthene 25%

Here, two β-H (ANTI) is present with respect to the α-Cl. Thus elimination can occur from both the β-H. But β_1 - H is eliminated as major because propyl group attached to it can stable the double bond more efficiently. Thus 3-menthene is produced as major product.



Menthyl Chloride

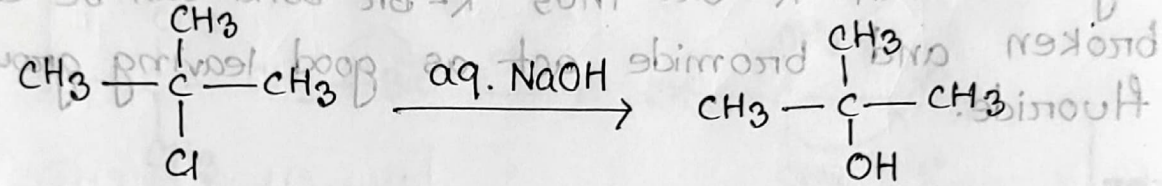
2-menthene (only product)

Here, only one β-H at anti position is present which can lead to 1,2-elimination with Cl. Thus only one product i.e. 2-menthene is obtained.

SUBSTITUTION REACTIONS

□ Reaction between 2-chloro-2-methylpropane with aq. NaOH gives tert-butanol. How do you prove that the reaction follows preferably S_N1 path showing an evidence.

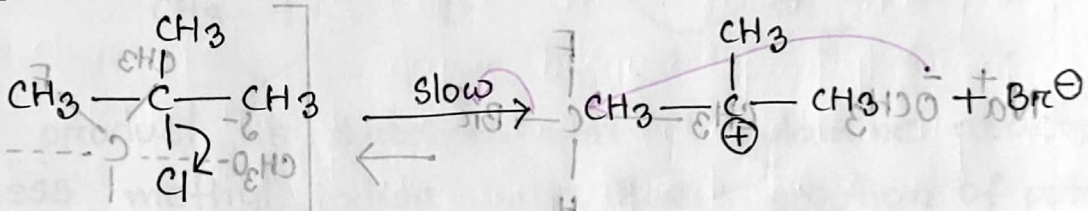
Ans:



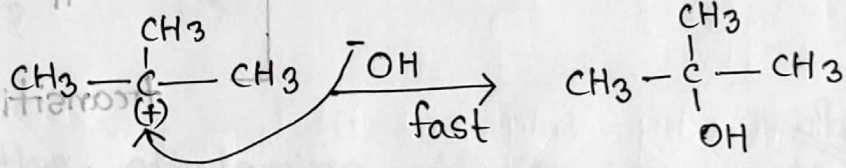
2-chloro-2-methylpropane → tert-butanol.

The above reaction proceeds as follows:

Step-1:

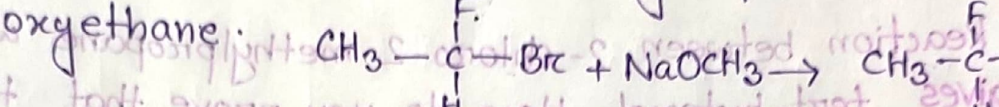


Step-2:



When the rate of the reaction is observed, it is found that reaction rate depends only one the first step which involves the formation of carbocation. If we increase/decrease the concentration of the substrate, the reaction rate will increase or decrease respectively. But increasing/decreasing the concentration of the neucleophile [OH^-] can not make any difference in the reaction rate. So, the rate depends on only substrate which indicates the nucleophilic substance unimolecular. Thus the reaction follows S_N1 pathway. [Also by stereochemistry: racimization + Inversion]

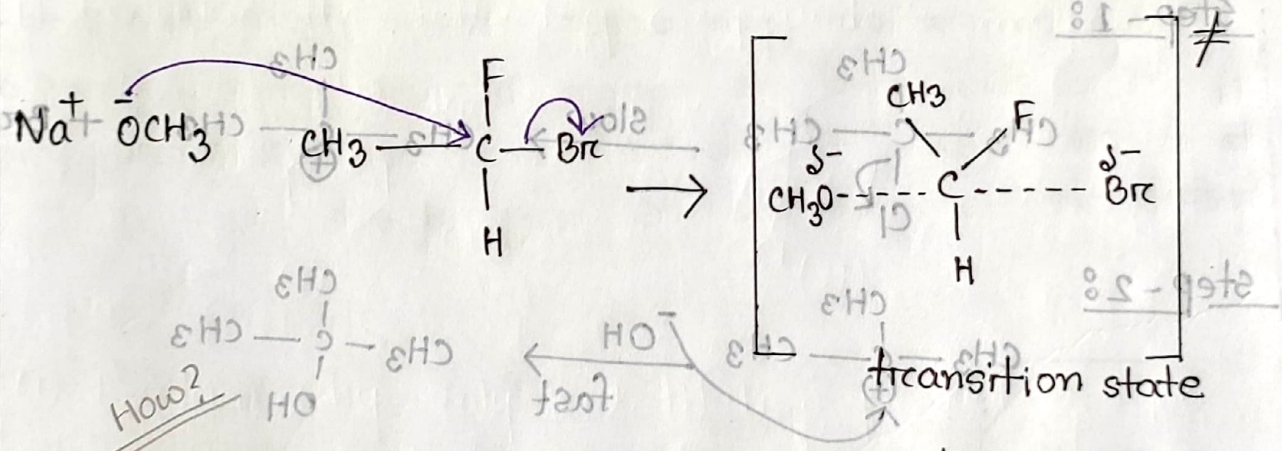
□ Under appropriate conditions, (S)-1-bromo-1-fluoroethane reacts with sodium methoxide to give pure (S)-1-fluoro-1-methoxyethane:



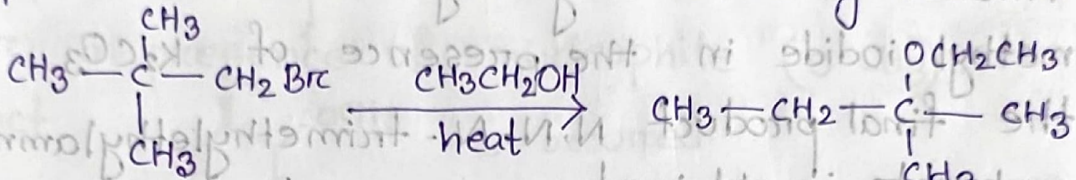
(i) Why is bromide rather than fluoride replaced?

Ans: The bond dissociation energy of R-F is much higher than R-Br. Thus R-Br bond can be easily broken and bromide act as good leaving group than fluoride.

(ii) Draw respective structures for the starting material, the transition state and the product.

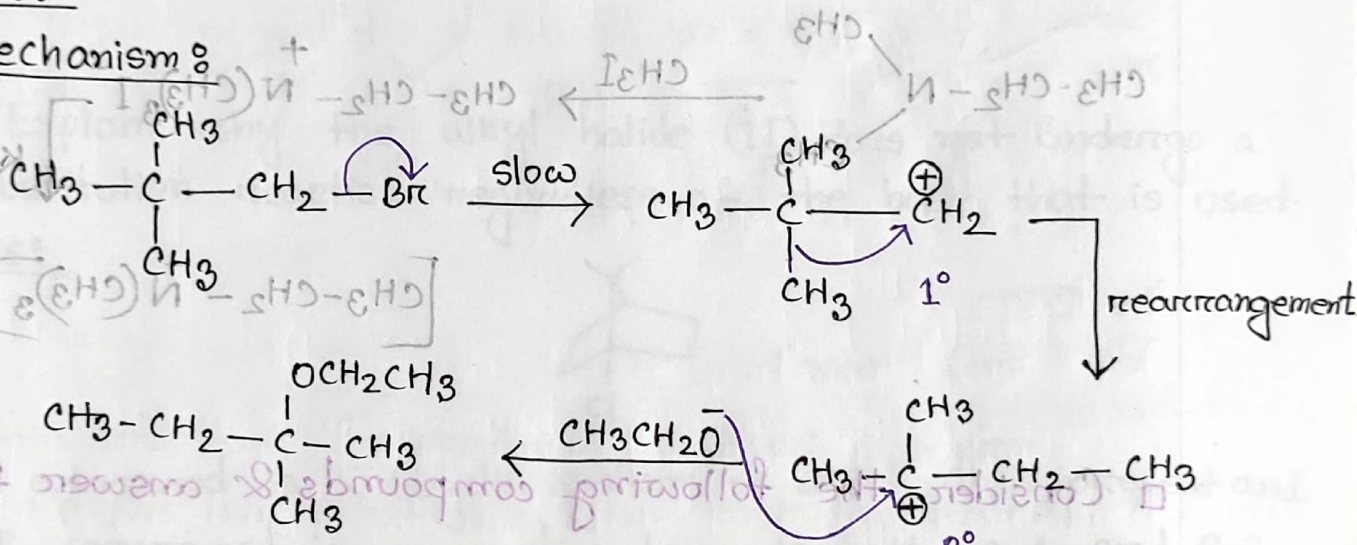


(b) Propose a mechanism for the following reaction:



Ans:

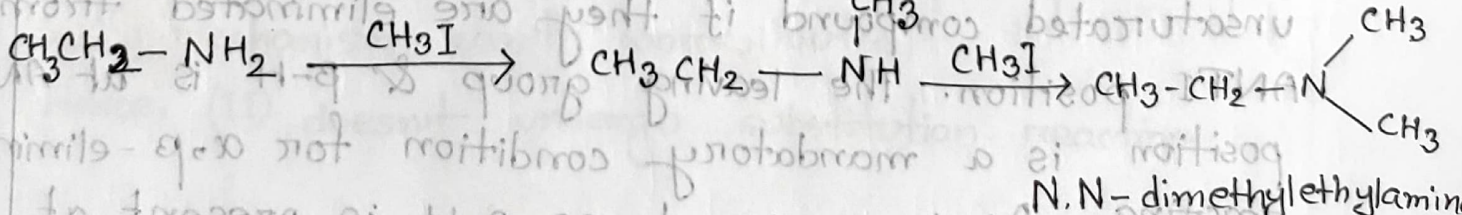
Mechanism:



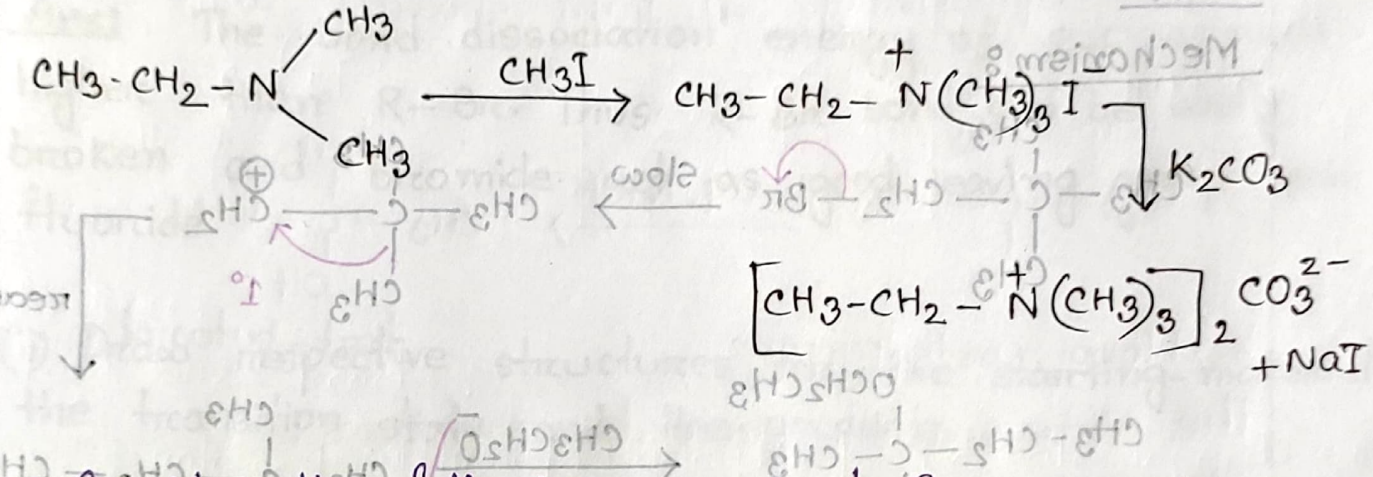
□ What product is obtained when ethylamine reacts with excess methyl iodide in a basic solution of potassium carbonate?

Ans:

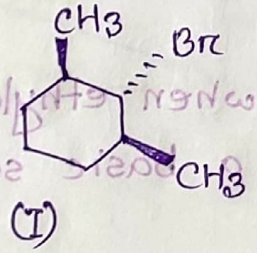
Firstly, the ethylamine attacks the methyl iodide, forming N-methyl ~~iodide~~ ethyl amine. However, due to the excess methyl iodide, the N-methylethylamine can further react with methyl iodide to form N,N-dimethylethylamine.



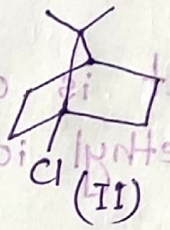
When, N,N-dimethylethylamine reacts with excess methyl iodide in the presence of K_2CO_3 solution, the final product N,N,N-trimethylethylammonium carbonate is obtained.



□ Consider the following compounds & answer the Ques:



(I)

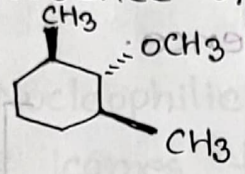


(II)

(i) Explain why only substitution product and no elimination product is obtained when compound (I) reacts with $NaOCH_3$.

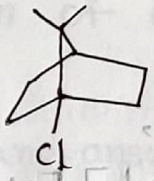
Ans:

In the elimination reaction, a leaving group from α-position and a H-form β position can form unsaturated compound if they are eliminated from ANTI position. The leaving group & β-H is at ANTI position is a mandatory condition for α,β-elimination reacting. As in compound no β-H is present at

ANTI position, α, β- elimination can not take place. Rather in presence of nucleophile (O^-CH_3), substitution product  is occured.

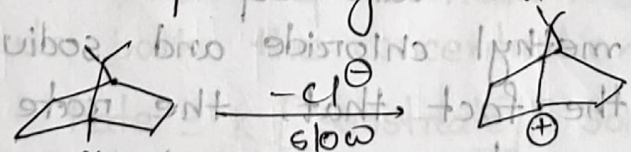
(ii) Explain why the alkyl halide (II) does not undergo a substitution reaction regardless of the base that is used.

Ans:



The compound is a bicyclic system, in which, Cl is present and this compound is unreactive towards both S_N1 and S_N2 reaction.

In S_N1 reaction, the first step is slow and rate determining step where carbocation is formed. But here carbocation can't form as it is sp^2 hybridized and planar in structure. But by bicyclo compound can't form planar structure as it ~~it~~ disrupts ring strain. Thus S_N1 is not possible.

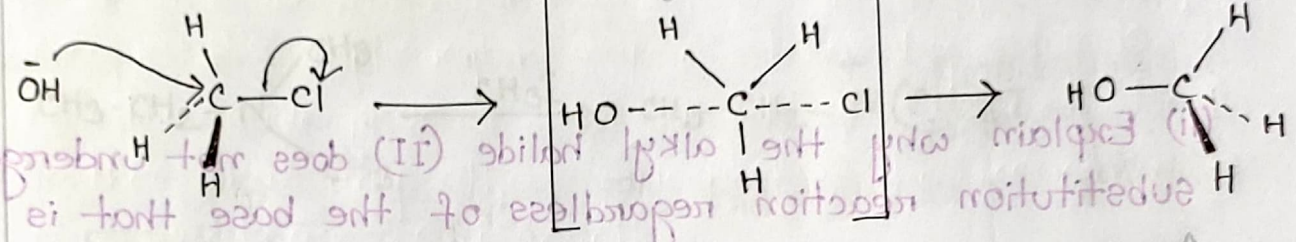


In S_N2 reaction, nucleophile comes from backside but in bicyclo compound, backside attack is perhaps not possible. So, S_N2 mechanism can't work out.

Hence, (II) doesn't undergo substitution reaction.

Q Reaction between methyl chloride with aq. NaOH gives methanol. How do you prove that the reaction follows S_N2 path showing an evidence.

Ans:



Evidence:

For S_N2 reaction,

rate $\propto$ [substrate] [Nucleophile] i.e. rate depends on the concentration of both substrate and nucleophile.

In the above reaction, discussing the kinetic rate profile would be enough as an evidence. If the concentration of methyl chloride is increased or decreased the rate of the reaction also increased or decreased respectively. Again, if the concentration of nucleophile i.e. OH^- is changed the reaction rate also changes proportionally. That means, the reaction rate depends on both the concentration of methyl chloride and sodium hydroxide. This accounts for the fact that, the rate determining step of the above reaction involves both reactants. Thus, the reaction is nucleophilic substitution bimolecular, S_N2. Also by observing stereochemistry, complete inversion is found which also indicates the mechanism pathway S_N2.

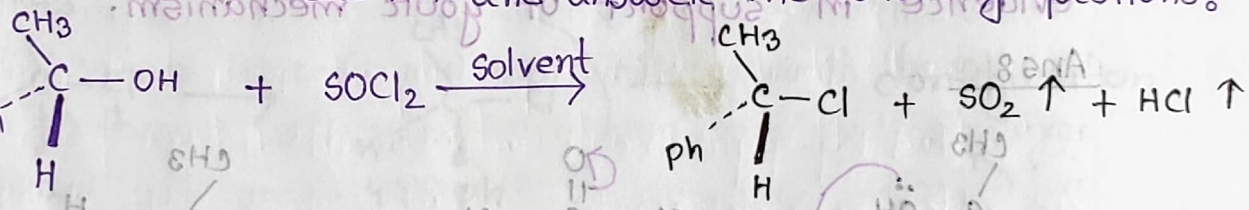
□ What is S_Ni reaction ? Is there any difference between S_Ni and S_N2 reaction ?

Ans:

S_Ni means Nucleophilic substitution intramolecular where the nucleophile comes from the solvent and in the reaction, retention of configuration occures.

Yes, there is a difference between S_Ni and S_N2 reactions ; S_Ni follows retention of configuration whereas S_N2 follows complete inversion of configuration.

□ Complete the reaction and answers the following questions:



(i) How will you mathematically represent the rate of the reaction ? Does the reaction follows S_Ni or S_N2 reaction? Give explanation in support of your answer.

Ans:

The mathematical representation of the rate of the reaction will include both the concentration of the substrate and solvent.

$$\text{Rate} = K[\text{substrate}][\text{solvent}]$$

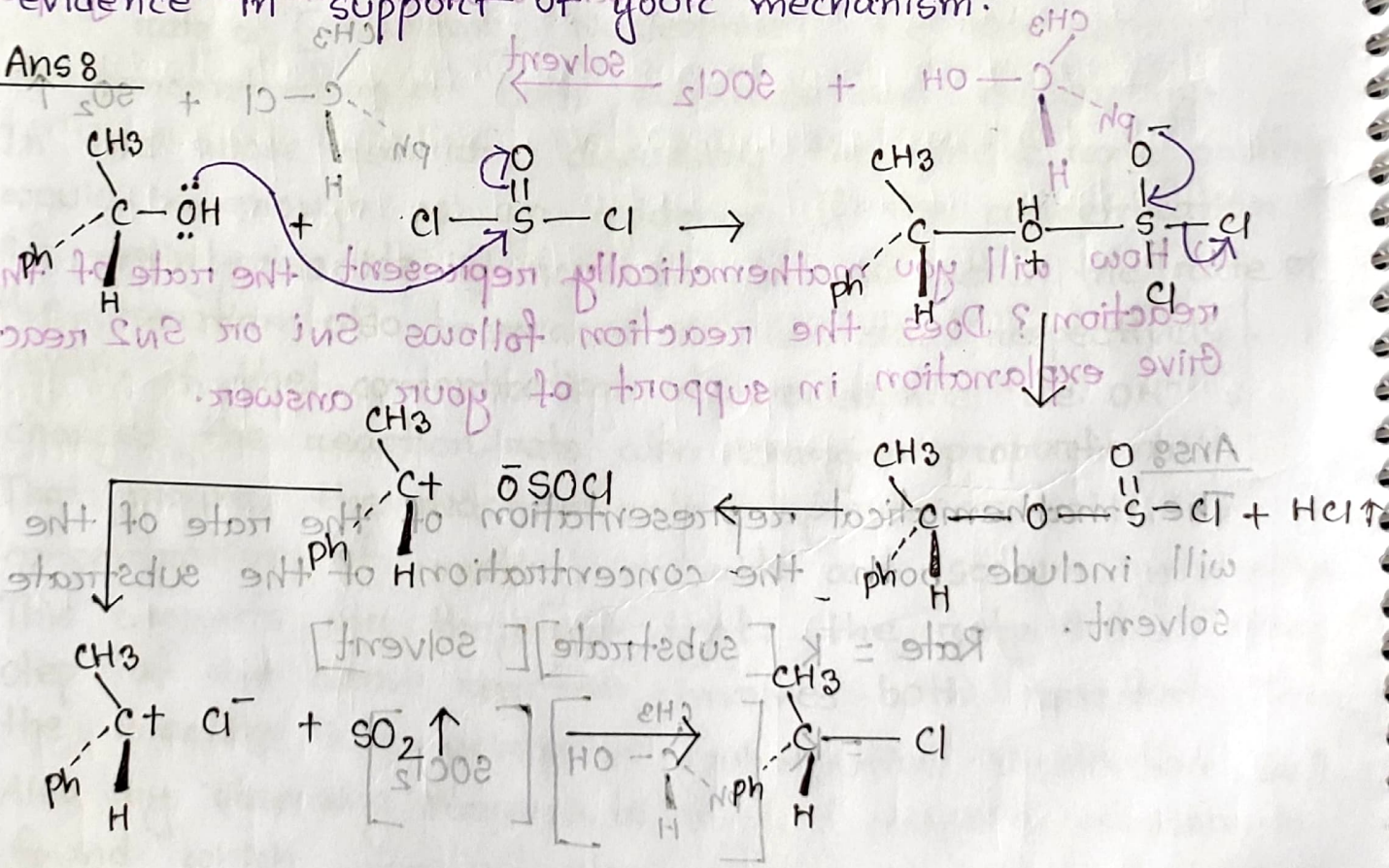
$$= K\left[\begin{array}{c} CH_3 \\ | \\ Ph-C-OH \\ | \\ H \end{array}\right][SOCl_2]$$

The mentioned reaction follows S_Ni rather than S_N2 reacton. Though both the reaction mechanism include the influence of concentration of bimolecules, there is

a stereo factor. that differs both the reaction mechanism. In S_N2 reaction, we find a product with complete inversion of configuration compared to the substrate. But in the above reaction no inversion occurs. Perhaps, only retention is observed where the place previously occupied by hydroxyl group is taken by chloride ion. As the nucleophile comes out of the solvent itself rather than any other nucleophile, the reaction is considered as S_Ni reaction.

(iii) Write the mechanism of the reaction and give an evidence in support of your mechanism.

Ans 8



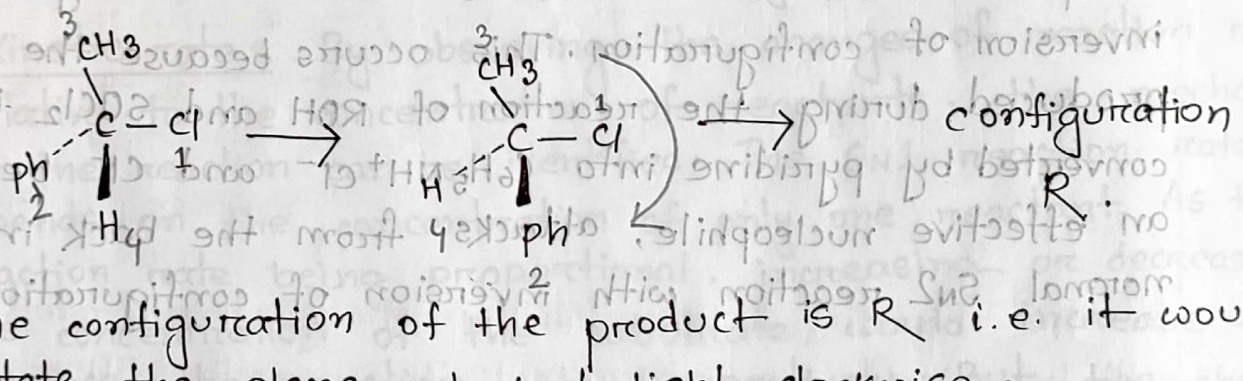
Evidence :

In S_Ni mechanism, the retension of configuration is observed.

From the mechanism, it is seen that the configuration would not change. Only, the position of OH^- has been taken by Cl^- ion. If we pass plane polarized light through the product obtained, it would rotate the light exactly in the same direction of the substrate. Thus retension is observed which proves the above mechanism.

(iii) What would be the stereochemistry of the final product?

Ans:



The configuration of the product is R i.e. it would rotate the plane polarized light clockwise.

(iv) What type of solvent will you choice to carry out the reaction?

Ans:

Polar solvent is more suitable for S_Ni reaction. The rate at which the alkyl chlorosulphite intermediate breaks down to the product, RCl is found to increase with the increasing polarity of the solvent. As polar solvent stabilizes the carbocation, the ion pair would form more prominantly which would speed up the S_Ni reaction.

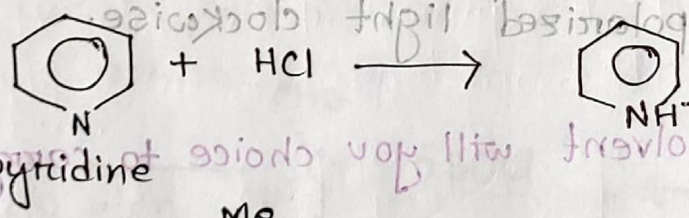
(v) What are the byproducts of the reaction?

Ans: Sulfur di-oxide (SO_2) and hydrogen chloride gas (HCl) are the by products of the reaction.

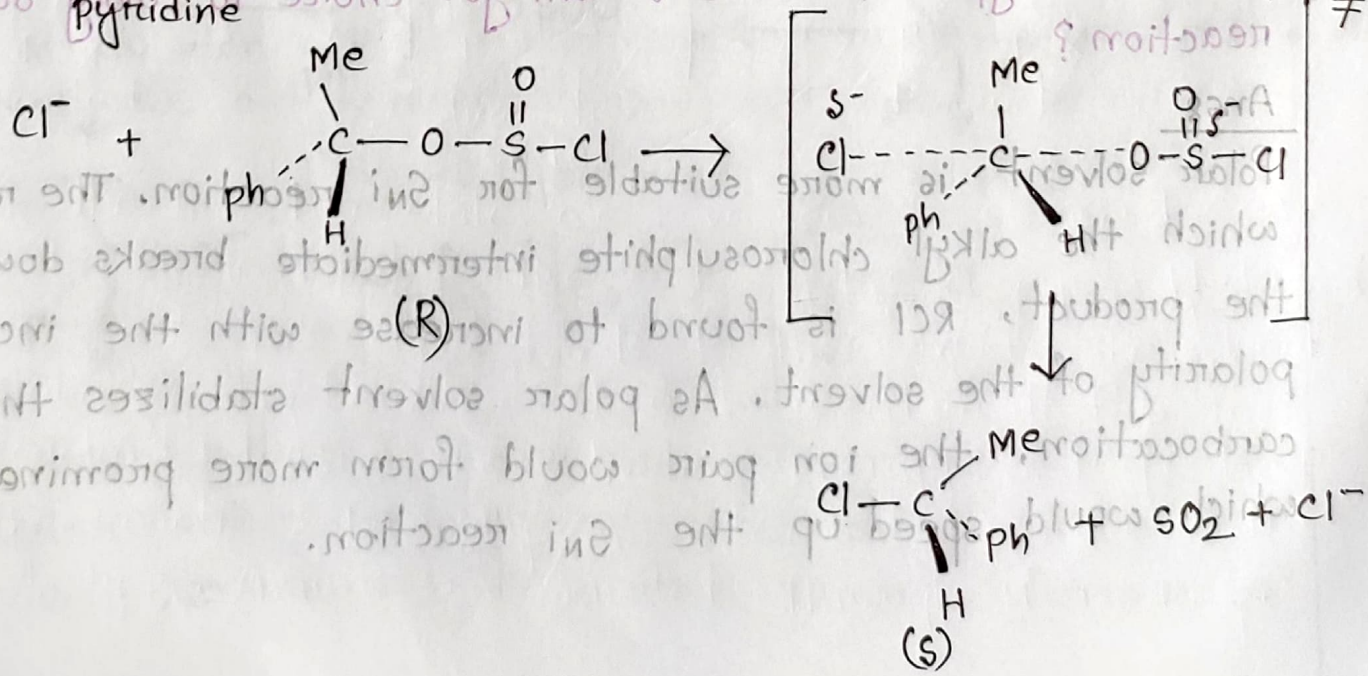
(vi) If the above reaction is carried out in presence of pyridine as a solvent then what will be the mechanism of the reaction? What will be the stereochemistry of the final product? Give explanation in favour of your answer.

Ans:

If the above reaction is carried out in presence of pyridine, the product RCl is now found to have undergone inversion of configuration. This occurs because the HCl produced during the reaction of ROH and $SOCl_2$ is converted by pyridine into $C_6H_5NH^+Cl^-$ and Cl^- being an effective nucleophile, attacks from the back in a normal S_N2 reaction with inversion of configuration:



pyridine



The product posses 'S' configuration which is exactly opposite to the configuration of substrate having configuration 'R'. It can rotate the plane polarized light in a direction opposite to the substrate. The exact opposite configuration accounts for the complete inversion followed by S_N2 mechanism.

□ How can you confirm that a reaction is occuring through S_N1 pathway?

Ans:

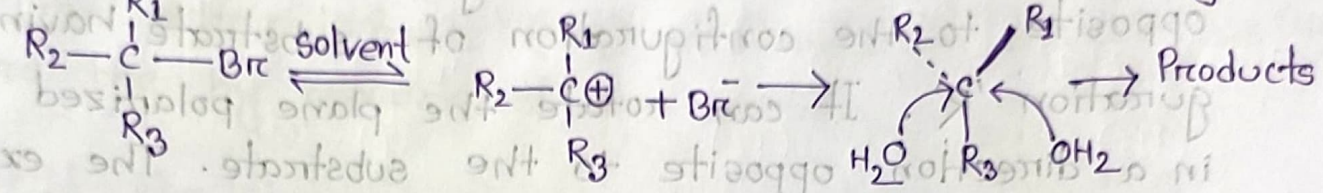
Some evidences for a reaction being S_N1 reaction are:

• Kinetic rate: By observing the changes of reaction rate relative to the concentration of reactants, the mechanism of the reaction can be identified. The S_N1 reaction rate depends on the concentration of only one reactant. As the reaction rate being proportional, increasing or decreasing the concentration of the substrate, would increase or decrease the reaction rate respectively. But the change of the concentration of nucleophile has no effect on the reaction rate. Through the process, S_N1 can be detected.

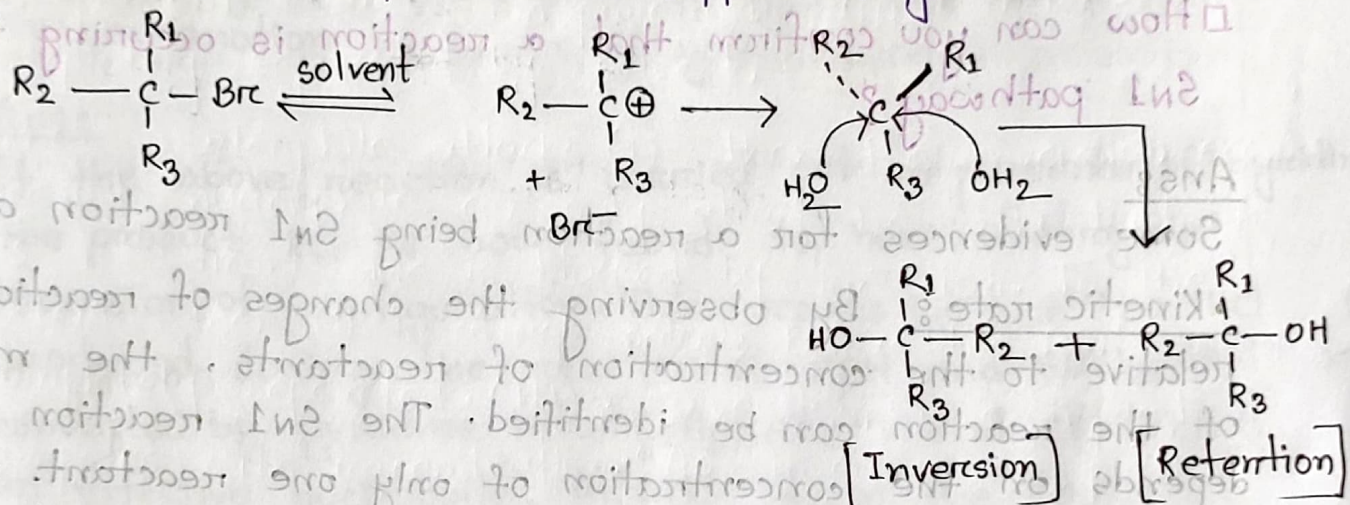
• Stereochemistry: S_N1 reaction involves racemization and some net inversion of configuration. By observing the stereo configuration of the product relative to the substrate, S_N1 mechanism can be identified.

• Effect of substituents: The morestable carbocation influences S_N1 reaction. If electron donating large substituents are attached to the carbocation, it would speed up S_N1 mechanism. Also, if any unexpacted rearranged product occurs, it accounts for S_N1 mechanism.

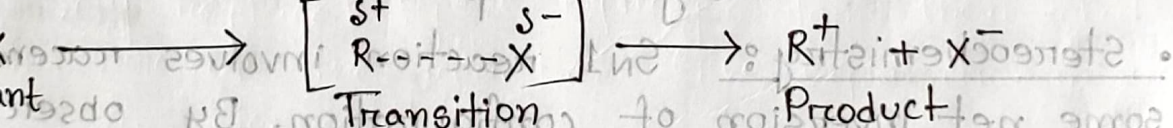
□ Complete the S_N1 reaction and answer the following ques:



(i) What type of solvent will you use to carry out the reaction? Give explanation in support of your answer.



As the equation follows S_N1 pathway. Polar protic solvent will be suitable to carry out the reaction. The reactant has dipole moment and forms dipole-dipole bonds to the solvent molecules.



In the transition state, we have seen a streched C–X bond and well developed positive and negative charges. If forms stronger dipole-dipole bonds to the solvent. The solvent stabilizes the transition state more than it does to reactant; lowers the activation energy and speeds up the reaction. Basically anion contributes more in the solvolysis. Anions are solvated through H-bonding.

Thus solvents capable of hydrogen bonding i.e. protic solvents are most suitable. Thus polar protic solvent like water, alcohol etc are suitable for the reaction.

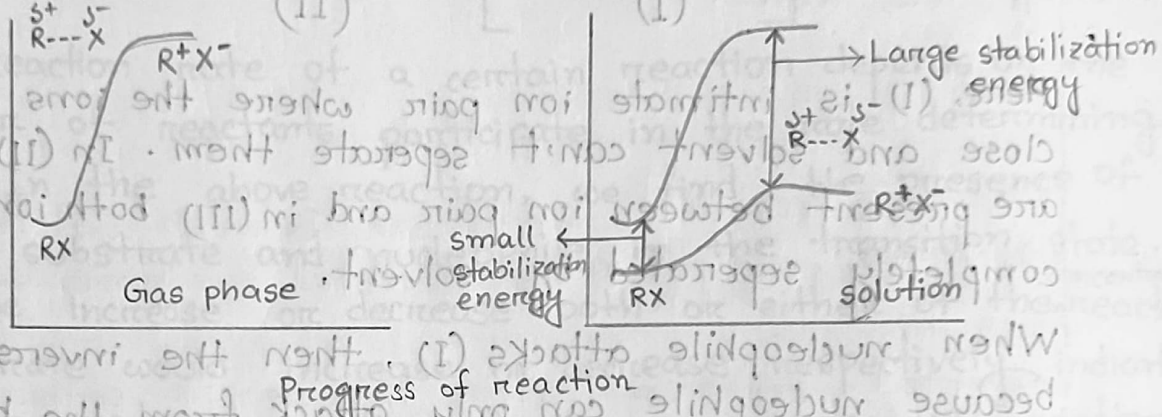


Fig: Effect of using polar protic solvent in S_N1 reaction.

(iii) We might expect that the equal amounts of the two products should be formed from the above reaction but actually a greater amount of (50-70)% of the product with the inverted configuration is obtained in most cases instead of racemic product." How can you account for the statement?

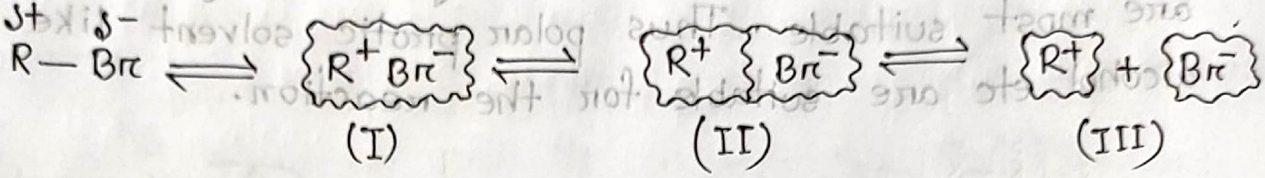
Ans:

In S_N1 reaction, carbocation is formed in the first step which is slow and rate determining. As carbocation is planar, nucleophile can attack from both side and give racemization but partial inversion also occures. Degree of inversion depends on two factors:

(i) stability of carbocation
(ii) Nucleophilicity

When stability of the carbocation is increased, then racemization is also increased and when strong nucleophile is present.

then the inversion is increased.



(I) (II) (III)

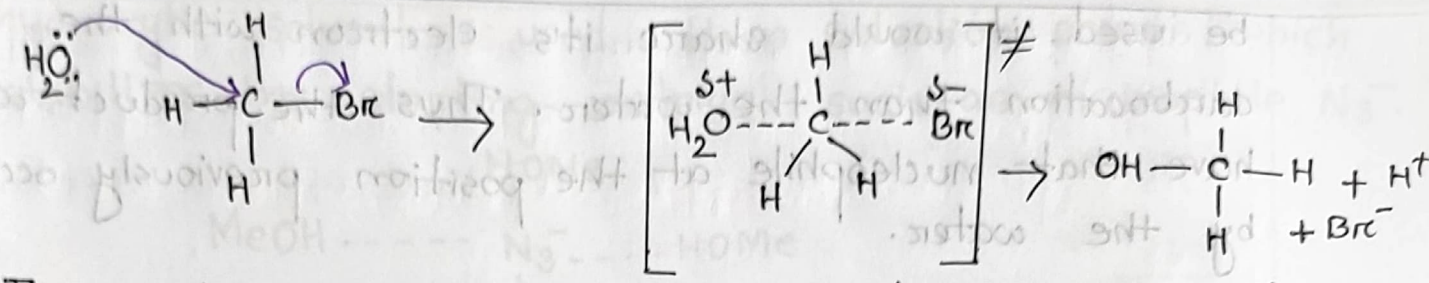
Here, (I) is intimate ion pair where the ions are very close and solvent can't seperate them. In (II) solvent are present between ion pair and in (III) both ions are completely seperated by solvent.

When nucleophile attacks (I), then the inversion occurs because nucleophile can only attack from the backside as the front side is blocked by Br^- ion. To the extent, then, that attack occurs before the ion pair has completely seperated, (III), Inversion of configuration competes with racemization. But in (III) attack from either side can occure. As inversion is predominant, about (50-70%) of inverted product is obtained instead of racemic product.

(iii) If the three alkyl groups of the alkyl halids are successively replaced by the hydrogen atoms then what will be the rate of the reaction? Give explanation in support of your answer.

Ans:

If the three alkyl groups are replaced by hydrogen atoms the substrate would be methyl bromide $[CH_3-Br]$. The reaction would follow S_N2 mechanism.



The reaction rate of a certain reaction depends on the number of reactants participate in the rate determining step. In the above reaction, we find the presence of both substrate and nucleophile in the transition state. If we increase or decrease both or either of the concentration of reactants, the rate would increase or decrease respectively indicating the proportional relation of the reactants with reaction rate. As, CH_3-Br is unhindered, nucleophile can easy attack from backside and help to follow S_N2 mechanism. As, rate determining step includes both substrate and ~~reaction~~ nucleophile, so

$$\text{Rate} \propto [\text{substrate}][\text{Nucleophile}]$$

$$\text{Rate} = K[\text{substrate}][\text{Nucleophile}]$$

where K = rate constant

(iv) If the above reaction is carried out in presence of a very strong nucleophile than that of water then what will happen? Give explanation.

Ans:

If a stronger nucleophile other than water is used, the reaction would follow again the same mechanism to form product. Here, the solvent water is used as a nucleophile which further gets deprotonated. But if any stronger nucleophile other than water would

be used, it would share it's electron with the carbocation than the water. Thus the product would have that nucleophile at the position previously occupied by the water.

□ What is concerted mechanism? Between S_N1 and S_N2 which one purely follows concerted mechanism - explain?

Ans:

Concerted mechanism refers to a reaction pathway where all bond breaking and bond formation steps occur simultaneously in a single step. S_N2 is a ~~eerte~~ concerted mechanism reaction where a nucleophile attacks an electrophilic centere, displacing a leaving group in a single step. But S_N1 reaction occures in two step: first step involves carbocation formation and secondly nucleophilic attack. Thats why S_N1 is not concerted reaction as like S_N2.

□ Why does polare non-hydroxylic DMF have marked effect on (increased 4.5×10^4) S_N2 reaction of MeI with N_3^- at 0°c compared the reaction with MeOH.

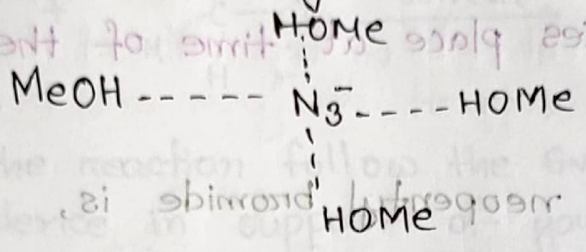
Ans:

Here the reaction is S_N2:

$$MeI + N_3^- \xrightarrow[0°C]{S_N2} Me\text{-}N_3 + I^{\ominus}$$

In S_N2 reaction, polare non-hydroxylic solvent is used which increase the rate of the reaction. When polare

hydroxylic solvent is used, then solvation occurs which stabilizes the starting material such as nucleophile N_3^-.



N_3^- is strongly solvated by polarc solvent MeOH. So, nucleophile can't easily attack on the substrate. But when polarc non-hydroxylic solvent DMF is used, it can't solvate N_3^- ion like MeOH. So, N_3^- in DMF act as a strong nucleophile which easily attack MeI and the rate of the reaction increased at 4×10^4 times.

□ Why does S_N2 reaction rate decreases as the following series is transversed?

CH_3-Br $MeCH_2-Br$ Me_2CH-Br Me_3C-Br.

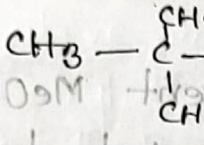
Ans:

The steric factor plays a vital role in the mechanism of S_N2 reaction. After the nucleophile attack, the resultant transition state will have five groups around the centre carbon. The more crowded the transition state relative to the starting material the higher its energy will be, and the slower therefore the transition state is formed. OH^- will find it progressively more difficult to attack the bromine carrying carbon as the latter becomes more heavily substituted. So, because of increasing substitute methyl group, we would expect the purely S_N2 reaction rate to decrease as the above series is transversed.

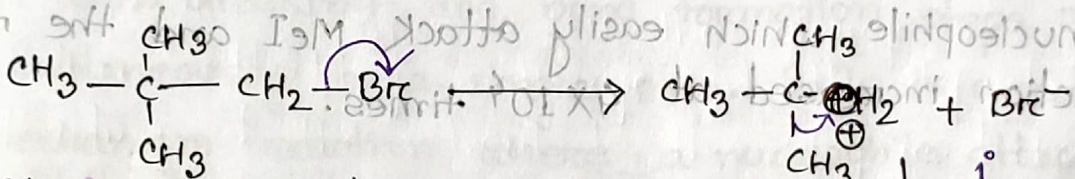
□ The reaction of neopentyl bromide with aq. NaOH gives 2-methyl-2-butanol. Give an evidence in support the rearrangement takes place at time of the reaction.

Ans:

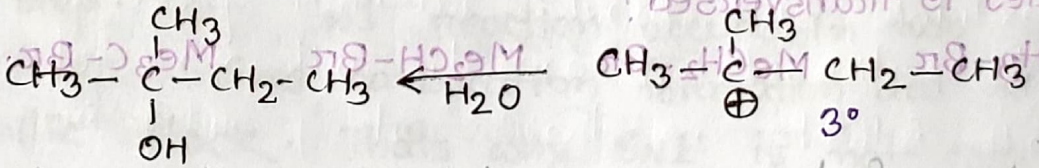
The structure of neopentyl bromide is,



Though the halide is 1°, larger groups present cause steric hindrence. Thus S_N2 reaction is very slow. So, it shows S_N1 reaction.



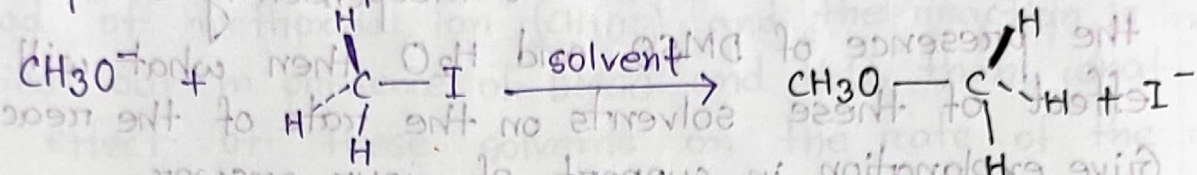
Neopentylbromide



2-methyl-2-butanol

We know, 3° carbocation is more stable than 1°, so re-arrangement can occur. The re-arrangement can be supported by kinetic rate of the equation. As, the reaction depends on the concentration of one reactant. it must be S_N1 and S_N1 is supported by rearrangement of more stable carbocation and unexpected product is obtained.

□ Complete the equation and answer the questions:



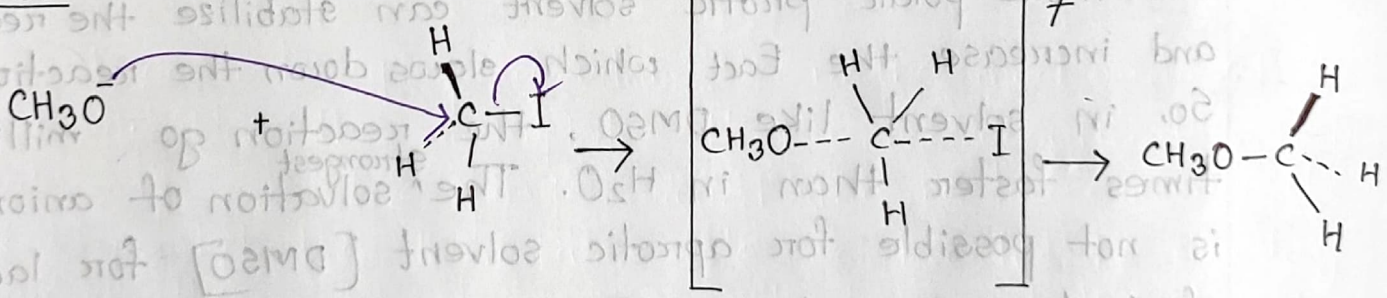
(i) Does the reaction follow the S_N2 mechanism? If so, give evidence in support of your answer.

Ans:

Yes, the reaction follows S_N2 reaction. The reaction pathway can be identified by observing the stereo configuration of the reactant and the product. As S_N2 supports complete inversion of configuration exactly like the mentioned equation, the substrate and the product of the reaction would rotate plane polarized light exactly in opposite direction. Thus, the reaction follows S_N2.

(ii) Propose a mechanism for the reaction.

Ans:



(iv) If the above reaction is independently carried out in the presence of DMSO and H_2O then what will be the effect of these solvents on the rate of the reaction? Give explanation in support of your answer.

Ans:

The above reaction follows S_N2 mechanism.

The alkyl halide has a dipole moment and forms weak dipole-dipole bonds to the solvent. The methoxide ion carries a full negative charge and forms very powerful ion-dipole bonds to the solvent. The transition state carries a full negative charge too but the charge has been divided between attacking methoxide and the departing halide group. Bonding of the solvent to this dispersed charge is much weaker than to the concentrated charge of the small methoxy ion. The solvent thus stabilizes the reactants more than it does the transition state. Thus, the more polar protic solvent can stabilize the reactants and increase the Eact which slows down the reaction. So, in solvent like DMSO, the reaction go million times faster than in H_2O. The strongest solvation of anions is not possible for aprotic solvent [DMSO] for lacking of hydrogen bonding. Thus, the rate of the reaction would increase with DMSO and decrease with the use of H_2O i.e. protic solvent.

(v) If CH_3OH is used as a nucleophile in the above reaction instead of methoxide ion (CH_3O^-) and the reaction is independently carried out in presence of DMSO and H_2O, then what will be the effect of these solvents on the rate of the reaction? Give explanation.

- How can you prove that S_N2 reactions are attended by inversion of configuration ?

□ Which reagent in each of the following pairs is more nucleophilic and why?

(i) $(CH_3)_3B$ and $(CH_3)_3N$ (ii) H_2O and H_2S.

Ans8

(i)

$(CH_3)_3\ddot{N}$ is more nucleophilic than $B(CH_3)_3$.

In $(CH_3)_3N$, N has a lone pair electron and three electron donating methyl groups bonded to the N atom. So the nitrogen can easily attack any electropositive center. On the other hand, $B(CH_3)_3$ is an electron deficient compound. The B has only six valance electron. Thus $B(CH_3)_3$ can't act as a nucleophile. So, $(CH_3)_3N$ is more nucleophilic.

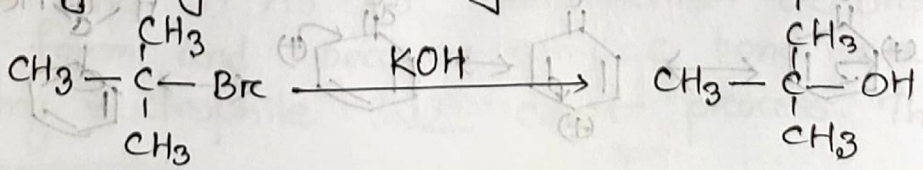
(ii) H_2S is more nucleophilic than H_2O.

Both S and O has two lone pair electron but oxygen is more electronegative than sulfer. So, it's very difficult for the oxygen atom to share it's electrons with others than S. So, H_2S is more nucleophilic.

□ Give the reaction mechanism of the hydrolysis of t-butyl bromide in presence of base.

Ans 8

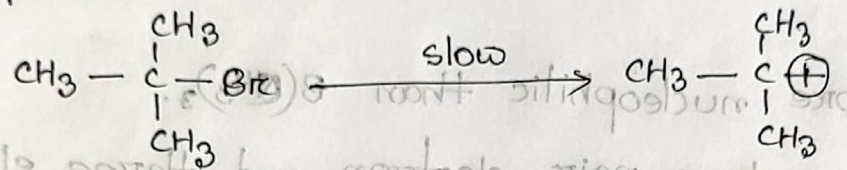
The hydrolysis of t-butyl bromide in presence of base:



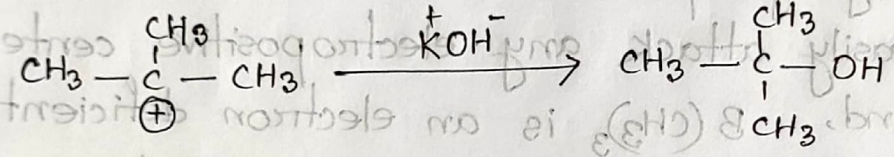
As the substrate is 3° halide, it undergoes S_N1 reaction which is two step of which 1st step is the rate determining one.

Mechanism:

step 01:



step 02:



□ Solvolysis of (+) phCHMeCl leads to 98% racemization in aquous acetone while (+) $C_6H_{13}CHMeCl$ leads only 34% racemization. Explain.

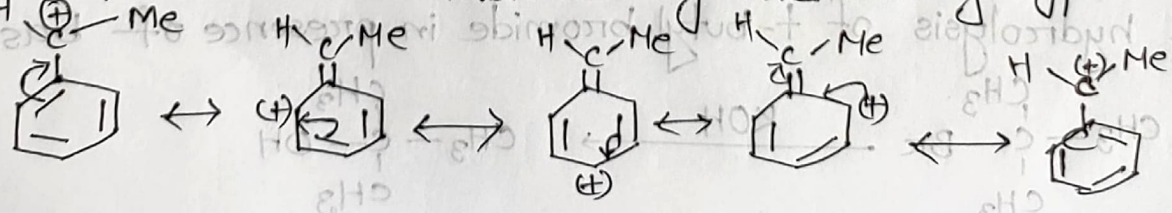
Ans:

(+) PhCHMeCl $\xrightarrow{\text{solvolysis}}$ 98% racemization

(+) $C_6H_{13}CHMeCl$ $\xrightarrow{\text{solvolysis}}$ 34% racemization.

Racemization depends on the stability of the carbocation. When the carbocation is more stable, then the racemization is more. Both PhCHMeCl and $C_6H_{13}CHMeCl$ gives S_N1 reaction while forming the product.

The phCHMeCl is stabilized by the benzyl type carbocation.



But no such resonance stabilization occurs in $C_6H_{13}CHMeCl$.

$$C_6H_{13}CHMeCl \xrightarrow{-Cl^{\ominus}} C_6H_{13}\overset{H}{\underset{H}{C}}\overset{Me}{\oplus}$$

Because of the resonance stabilization of the benzyl cation, PhCHMeCl undergoes maximum racemization.

□ Compared to allyel chloride, vinyl chloride is very unreactive towards nucleophilic substitution reaction. Explain. / Or, Allyl halides are more reactive than n-propyl halides in S_N1 reaction. Explain [* Same reason *]

Ans:

$CH_2=CH-CH_2-Cl$ — Allyl chloride

$CH_2=CH-Cl$ — Vinyl chloride

In allyl chloride, the carbocation formed when the Cl^- leaves the substrate is stabilized by the resonance of the double bond. As carbocation stabilization occures, it promotes S_N1 mechanism significantly.

$$CH_2=CH-CH_2-Cl \xrightarrow{-Cl^{\ominus}} CH_2=CH-\overset{\oplus}{C}H_2 \leftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$$

$$CH_2 \equiv CH \equiv CH_2$$

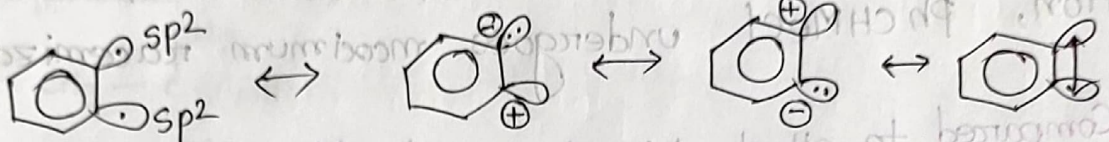
Due to lacking of strong steric hindrence, some S_N2 product can also occurs.

But in $CH_2=CH-Cl$, the C-Cl bond is very strong having a very short bondlength between sp^2 hybridized carbon and Cl. As no stabilization occurs, carbocation can't form and because of C=C bond repulsion against entering nucleophile, S_N2 can't process. Thus, vinyl chloride is unreactive.

□ Benzyne is very unstable and reactive - why?

Ans:

In benzyne, an additional bond is formed between two sp^2 hybridized C atom by sidewise overlap. The new bond lies along the side of the ring.



But the sidewise overlap is not very strong. So the weak π bond can easily disrupted to react with others accounting for the unstability and reactivity of benzyne.

□ What is the salient difference between S_N1 and S_N2 pathway?

Ans:

Some differences are:-

- S_N1 is a two step reaction, whereas S_N2 is one step reaction.
- S_N1 follows first order Kinetics and S_N2 follows second order Kinetics.
- In S_N1 the first step is slow and determining in which carbocation is formed. On the other hand, in S_N2 reaction transition state is formed.
- In S_N1 partial racemization occurs but S_N2 accounts for complete inversion of the product.
- As S_N1 suggests for carbocation formation, re-arrangement occurs but no re-arrangement occurs in S_N2.
- Polar protic solvent is used in S_N1 and polar aprotic solvent is used for S_N2 reaction.

□ How do you make kinetic distinction between the operation of the S_N1 and S_N2 modes.

- By adding a competting nucleophile, the distinction can be made. As S_N1 doesn't depend on nucleophile, the rate of the reaction would be the same. But if it causes change in the overall rate of the reaction, it would be considered as S_N2 reaction.

□ The rate of solvolysis of the terctiarcy halide Me_3CBrc is found to be 3×10^4 times fasterc in 50% aq. ethiand than in ethanol - explain.

Ans8

The solvolysis of terctiarcy halide, Me_3CBrc is S_N1 mode. We know that carcbocation is forcmed in S_N1 reaction. The polarc solvent stabilizes the trcansition state. The anions (Br^-) arce highly solvated due to hydrogen bonding occurced by protic solvent. Thus when 50% waterc is used with ethanol, it act as a stabilizing agent forc carcbocation and the reaction gets fasterc than ethanol alone.

□ Forc the S_N2 mode, increasing solvent polarcity is found to result in slight decrease in reaction rate. Why?

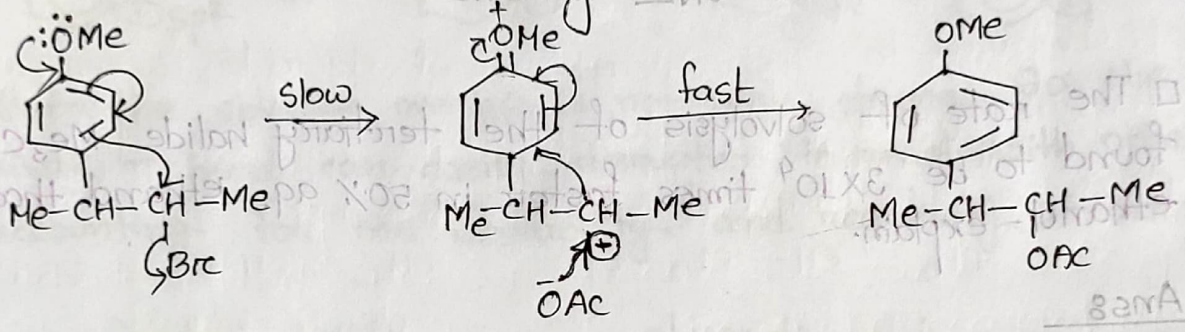
Ans8

[Discussed laterc]

□ The π- orbital system of benzene ring can act as a neighbouring group in nucleophilic substitution. Explain.

Ans 8

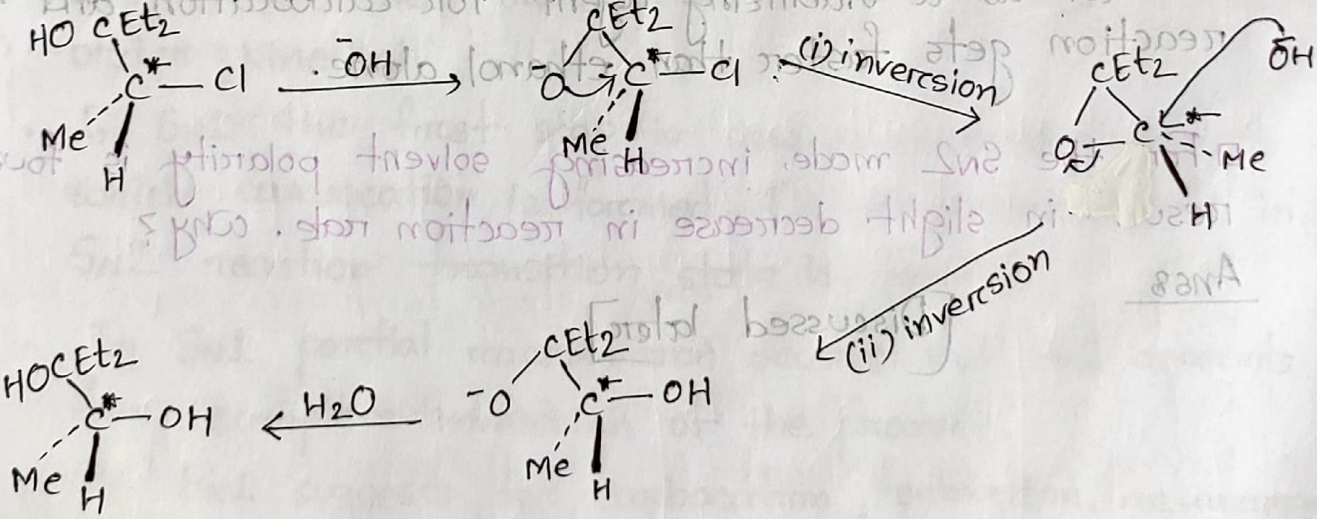
The π- orbital of the benzene can act as a neighbouring group and preventing the backside attack by nucleophile. So, nucleophile can only attack from the front side and retention of the configuration occurs.



□ The alkaline hydrolysis of 1,2- Chlorohydrin ($MeCHClCEt_2OH$) is found to yield the 1,2-diol ($MeCHOHCEt_2OH$) with retention of configuration. Explain.

Ans 8

Alkaline hydrolysis of chlorohydrin -



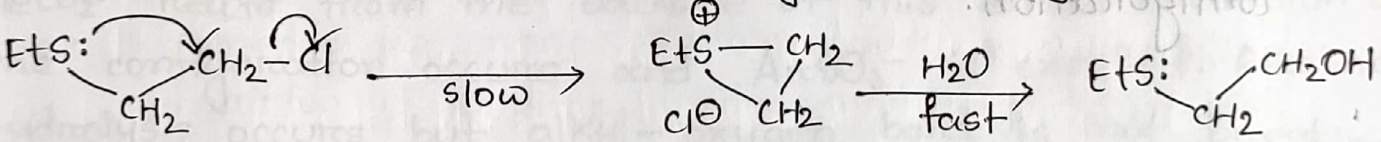
[Retention]

Initial attack by base on the substrate, yields alkoxide anion. Internal attack by the RO^- yields epoxide with complete inversion of configuration at C*. The carbon atom undergoes ordinary S_N2 attack by OH^- with second inversion of configuration at C*. The second alkoxide ion abstracts a proton from the solvent to yield the product 1,2-diol with the same configuration as the starting material. Thus retention of the configuration is found.

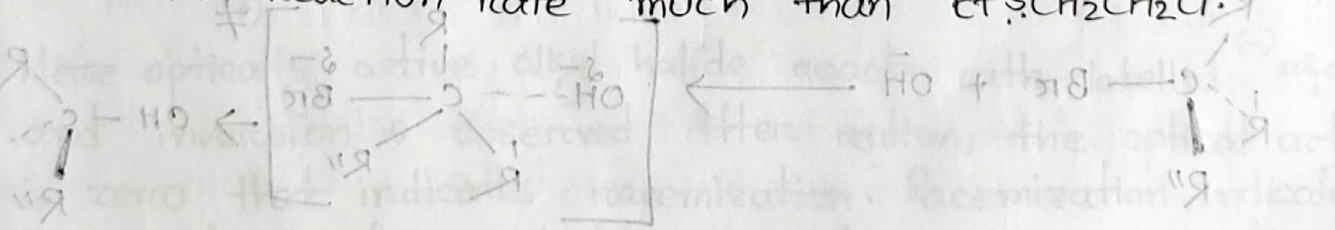
□ $Et\ddot{S}CH_2CH_2Cl$ is found to undergo 10^4 times faster substitution than $Et\ddot{O}CH_2CH_2Cl$ under comparable condition. Explain.

Ans:

In $Et\ddot{S}CH_2CH_2Cl$, S acts as a neighbouring group:



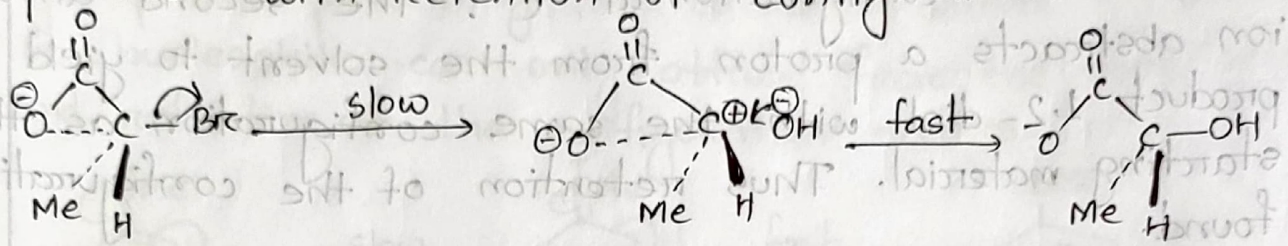
The sulphur has a lower electronegativity than oxygen. Thus internal nucleophilic attack occurs by donating electron pairs of S. On the other hand, O is more electronegative atom, so oxygen can't donate easily it's electron pair. Thu hydrolysis of $Et\ddot{O}CH_2CH_2Cl$ proceeds via ordinary attack by an external nucleophile which slows down the reaction rate much than $Et\ddot{S}CH_2CH_2Cl$.



□ Explain the fact that hydrolysis of 2-bromopropenoate anion at low [OH⁻] proceeds with retention of configuration.

Ans:

The hydrolysis of 2-bromopropeonate anion at low OH^- proceeds with retention of configuration:

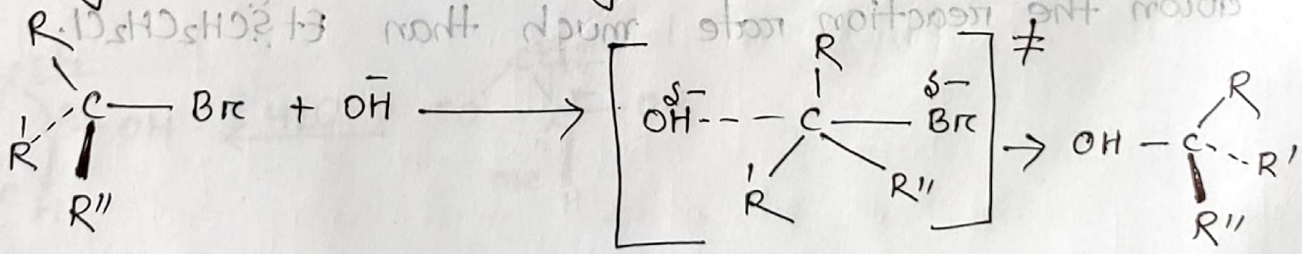


In this reaction, Oxygen acts as a neighbouring group. which form epoxide. At low concentration of nucleophile, backside attack i.e. S_N2 mechanism can't take place. Thus OH^- occupies the place previously occupied by the Br^- ion via S_N1 mechanism with the retention of configuration.

□ How can you prove that S_N2 reactions are attended by inversion of configuration?

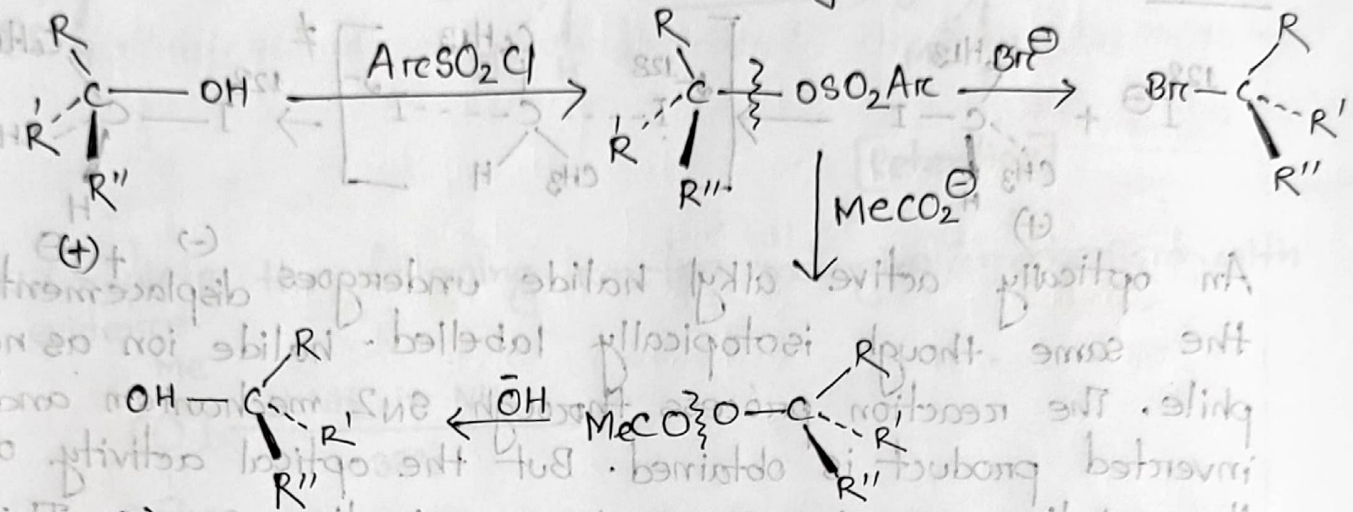
Ans:

In S_N2 reaction, nucleophile attack from the backside of the leaving group and transition state is formed. Finally inversion of configuration occurs.



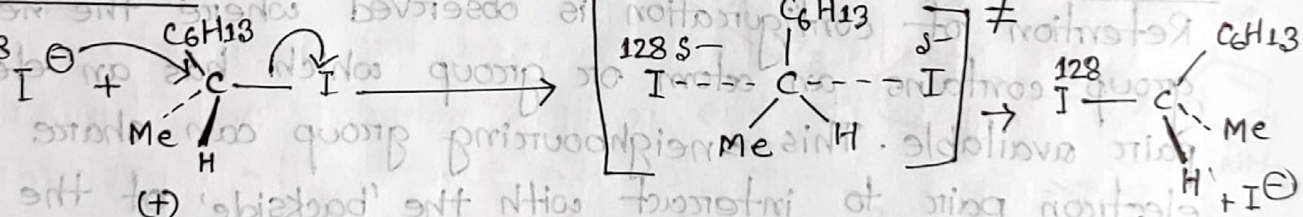
Evidence 1:

Determination of the relative configuration:



In this reaction (+) alcohol reaction with tosyl chloride and C-O bond breaking is not involved. Thus tosylate must have the same configuration as the original alcohol. When $MeCO_2^{\ominus}$ reacts from the backside of this ester, inversion of the configuration occures and $ArcSO_3^-$ is expelled. Alkaline hydrolysis occures but alkyl-oxygen bond is not breaking. Finally (-) is found. This accounts for the inversion of configuration.

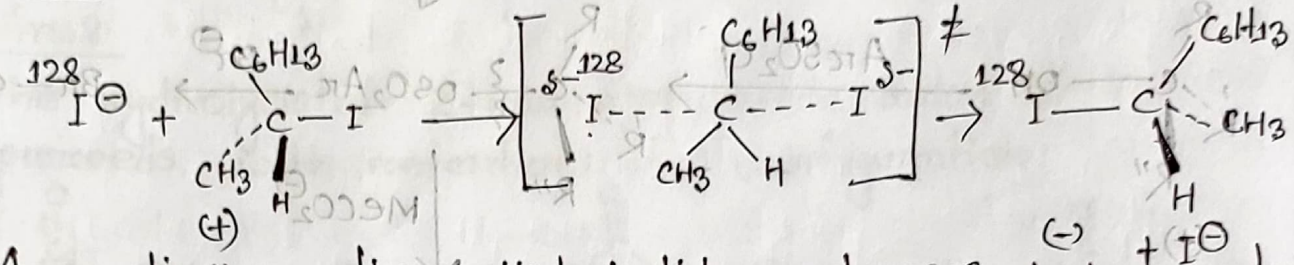
Evidence 2:



Here optically active alkyl halide reacts with labelled $^{128}I^{\ominus}$ and inversion is observed. After reaction, the optical activity is zero that indicates racemization. Racemization indicates the mixture of inverted compounds. Thus S_N2 reactions are attented by ~~racimiz~~ inversion.

Ques: Why the optical activity is zero for evidence two?

Ans:

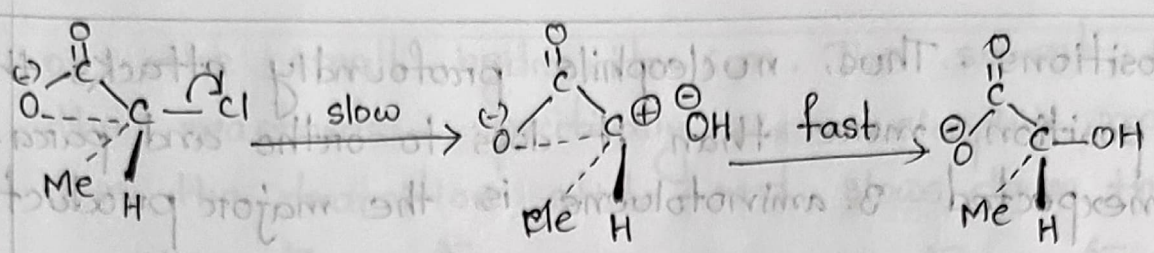


An optically active alkyl halide undergoes displacement by the same though isotopically labelled halide ion as nucleophile. The reaction process through S_N2 mechanism and inverted product is obtained. But the optical activity of the solution remains zero i.e. racemization occurs. This happens because inversion of configuration of a molecule of (+) substrate results in a molecule of its mirror image (−). But the produced $I^\ominus$ further attack on the (−) product which by inversion produce the previous (+) alkyl halide. Since two reactions occurs simultaneously, racemic mixture with zero optical activity is obtained.

□ In substitution reaction, neighbouring group participation leads to retention of configuration - Explain.

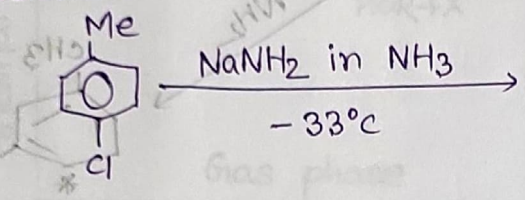
Ans:

Retention of configuration is observed where the neighbouring group contains an atom or group which has an electron pair available. This neighbouring group can share its electron pair to interact with the 'backside' of the carbon atom undergoing substitution. It prevents the nucleophile attacking from the backside. Thus nucleophilic attack can only take place from the front side, leading to retention of configuration.

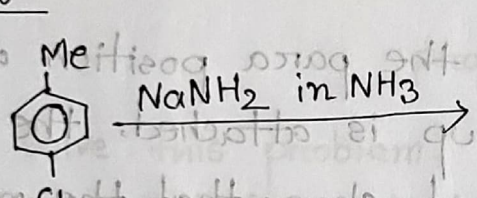


[Retention]

□ Complete the following reaction and give mechanism with evidence.



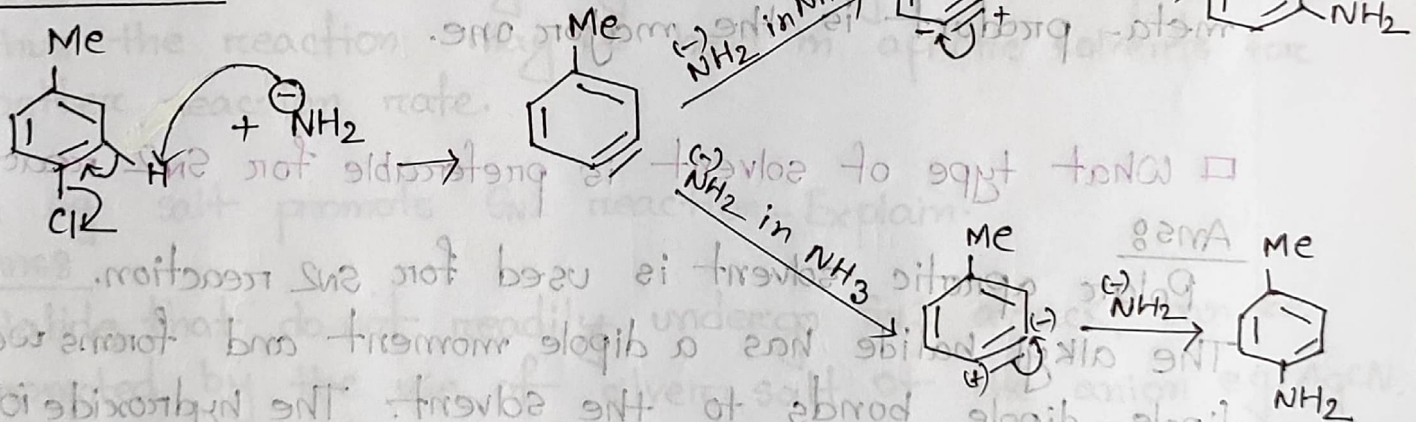
Ans:



expected (38%)

unexpected - (62%)

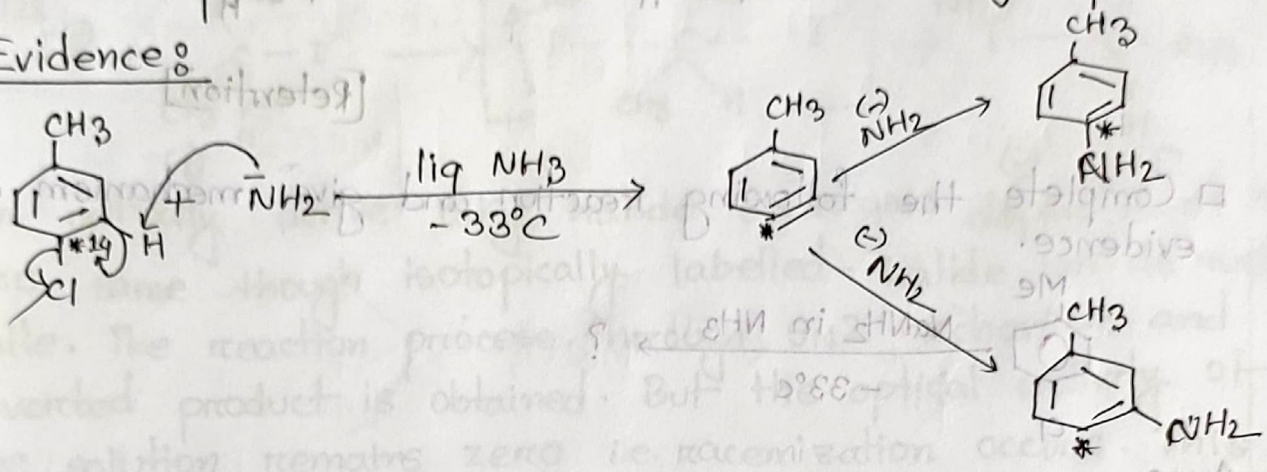
Mechanism:



In this reaction, arcyne intermediate is formed. When nucleophile gets attached with arcyne two product is formed. 3- Aminotulene is the major product because Me group is an electron donating group and increases electron density at the ortho and parca position; meaning that meta positive remains morce electropositive than rest of the

two positions. Thus, nucleophile profoundly attacks the meta position more than it does to ortho and para. So the unexpected 3-aminotoluene is the major product.

Evidence:



If we label the carbon of the para position with ^{14}C with which the leaving group is attached, the observation would be much easier. If product shows that the minor product is the one in which the Amino group is attached with ^{14}C i.e. the para carbon. Thus it proves, the unexpected meta-product is the major one.

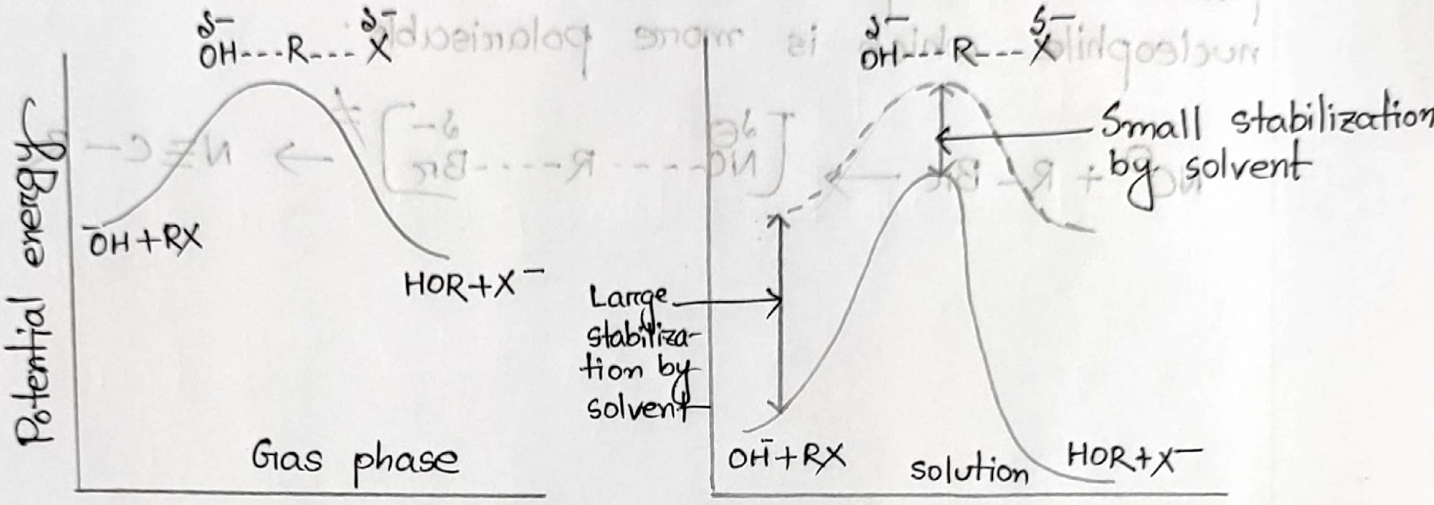
□ What type of solvent is preferable for S_N2 reaction?

Ans:

Polar aprotic solvent is used for S_N2 reaction.

The alkyl halide has a dipole moment and forms weak dipole-dipole bonds to the solvent. The hydroxide ion carries full negative charge and forms powerful ion-dipole bonds to the solvent. The transition state carries a full charge too but the charge is dispersed between two species. Bonding of the solvent to this dispersed charge is much weaker than to concentrated charge

of the small hydroxide ion. The solvent thus stabilizes the reactants - specifically the nucleophile more than it does the transition state and slows down the reaction.



To solve this problem, aprotic solvents like DMSO, DMF etc can be used. As anionic solvation via hydrogen bonding slows slows down the S_N2 reaction more. As aprotic solvent can't form hydrogen bonding like the protic ones, thus the reaction may be held in aprotic solvents for better reaction rate.

□ Ag salt promots S_N1 reaction - Explain.

Ans8

Halids that do not readily undergo S_N1 attack can be promoted by the use of silver salt of the anion e.g. AgCN. As $Ag^{\oplus}$ promotes $R^{\oplus}$ formation by precipitation of AgHal.

$$R-Br + Ag^{\oplus}CN^{\ominus} \xrightarrow{\text{slow}} AgBr\downarrow + R^{\oplus} + [CN]^{\ominus} \xrightarrow{\text{fast}} R-CN \;\downarrow\; R-\overset{\oplus}{N}\equiv\overset{\ominus}{C}$$

But in absence of such promotion by $Ag^{\oplus}$ i.e. with $Na^{\oplus}$ $CN^{\ominus}$, the resulting SN2 reaction is found to proceed with preferential attack on the atom in the nucleophile which is more polarisable:

$$NC^{\ominus} + R-Br \longrightarrow \left[{}^{\delta\ominus}_{NC}\text{----}R\text{----}{}^{\delta-}_{Br} \right]^{\neq} \longrightarrow N\equiv C-R + Br^{\ominus}$$

IR SPECTROSCOPY

- IR basically describes the vibrational characterc of a molecule In terms of wavenumberc, the vibrational infrared extends from about 4000 - 400 cm^{-1}.

  Functional group region → 4000 - 1400 cm^{-1}
  Fingerprint region → 1400 - 400 cm^{-1}.

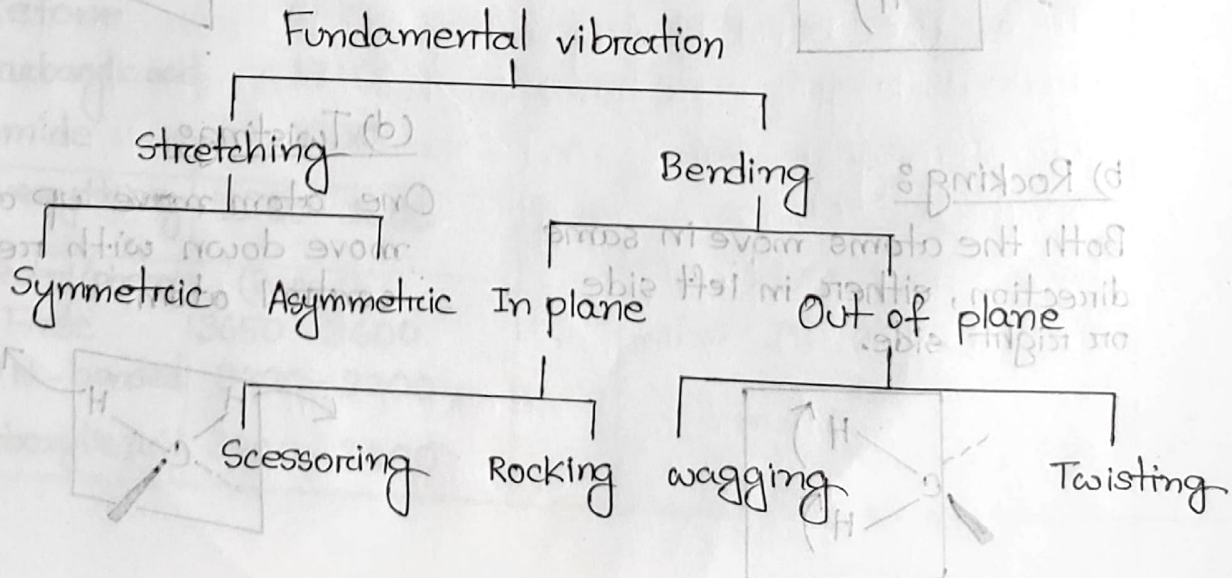
- **Conditions for a molecule to be IR active:**

a) Molecule must contain a bond/bonds with an electric dipole moment that is changing as a function of time.

b) The frequency of the radiation exactly should match the bond motion.

i.e H_2, Cl_2, O_2 etc are IR inactive with zero dipole moment. whereas HCl, R-OH etc are IR active.

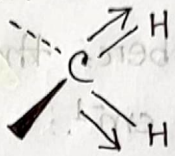
- **Types of vibrations:**



• **Stretching vibrations:** Only distance changes but bond angle remains same.

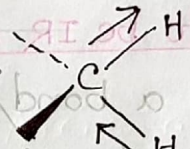
a) **Symmetric stretching vibrations**

Both the atoms stretched or compressed in same direction.



b) **Asymmetric stretching vibrations**

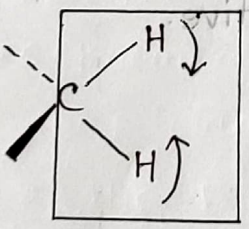
One atom undergoes stretching and other undergoes compression.



• **Bending vibrations**

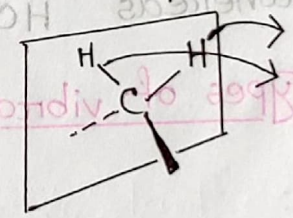
a) **Scissoring:**

Both the atoms move towards each other like scissor.



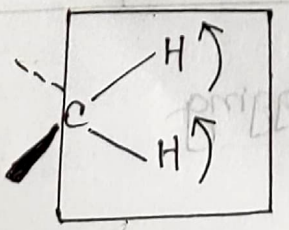
(c) **Wagging:**

Both the atom move up and down with respect to central atom.



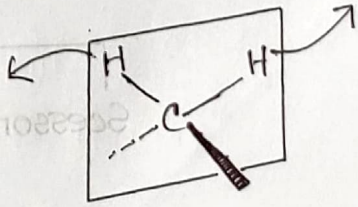
b) **Rocking:**

Both the atoms move in same direction, either in left side or right side.



(d) **Twisting:**

One atom move up and other move down with respect to central atom.



• Table for frequency (cm^{-1}) :

	Type	Frequency (cm^{-1})
C-H	Alkanes:	
	(i) Stretch	3000-2850
	(ii) Bend	1450 – 1375
	Alkene	3100 – 3000
	Aromatic	3150 – 3050
	Alkyne	3300
	Aldehyde	2900 – 2800
C=C	Alkene	1650 (1680–1620)
	Aromatic	1600 & 1475
C≡C	Alkyne	2150 (2250 – 2100)
C=O	Anhydride	1810 & 1760
	Acid chloride	1800
	Ester	1735
	Aldehyde	1725
	Ketone	1715
	Carboxylic acid	1710
	Amide	1690
C-O	Everything	1300-1000
O-H	Alcohol/phenol:	(Broad)
	(i) Free	3650 – 3600
	(ii) H-bonded	3400-3200
	Carboxylic acid	3400 - 2400

	Type	Frequency (cm^{-1})
N-H	1° & 2° Amines/ Amides:	
	(i) Stretch:	3500-3100 (sharp)
	(ii) Bend:	1640-1550
	* 3° Amine/Amide has no H	no Absorption
C-N	Amines	1350-1000
N=O	Nitro ($R-NO_2$)	1500 & 1350
S-H		2550 (weak)
C-X	C – Cl	750
	C – Br	600
	C – I	500

• Wave number of a vibration:

wave number, $\bar{\upsilon} = \frac{1}{2\pi c}\sqrt{\frac{K}{\mu}}$ where, K = Force constant

μ = Reduced mass

Here, $\mu = \frac{m_1 m_2}{m_1 + m_2}$

(i) $\bar{\upsilon} \propto \sqrt{K}$; ie. If the force constant of a bond is large then it will have higher wavenumber. Thus the IR value would be large.

(ii) If the reduced mass increases, then the wavenumber decreases because $\bar{\upsilon} \propto \frac{1}{\sqrt{\mu}}$. Thus IR value would be small.

□ Arrange Alkane, Alkene and Alkyne according to their IR value.

- Alkane contains carbon-carbon single bond ; C–C
  Alkene contains carbon - carbon double bond ; C=C
  Alkyne contains carbon - carbon triple bond ; C≡C.

Thus the bond is the strongest for Alkyne and weakest for Alkane because of the number of bonding between them.

we know,

wavenumber, $\bar{\upsilon} \propto \sqrt{K}$;

Thus the bond having a higher force constant would produce larger wave number. Force constant is higher for strongest bond and lower for weak bonds. Thus Alkyne would have larger IR value and Alkane would have the lowest.

According to IR value,

Alkane < Alkene < Alkyne

1200 cm^{-1} 1650 cm^{-1} 2150 cm^{-1}

□ Why the IR value decreases in a pattern: C-Cl > C-Br > C-I?

- IR value depends on the reduced mass of the atoms holding a bond.

$$\bar{\upsilon} \propto \frac{1}{\sqrt{\mu}}.$$

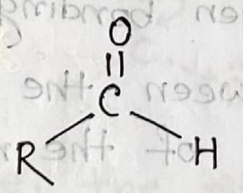
As wave number is inversely proportional to the reduced mass, increasing mass would lower the IR value gradually. From Cl to I the molecular mass increases. Thus reduced mass also increases and hence the wave number, i.e. IR value decreases.

□ Give explanation for increasing or decreasing the IR value of the compounds with respect to Aldehyde:

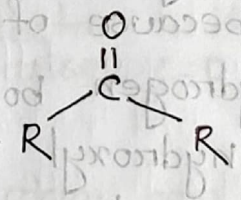
Anhydride	Acid chloride	Anhydride	Ester	Aldehyde	Ketone	Carboxylic Acid	Amide
1810	1800	1760	1735	1725	1715	1710	1690

← cm^{-1} →

• Aldehyde Vs Ketone:



Aldehyde

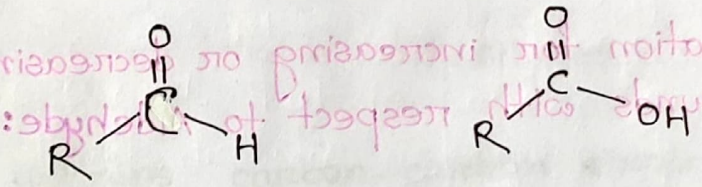


Ketone

The C=O bond frequency is higher for aldehyde than ketone. In a ketone, two alkyl groups are attached with the carbonyl carbon. As alkyl groups are electron donating, they increase the electron density on the carbonyl carbon. This increased electron density weakens the C=O

bond as the electron withdrawing oxygen can efficiently attracks the electron towards itself. Thus the C=O bond in ketone results in a lower stretching frequency. Whereas, in aldehyde, there is only one alkyl group attached to the carbonyl carbon which can not necessarily increase the electron density as two alkyl groups of ketone. Thus bond weakening of C=O is lower than ketone and appears at higher frequency.

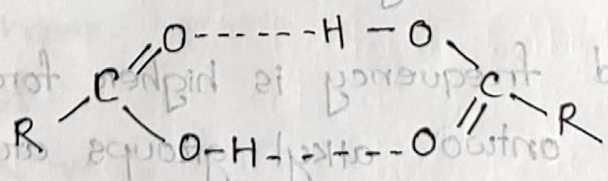
- Aldehyde Vs carboxylic acid:



Aldehyde carboxylic acid

The C=O frequency is lower for carboxylic acid than aldehyde.

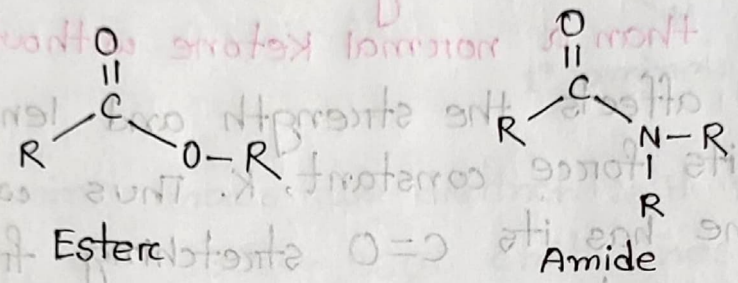
The lowering of IR frequency of C=O bond of carboxylic acid is because of hydrogen bonding. Carboxylic acid forms hydrogen bonds between the carbonyl oxygen and the hydroxyl hydrogen of the neighbouring acid and appears as a dimer.



This dimerisation weakens the bond of C=O and lowers the force constant, K, resulting in lower frequency of carboxylic acid.

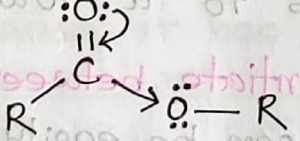
In, Aldehyde, no such hydrogen bonding is observed. So the IR value is higher for aldehyde than carboxylic acid.

• Ester Vs Amide :



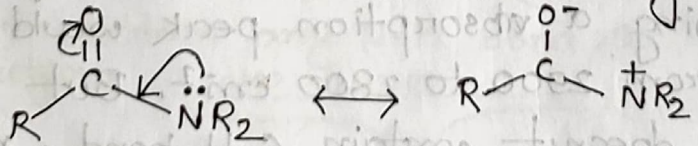
The C=O frequency is higher for ester than Amide.

The carbonyl group in ester is less conjugated than in amide. This means that the lone pair on the oxygen atom in an ester is less delocalized into the carbonyl π system. When the single bonded oxygen atom pulls the electron density from the carbon, the oxygen atom of the carbonyl group donates its lone pairs to strengthen the C=O bond. Thus this stronger bond appears at higher frequency.



[Electron withdrawing effect raises C=O frequency]

Again, in amide, the nitrogen atom can donate its lone pair electron into the carbonyl π system, resulting in significant resonance stabilization. This delocalization weakens the C=O bond in amides, lowering IR frequency.



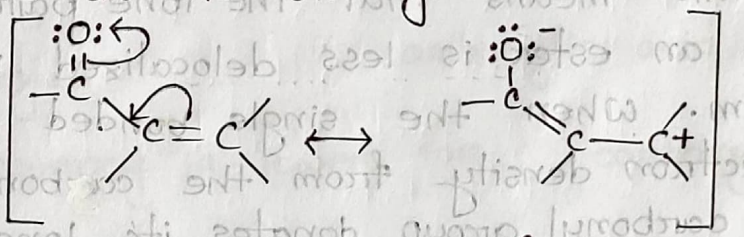
[Resonance effect lowers C=O frequency]

Though oxygen and nitrogen both are electronegative but nitrogen is less electronegative than oxygen atom, so it

can easily accomodate a positive charge which leds to resonance. Thus IR frequency to C=O bond is largere fore ester than fore amide.

□ Why the C=O frequency is lowere fore a ketone conjugated with a C=C than a normal ketone without conjugation?

- Resonance affects the strength and length of a bond and hence its force constant, K. Thus whereas a normal ketone has its C=O stretching frequency at 1715 cm^{-1}, a ketone that is conjugated with a C=C double bond absorbs at a lowere frequency, neare 1675 to 1680 cm^{-1}. That is because resonance lengthens the C=O bond distance and gives it more single bond characterc.



Resonce effect thus reduces the force constant and the absorption moves to a lowere frequency.

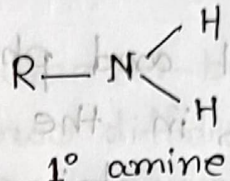
□ How can you differentiate between an aldehyde and a ketone?

- Aldehyde and ketone can be easily distinguished by IR spectroscopy. Though they have a little difference between their C=O bond stretching but the main difference is because of C-H bond stretching. Aldehyde contains a C-H bond which represents its functional group. Thus fore C-H bond stretching a absorption peak would appeare at a region ranges from 2900 to 2800 cm^{-1}. But as the functional group of ketone doesn't contain C-H bond, no peak would appeare in this region. That's how aldehyde and ketone can be distinguished.

□ How can you differentiate between 1°, 2° and 3° amine?

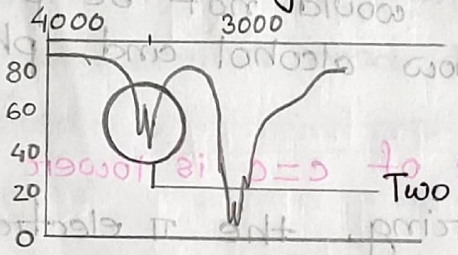
- By using IR spectroscopy, 1°, 2° and 3° amines can be distinguished.

Primary amine (1°):



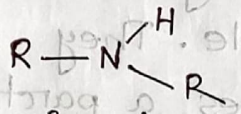
1° amine.

Primary amine shows two distinct N-H stretching bands in the range of 3300 - 3600 cm^{-1}. These are due to symmetric and assymmetric stretching vibrations of the two N-H bonds.



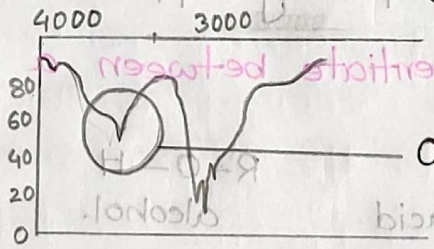
Two peaks for 1° amine.

Secondary amine:



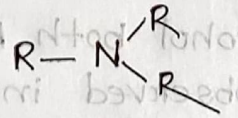
2° amine.

Secondary amine shows only one N-H stretching band in the same region because it has only one N-H bond.

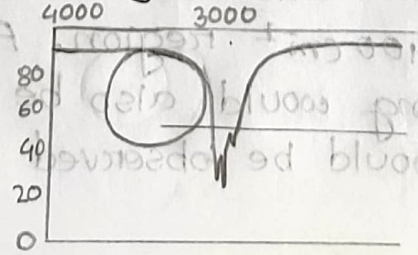


One peak for 2° amine

Tertiary amine:

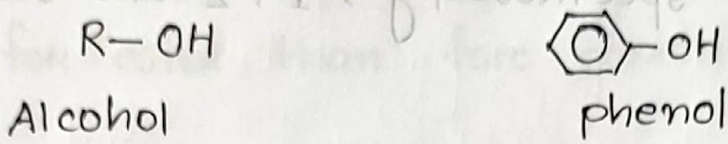


It doesn't contain any N-H bond; so it doesn't show any absorption in this region.



No peak for 3° amine

□ How can you distinguish between an alcohol and phenol?



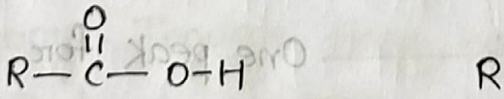
Alcohol phenol

For both alcohol and phenol, we would find a O–H bond stretching in the 3300 – 3600 cm^{-1} region. But as the phenol contains a benzene ring, a value of vibration frequency for C=C bond can be found at 1600 cm^{-1} region and also the C–H bond stretching of the aromatic ring would be found at 3150 – 3050 cm^{-1} region. These values would not be present in an aliphatic alcohol. This is how alcohol and phenol can be distinguished.

□ Why is the value of C=C is lower for benzene than alkene?

– In a benzene ring, the π electrons of the ring never remains stable. They keep rotating all the time. This resonce causes a partial double bond character between the carbon atoms than a stable double bond. Thus resonance weakens the carbon – carbon bond than in alkene and the frequency appears at a lower region.

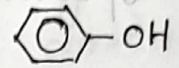
□ How can you differentiate between a carboxylic acid and an alcohol?



carboxylic acid alcohol.

Carboxylic acid and alcohol both have a O–H bond. But the O–H bond stretching observed in carboxylic acid in 3400 – 2400 cm^{-1} region, whereas in alcohol the value ranges from 3600 to 3400 cm^{-1} region. Again for carboxylic acid a C=O stretching would also be obseved at 1710 cm^{-1} but no such peak would be observed for alcohol.

□ How can you distinguish between an alcohol and phenol ?

– 

R–OH

Alcohol

phenol

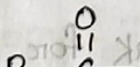
For both alcohol and phenol, we would find a O–H bond stretching in the 3300 – 3600 cm^{-1} region. But as the phenol contains a benzene ring, a value of vibration frequency for C=C bond can be found at 1600 cm^{-1} region and also the C–H bond stretching of the aromatic ring would be found at 3150 – 3050 cm^{-1} region. These values would not be present in an aliphatic alcohol. This is how alcohol and phenol can be distinguished.

□ Why is the value of C=C is lower for benzene than alkene?

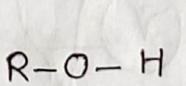
– In a benzene ring, the π electrons of the ring never remains stable. They keep rotating all the time. This resonce causes a partial double bond character between the carbon atoms than a stable double bond. Thus resonance weakens the carbon – carbon bond than in alkene and the frequency appears at a lower region.

□ How can you differentiate between a carboxylic acid and an alcohol ?

–



carboxylic acid



alcohol.

Carboxylic acid and alcohol both have a O–H bond. But the O–H bond stretching observed in carboxylic acid in 3400 – 2400 cm^{-1} region, whereas in alcohol the value ranges from 3600 to 3400 cm^{-1} region. Again for carboxylic acid a C=O stretching would also be obseved at 1710 cm^{-1} but no such peak would be observed for alcohol.

□ How can you differentiate between intermolecular and intramolecular H-bonding?

- In intramolecular H-bonding, the H-bond is formed within the molecule; leads to a stronger and more stable H-bond. Thus the spectrum^ necessarily differs so much.

whereas in an intermolecular H-bonding, the hydrogen bond is formed between two different molecules. This bond formation moves the electron density away from the electronegative atom and weakens the bond. Also the force constant, K becomes smaller and the frequency of the bond appears at a lower region. By noticing the changing frequency of a bond, intramolecular and intermolecular H-bonds can be identified.

NMR SPECTROSCOPY

- The atomic nuclei having a spin is nmrc active.

- **Conditions for a neucleus to be NMR active:**
  (i) odd mass
  (ii) odd atomic number
  (iii) both odd mass and atomic numbers.

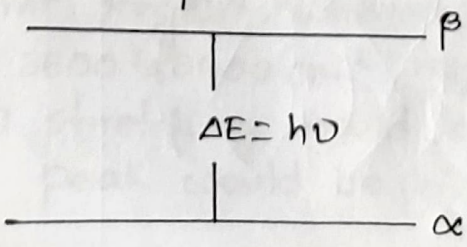
Example: $^{14}_{7}N$, $^{13}_{6}C$, $^{23}_{11}Na$ etc.

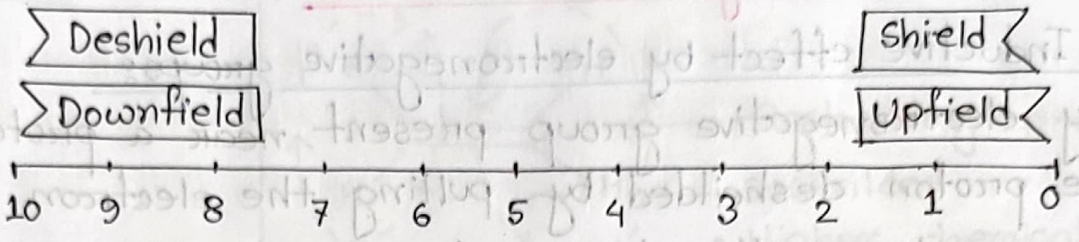
Here, only proton-nmrc would be discussed.

$^{1}_{1}H$ has two spin: $+1/2$ (α)↑ & $-1/2$ (β)↓

- **Principal of HNMR spectroscopy:**

Atomic nuclie of many atoms possess a spin property. When placed in a magnetic field is applied, the nucleus begins to precess about its own frequency. This produces an electric field around the nucleus. when the frequency of the oscillating electric field component of the incoming radiation matches the frequency of the electric field generated by the precessing nucleus, the two fields can couple and the energy is transferred from low energy state (α) to the higher energy state (β). After absorbing energy, the excited nuclei return to their lower energy state and emits a radiation known as NMR spectrum.





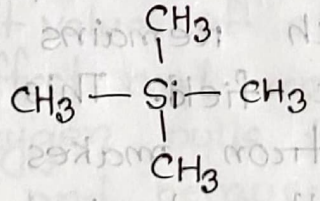
• **Sheilding:** Sheilding is a phenomegone that the electron density arround a neucleus resist to experience external magnetic field.

• **Desheilding:** Deshelding is when the electron density around a nucleus decreases, resulting in exprecience more of the external magnetic field.

• **Chemical shift:** The chemical shift expresses the amount by which a proton resonance is shifted from the relative standard value of TMS.

$$\delta = \frac{\text{shift in Hz}}{\text{spectrometer frequency in MHz}}$$

• **TMS:** The standard reference substance that is used universally in NMR spectroscopy is called tetramethylsilane or TMS.



• When protons are sheilded they remains at upfield with low frequency.

• When proton becomes deshielded by any electronegative effect they goes to down field with high frequency.

• Factors affecting chemical shift:

(i) Inductive effect by electronegative groups:

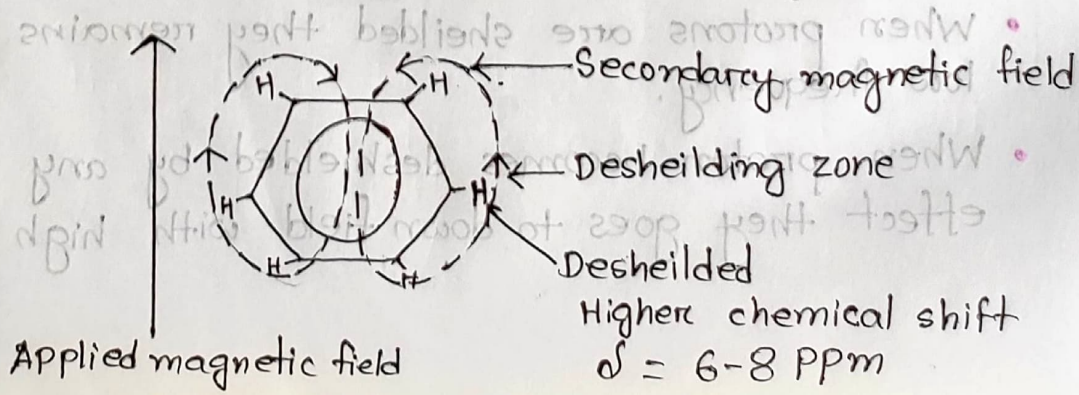
Any electronegative group present near a proton makes the proton desheilded by pulling the electron density away. Hence the desheilded proton moves to downfield and produce higher frequency.

Ex: In CH_4, the protons are highly sheided by the electron cloud of carbon. [$\delta = 0.9$]
whereas in CH_3F, the fluorine atom being highly electronegative pulls the electron density away; making the hydrogen desheilded and causing the frequency to be higher/ the chemical shift value, δ to be larger. [$\delta = 4.26$]. For the same reason the value of CH_3OH is 3.26. As F is more electron negative than O, it can desheild more and appear at higher frequency than oxygen.

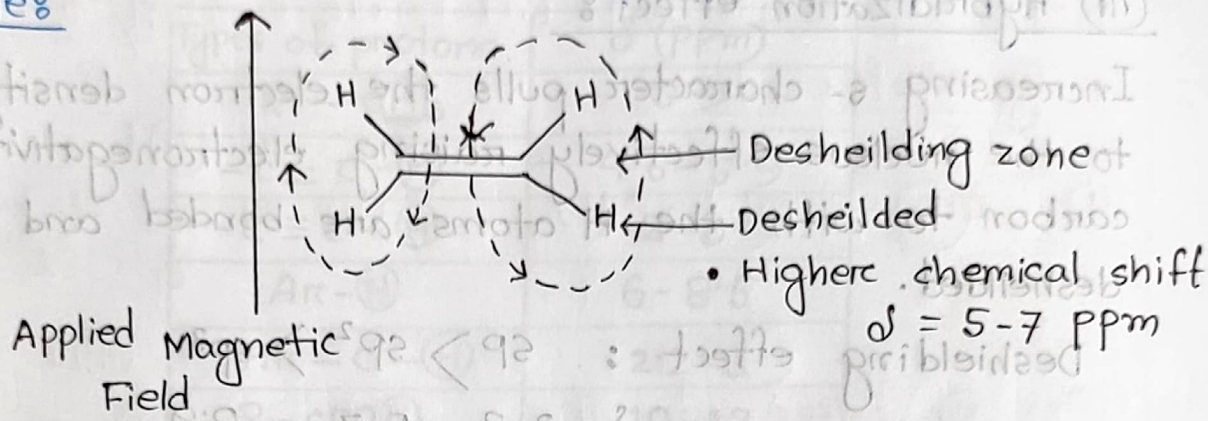
(ii) Magnetic Anisotropic Effect:

This effect is observable for multiple bonds.

□ Benzene: When benzene is placed in a magnetic field, the p electrons produces another secondary field which remains at alignment with the applying magnetic field. This anistoropic effect generated by p electron makes the benzene proton desheided and shift the chemical value greater than expep expected.



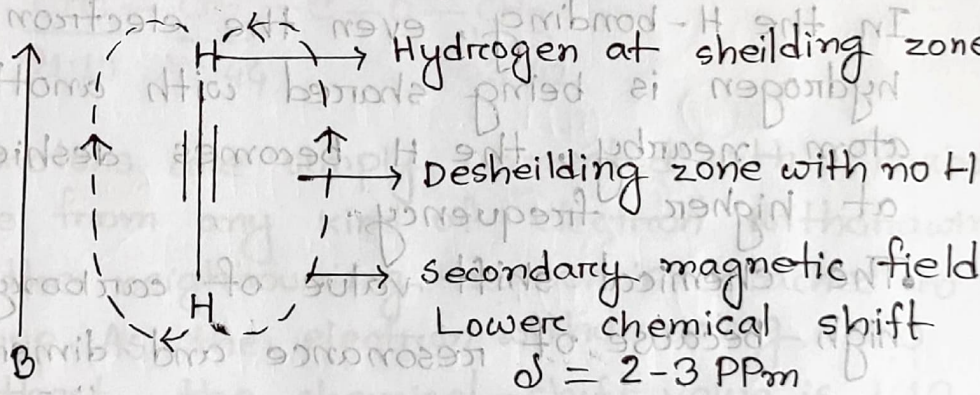
□ Alkene:



- Higher chemical shift
- δ = 5-7 PPm

□ Alkyne:

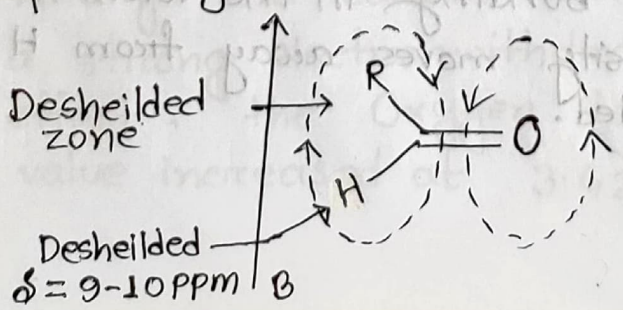
Though alkyne has p electrons, it can not produce high chemical shift due to magnetic anisotropic effect. The magnetic field generated by induced circulation of the p electrons has a geometry such that the hydrogens are shielded.



Lower chemical shift
δ = 2-3 PPm

□ Aldehyde:

The chemical shift value of aldehyde is very high due to both inductive effect and magnetic anisotropic effect. The carbonyl oxygen pulls the electron density away from the proton and because of the secondary magnetic field produced by p electron align with external field the proton get more deshielded and appear at higher frequency



(iii) Hybridization effect:

Increasing s- character pulls the electron density closer to the nucleus effectively raising electronegativity of the carbon to with the H atoms are bonded and H become desheilded.

Deshielding effect : $sp > sp^2 > sp^3$

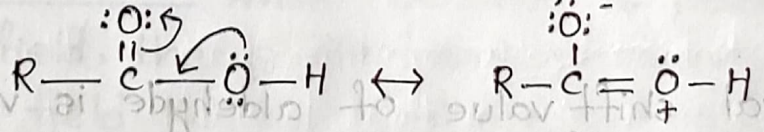
δ 2-3 5-7 0·9

The desheilding of sp^2 should be lower than sp but because of magnetic anisotropic effect sp^2 hybidized H shows larger chemical shift value.

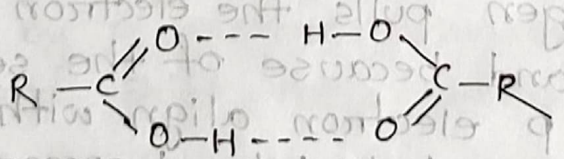
(iv) H-bonding:

In the H-bonding, even the electron cloud around the hydrogen is being shared with another electronegative atom nearby, the H becomes deshielded and appear at higher frequency.

- The chemical shift value of carboxylic acid is rediculously high because of resonance and dimerization.



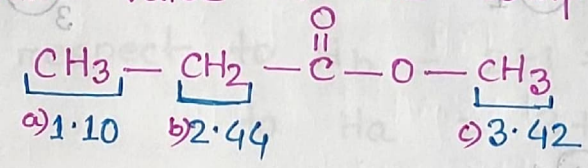
Because of this resonance the hydrogen gets desheilded.



Because of the hydrogen bonding in the formation of dimer, the electron density moves away from H and makes it more desheilded.

Types of protons	δ (ppm)
R – C(H)3	0·9
-C=C-(H)	4·6-5·9
-C≡C-(H)	2-3
Ar-(H)	6-8·5
Ar-C-(H)	2·2-3
R-COO(H)	10-12
R-COO-C(H)	3·7-4·1
Ar-O(H)	4-12
R-CO-(H)	9-10

- Explain the nmr value of the compound :



a) Here, the protons are simply connected with a carbon that is free from any kind of electron withdrawing group. But the neighbouring carbon is attached to a carbonyl group. As the electron withdrawing C=O group is furthest, the chemical shift value is 1·10 (very close to 0·9)

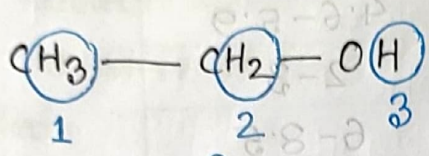
b) Here, the carbon is attached to a carbonyl carbon. The oxygen atom pulls the electron density also away from the neighbouring carbon which makes the hydrogen deshield and it's value rises to 2·44 ppm

c) Finally, the carbon atom is directly connected to a strong electronwithdrawing group. By inductive effect, the Oxygen deshields the protons and value increased at 3·42 ppm.

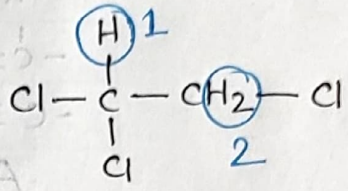
• Spin - spin splitting (n+1) Rule/coupling/Multiplicity rule:

• Types of protons:

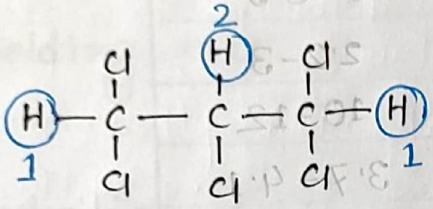
(i)



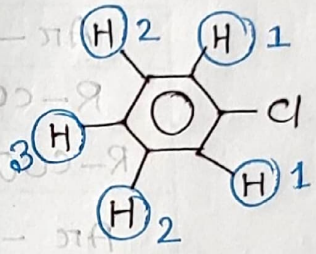
(ii)



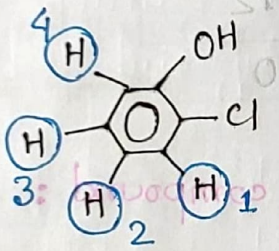
(iii)



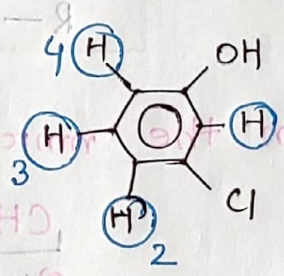
(iv)



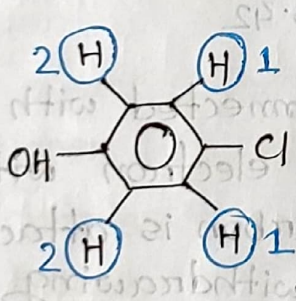
(v)



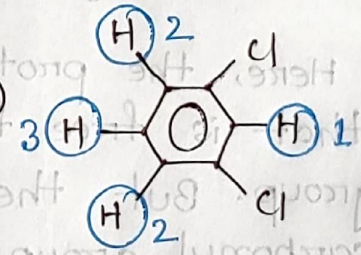
(vi)



(vii)



(viii)



- Protons with same number are consided as equivalent protons.
- Protons with different number are non equivalent protons.
- Number of types of proton = Number of NMR peak.

• Splitting of proton - nmrc with respect to neighbouring protons

(i) $CH_3(Ha) - CH_2(Hb) - Cl$

Here, two type of protons are present.
Ha has two hydrogen at its neighbouring carbon.
Thus the splitting of Ha = 2+1 = 3 (triplet)

Hb has three hydrogen at it's neighbouring carbon.
splitting = 3+1 = 4 (quadrate).

(ii) $CH_3(Ha) - CH_2(Hb) - CH_3(Ha)$

For Ha with respect to Hb = 2+1 = 3
For Hb " " to Ha = 3+3+1 = 7 [three from each carbon]

(iii) $CH_2(Ha) - CH_2(Hb) - OH(Hc)$

For Ha with respect to Hb = 2+1 = 3
For Hb " " to Ha = 3+1 = 4
For Hc = 1

[** Here Hc won't be count for splitting of Hb, because the alcoholic proton is exchangable. **]

[As, the proton of O-H is exchangable, it won't be considered with respect to Hb. Thus it appear as singlet]

• Also the protons of NH_2, COOH exchange so frequently with the neighbouring proton (of other molecule) that it can't sense the proton of adjacent carbon and appear as singlet.

The splittings are known as :

Number	Name	Number	Name
1	Singlet	5	Quintet
2	Doublet	6	Sextet
3	Triplet	7	Septet
4	Quartet	8	Octet

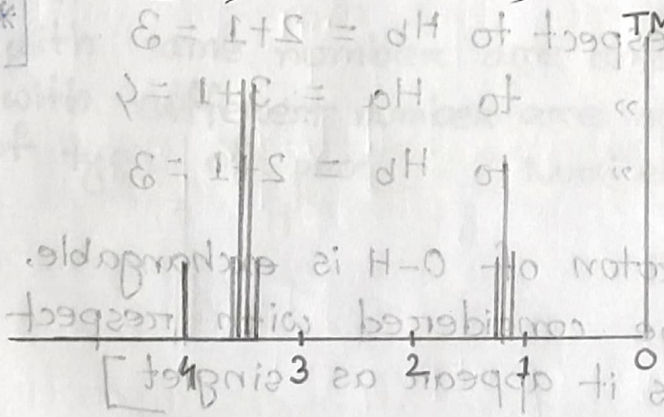
** Generally more than number 5 are termed as multiplets.

These splitting varies the intensity ratios according to pascall's triangle :

Singlet → 1
Doublet → 1 : 1
Triplet → 1 : 2 : 1
Quartet → 1 : 3 : 3 : 1

Example: CH_3-CH_2-OH

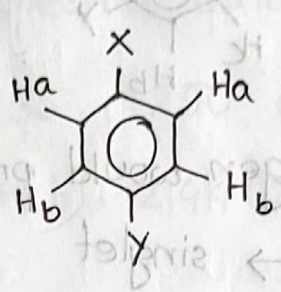
δ = 1.2 (3H, t), 3.5 (2H, q), 4.0 (1H, s)



* H of aldehyde is not exchangable. Thus it gives triplet in $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$. But CH_2 would experience 3+1+1 = 5, splitting

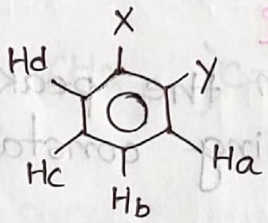
- How can you differentiate between ortho, parca and meta substituents by NMR spectroscopy?

Parca substituent:



Here, in the parca substituent, Hb would experience doublet spliting because of one neighbourcing Ha. Another Hb would also experience the same doublet spliting. This phenomenone is known as doublet of doublet. Two Ha would also experience the same doublet spliting because of the same reasons. By observing doublet of doublet spliting relative to TMS, parca substituents of the benzene ring can be identified.

Ortho substituents:

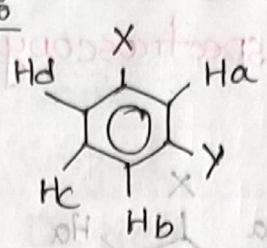


In case of ortho substituents, there are four distinct hydrogen.

Ha → Doublet
Hb → Triplet
Hc → Triplet
Hd → Doublet

But these four peaks are so close that they often appear as multiplet. When δ 7.2-7.5 (4H, m) value is observed in the nmr spectrum, ortho substituent would be identified.

Meta substituents:



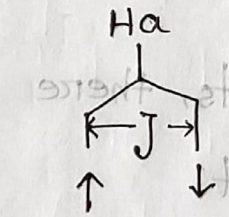
Four distinct hydrogen would produce four different splitting

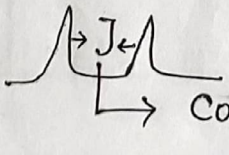
Ha → singlet
Hb → Doublet
Hd → Doublet
Hc → Triplet.

They don't appear as multiplets as of ortho substituents. By observing three types of splitting the presence of substituents in the meta position of a benzene ring would be identified.

• Coupling constant:

The distance between the peaks in a simple multiplet is called the coupling constant J.





• Deduce the structure of the following compounds:

⊙ An organic compound of molecular formula $C_4H_8O_2$ shows the following spectral data :

IR : 2960, 2850, 1735, 1250 cm^{-1}

^1H-NMR : δ 1·10 (3H, t), 2·44 (2H, q), 3·42 (3H, s)

Answer:

IR Analysis:

υ max (cm^{-1})	Inference
2960	Asymmetric C-H stretching vibration of $-CH_3$
2850	Symmetric C-H stretching vibration of CH_3
1700	$>C=O$ stretching vibration
1250	C-O stretching vibration.

Thus, presence of $-CH_3$, $>C=O$ and $>C-O-$ is confirmed from IR spectrum.

H-NMR Analysis:

δ (ppm)	Inference
1·10 (3H, t)	CH_3-CH_2-
2·44 (2H, q)	$CH_3-CH_2-\overset{O}{\overset{\|\|}{C}}-$
3·42 (3H, s)	$-\overset{O}{\overset{\|\|}{C}}-O-CH_3$

Here, the chemical shift value 1·10 (3H, t) accounts for such three protons which is connected to a carbon having only two hydrogen atom and no electron withdrawing

group. These protons are shielded and remains at upfield. Two protons produce quartet spliting and its value is at comparatively downfield because it may have electron withdrawing group like C=O nearby. The last electron shift value of three protons resembles for such a condition where the proton bearing carbon is directly connected to a electron withdrawing group which cause the protons to be highly deshielded and appeare at downfield with high chemical shift value.

Thus, ressembling all the spectral analysis and molecular formula, it is observed that mentioned compound should be methylpropanoate.

$$CH_3 - CH_2 - \overset{\overset{O}{\|}}{C} - O - CH_3$$

Methyl propanoate.

⊙ An organic compound of molecular formula $C_5H_{10}O$ shows the following spectral data:

IR : 2960, 2845, 1718, 1460, 1370 cm^{-1}

^{1}H-NMR: δ 2·3 (2H, t), 2·1 (3H, s), 1·6 (2H, m), 0·9 (3H, t)

MS (m/z): 86, 71, 58, 43, 29

Answer:

IR Analysis:

υ_{max} (cm^{-1})	Inference
2960	Asymmetric C-H stretching vibration of CH_3
2845	Symmetric C-H " " " CH_3
1718	$>C=O$ bond stretching
1460	Asymmetric C-H bending vibration of CH_3
1370	Symmetric C-H " " of CH_3

Thus, CH_3, C=O groups are present in the molecule.

H- NMR Analysis:

δ (ppm)	Inference
2.3 (2H, t)	$-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-$
2.1 (3H, S)	$-\overset{O}{\overset{\|}{C}}-CH_3$
1.6 (2H, m)	$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-$
0.9 (3H, t)	CH_3-CH_2-

Analyzing all the spectral analysis and molecular formula the structure seems to be,

$$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$$

[2 - propanone]

⊙ A compound having a molecular formula $C_9H_{10}O_2$ has the following spectral data:

IR : 3100, 2950, 2900, 1719, 1600, 1500, 1000 cm^{-1}

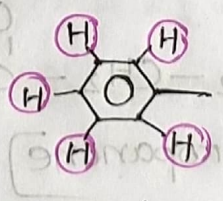
^1H NMR: 1·2 (t, 3H), 4·5 (q, 2H), 7·5 - 8·5 (m, 5H)

Answer:

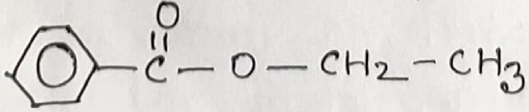
IR Analysis:

$\bar{\nu}$ max (cm^{-1})	Inference
3100	C-H bond stretching for aromatic ring
2950	Asymmetric C-H bond stretching for CH_3
2900	Symmetric C-H " " " CH_3
1719	$>C=O$ bond stretching
1600	$>C=C<$ bond stretching for aromatic ring
1500	$>C=C<$ bond " " " "
1000	C-O bond stretching.

H-NMR Analysis:

δ (ppm)	Inference
1·2 (t, 3H)	CH_3-CH_2-
4·5 (q, 2H)	CH_3-CH_2-O-
7·5 - 8·5 (m, 5H)	

From, the IR and N-MR Data it can be observed that the structure of the compound would be:



[Ethylbenzoate]

⊙ An organic compound of molecular formula C_8H_8O shows the following spectral data;

IR : 3080, 2960, 2850, 1690, 1600 cm^{-1}

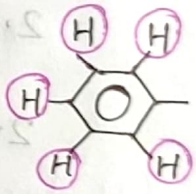
^1H-NMR : δ 7.2-7.3 (5H, m), 2.2 (3H, s)

Answer:

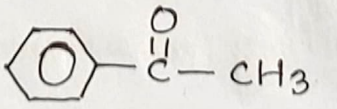
IR Analysis:

$\bar{\upsilon}$ (cm^{-1})	Inference
3080	C-H bond stretching for aromatic ring
2960	Asymmetric C-H bond stretching vibration for CH_3
2850	Symmetric C-H " " " " CH_3
1690	C=O bond stretching vibration
1600	C=C bond stretching for aromatic ring.

H- NMR Analysis:

δ (ppm)	Inference
7.2-7.3 (5H, m)	Monosubstituted benzene ring. 
2.2 (3H, s)	$-\overset{O}{\overset{\|}{C}}-CH_3$

The structure of the compound would be,



Acetophenone

⊙ A compound with molecular formula $C_5H_8O_3$ gave the following spectral data:

(i) IR : 3000–2500 (bn), 1715 (S), 1342 (w) cm^{-1}

(ii) ^1H-NMR: δ 2·12 (S, 3H), 2·25 (t, 2H), 2·60 (t, 2H), 11·1 (S, 1H)

Answer:

IR Analysis:

$\bar{\upsilon}$ (cm^{-1})	Inference
3000–2500	O–H stretching vibration for carboxylic acid
1715	C=O stretching vibration
1342	C–O stretching "

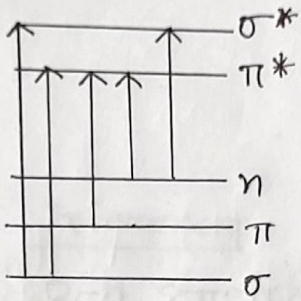
H-NMR Analysis:

δ (ppm)	Inference
2·12 (S, 3H)	$CH_3-\overset{O}{\overset{\|}{C}}-$
2·25 (t, 2H)	$-CH_2-CH_2-$
2·60	

UV SPECTROSCOPY

UV spectroscopy provides basic concepts for functional group.

Excitation level:



Increasing energy

$\sigma \rightarrow \sigma^*$ → Alkanes

$\sigma \rightarrow \pi^*$: Carbonyl compound

$\pi \rightarrow \pi^*$: Alkenes, Alkynes etc

$n \rightarrow \sigma^*$: oxygen, nitrogen

$n \rightarrow \pi^*$: generally forbidden

Energy of transition, $\Delta E = E_2 - E_1 = \frac{hc}{\lambda}$;

i.e. higher energy indicates lower wavelength.

Highest energy & Lowest wavelength : $\sigma - \sigma^*$

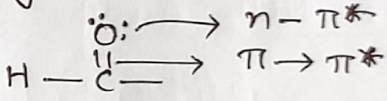
Lowest energy & Highest wavelength : $n \rightarrow \pi^*$

$CH_3 - CH_3$

$\sigma \rightarrow \sigma^*$

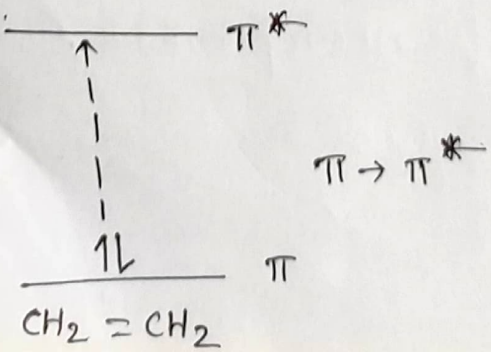
$CH_2 = CH_2$

$\pi \rightarrow \pi^*$

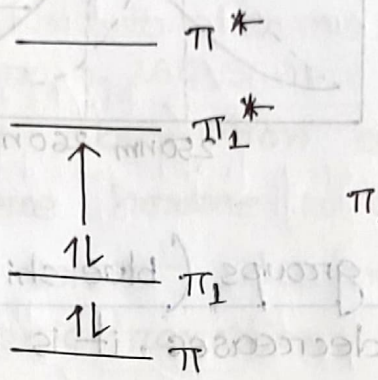


□ Among $CH_2 = CH_2$ & $CH_2 = CH - CH = CH_2$ for which one the wavelength would be higher?

- The wavelength would be higher for $CH_2 = CH - CH = CH_2$. In $CH_2 = CH_2$ there would be observed a $\pi \rightarrow \pi^*$ transition. The energy difference between these levels are quite high, resulting in a smaller wavelength.



$CH_2 = CH - CH = CH_2$ is a conjugated diene. Because of the conjugation two new energy levels are produced between π & π^*. As the transition between new π_1 and π_1^* requires less energy, their wavelength would be higher.



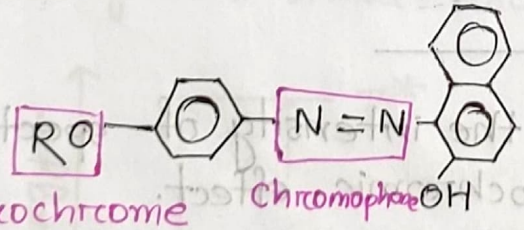
$CH_2 = CH - CH = CH_2$

- **Chromophore:** Any functional group which is responsible for imparting color to the compound is called chromophore.

  C=C, C=O, NO_2 etc works as a color bearer.

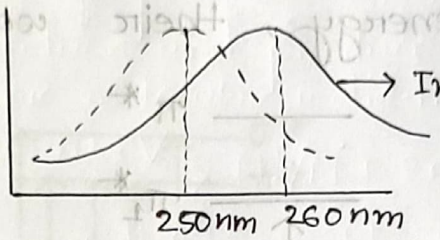
- **Auxochrome:** Any functional group which doesn't show color alone but helps the chromophore to increase its intensity is called auxochrome.

  -OH, $-NH_2$, -OR etc are auxochromes.

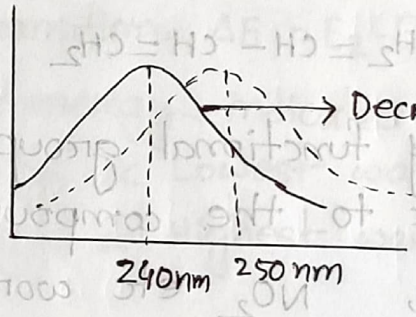


In this compound, N=N is surrounded by benzene ring causing conjugation. Futher, -OR, -OH also help in conjugation which reduces the energy gap and increases the wavelength and appear as bright colors.

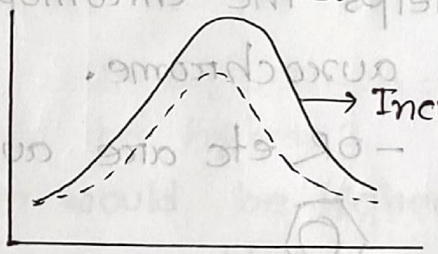
(i) Bathochromic groups (Red shift): If the wavelength of a spectrum increases i.e. UV value increases, it is termed as red shift.



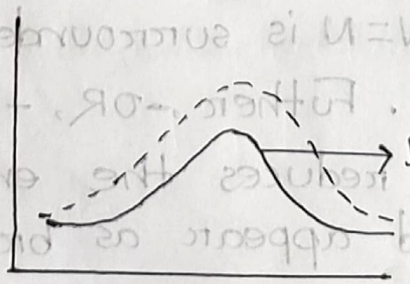
(ii) Hypsochromic groups (blue shift): If the wavelength of a spectrum decreases, it is known as blue shift.



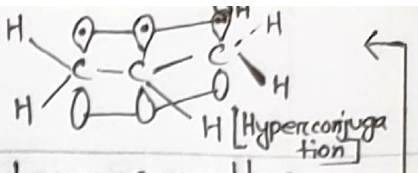
(iii) Hyperchromic effect: If the intensity of spectrum increases, it is called hyperchromic effect.



(iv) Hypochromic effect: If the intensity of spectrum decreases, it is called hypochromic effect.



The interaction is thought to result from overlap of C-H bonding orbitals with p electrons.



- $>C=CH-CH_3$, has a $-CH_3$ group which decreases the energy gap of $\pi \rightarrow \pi^*$ because of hyperconjugation. Thus the wavelength increases. Methyl group produces a bathochromic effect. Methyl group doesn't have unshared electrons.

□ When acetone is added to hexane, the $\lambda max = 279$ nm, whereas in H_2O $\lambda max = 264.5$ nm. Explain.

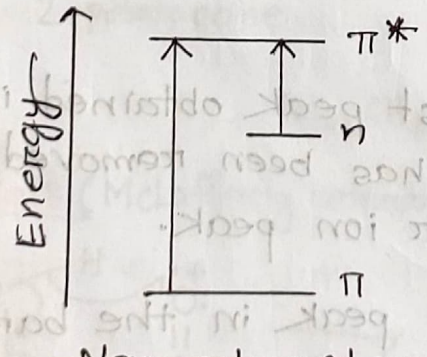
- The difference in absorption occurs due to the solvatochromism. Here hexane is a non-polar solvent and H_2O is a polar solvent. Acetone primarily undergoes two types of electronic transition:

$$n \rightarrow \pi^*$$

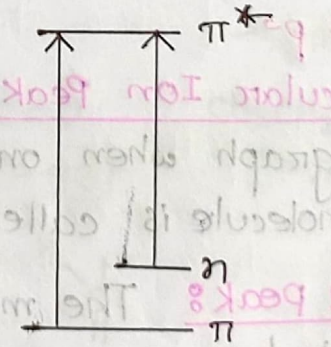
$$\pi \rightarrow \pi^*$$

In hexane, non-polar nature of hexane doesn't interact significantly with aceton's electronic structure. Thus both $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ occurs at a lower energy and the wavelength is higher.

In water, the polarity of water strongly interacts with acetons electronic structure forming H-bond. Thus $n \rightarrow \pi^*$ experiences a significant blue shift; increasing the energy gap and decreasing the λmax value.



Non-polar solvent



polar solvent

→ Acetone forms H-bond with water in ground state. Thus water stabilizes the ground state more which leads to higher energy gap in the excited state.

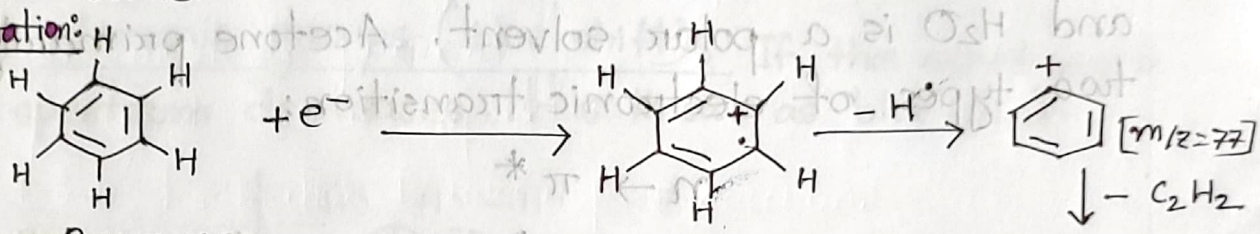
Mass spectroscopy

Principal: When a highly mobile electron strikes an atom/molecule it removes an electron and produce a positive charge of the atom/molecule or molecular radical.

$$M + e^- \xrightarrow{70\ eV} M^{+\cdot} + 2e^-$$

As electrons are very light weight, the molecular weight remains same.

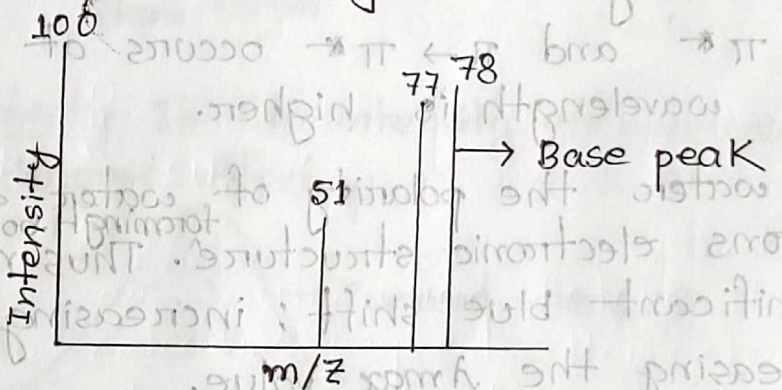
Fragmentation:



Benzene:

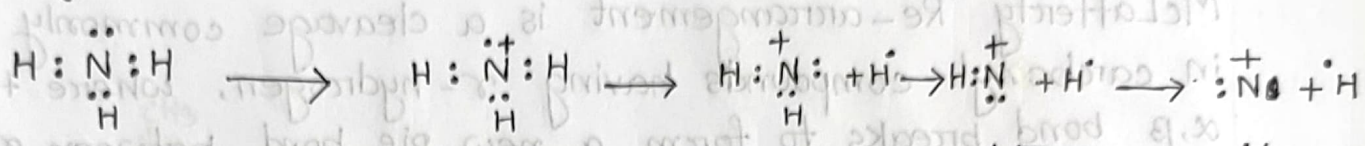
M/z = Mass : charge

Bar graph:

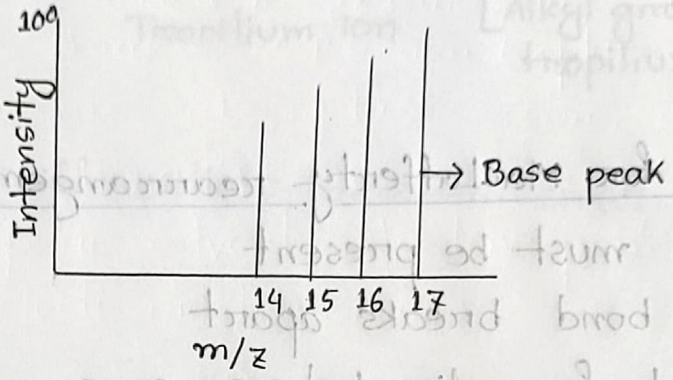


- **Molecular Ion Peak:** The first peak obtained in the bar graph when one electron has been removed from the molecule is called molecular ion peak.
- **Base peak:** The most intense peak in the bar graph is called base peak.

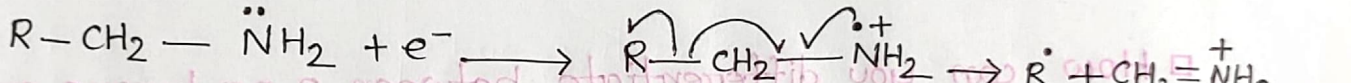
Fragmentation of NH_3:



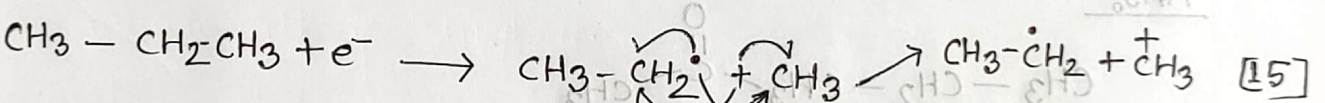
m/z = 17 16 15 14



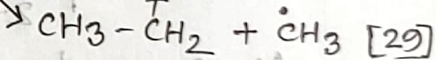
Fragmentation of amines:



m/z = 30, base peak

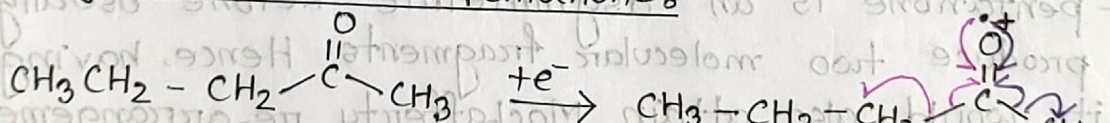
- 

[15]

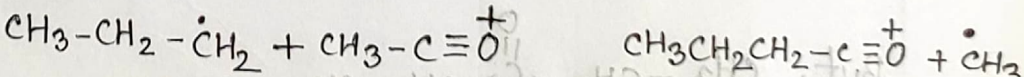


[29]

Fragmentation of 2-Pentanone:

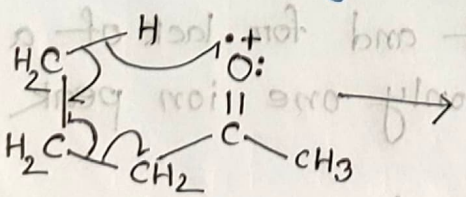


2-pentanone

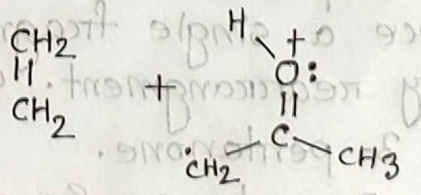


[m/z = 43] [m/z = 71]

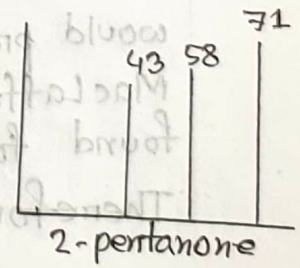
Specials (McLafferty rearrangement)



2-pentanone.



m/z = 58



• McLafferty Re-arrangement:

McLafferty Re-arrangement is a cleavage commonly occurs in carbonyl compounds having γ-hydrogen, where the α,β bond breaks to form a new pie bond between γ & β carbon.

conditions for MacLafferty rearrangement:

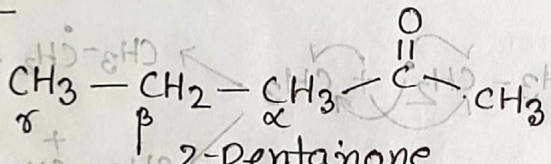
(i) γ-H must be present

(ii) α-β bond breaks apart

(iii) π bond formation between β-γ carbon.

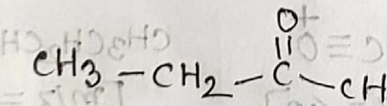
□ How can you differentiate between 2-pentanone and 3-pentanone?

Ans:



2-pentanone

As 2-pentanone is an asymmetric ketone usually it would produce two molecular fragments. Hence, having a γ-H it would also attain MacLafferty re-arrangement. [Show fragmentation] Thus three ion peak could be obtained.

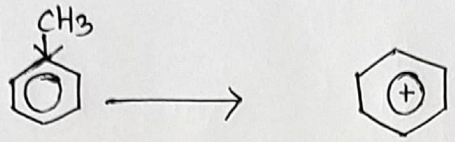
But,  having a symmetric structure

3-pentanone

would produce a single fragment and for lack of a γ-H, MacLafferty rearrangment. So only one ion peak is found for 3-pentanone.

Therefore, 2-pentanone & 3-pentanone can be distinguished.

Tropilium ion:



Tropilium ion

[Alkyl group produces tropilium ion]

□ How can you differentiate between CH_3Cl & CH_3Br ?

STEREOCHEMISTRY

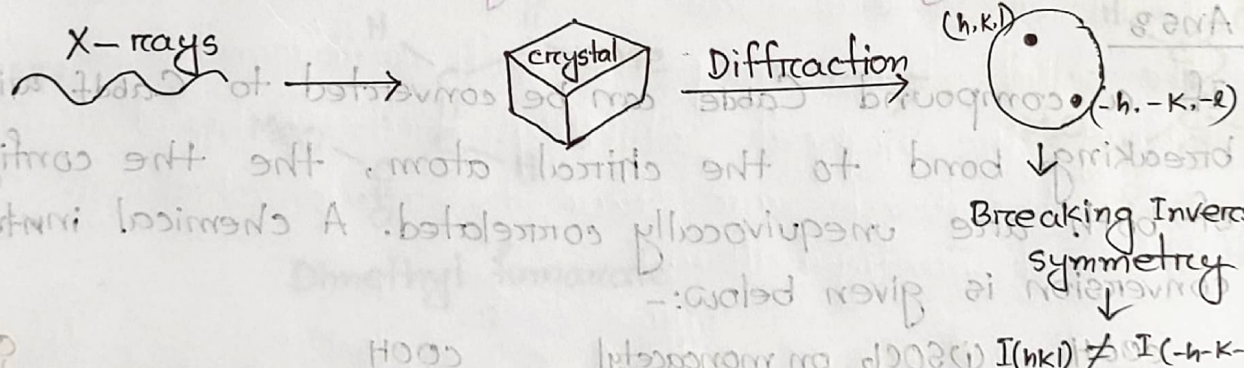
□ Define absolute and relative configuration. How can you determine absolute configuration?

Ans:

In three dimentional arrcangement of a molecule, the actual arrcangement of different atom orc groups that is determined by X-ray, electron diffraction is called absolute configuration.

Relative configuration is the arrcangement of atoms orc group of atoms that is described relative to otherc atoms orc groups of atoms in the molecule.

The absolute configuration can be determined by Bijvoet method. He studied a single crystal of sodium rubidium (+) tetrtrate tetrahydrcate with X-ray diffraction which excites a single heavy atom rubidium.



The absolute configuration of a chiral crystal is determined from the intensity differences of pairs of diffraction spots (reflections) that arre related by an inversion symmetry

□ Name the method for the determination of relative configuration.

Ans:

Some methods are:

(i) X-ray structure analysis

(ii) Chemical interconversion not affecting bonds to the chiral atom.

(iii) Methods based on symmetry properties

(iv) optical rotations

(v) The study of quasi-racemic compound

(vi) Enzyme studies.

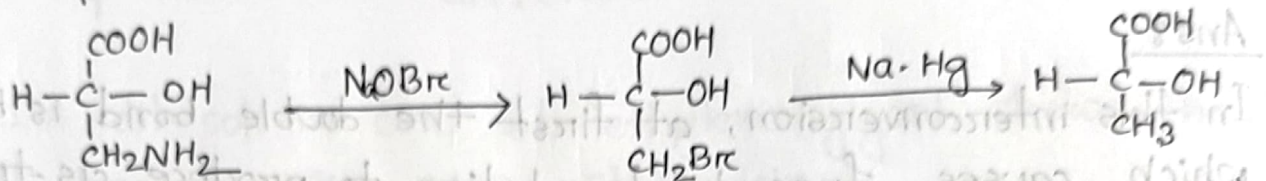
□ Discuss the correlation methods for the determination of relative configuration.

□ Discuss the correlation method of chemical interconversion not affecting bonds to the asymmetric atom.

Ans:

If a compound Cabde can be converted to Cabdf without breaking bond to the chiral atom, the the configuration of both are unequivocally correlated. A chemical interconversion is given below:-

Here, though the reactant is converted into another compound but the bond of chiral center is not broken.



R-(+)-Isoserine

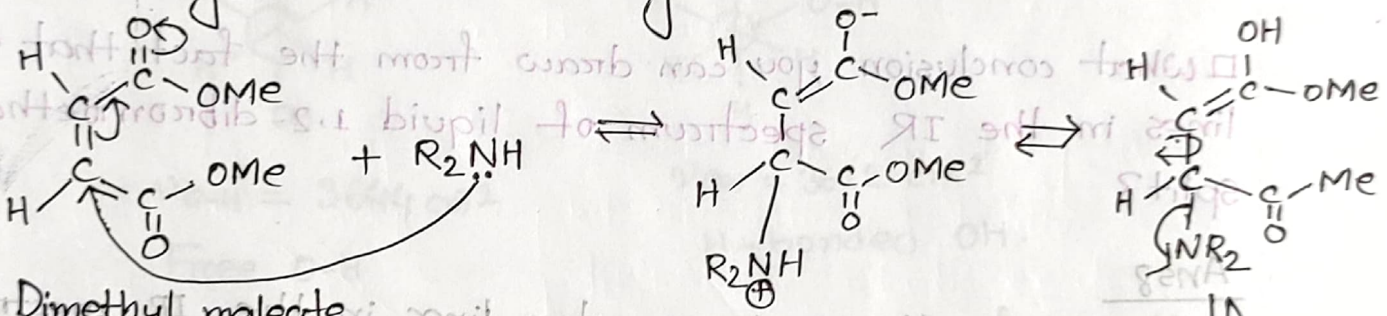
S-(-)- Bromolactic acid

R-(-)- Lactic acid

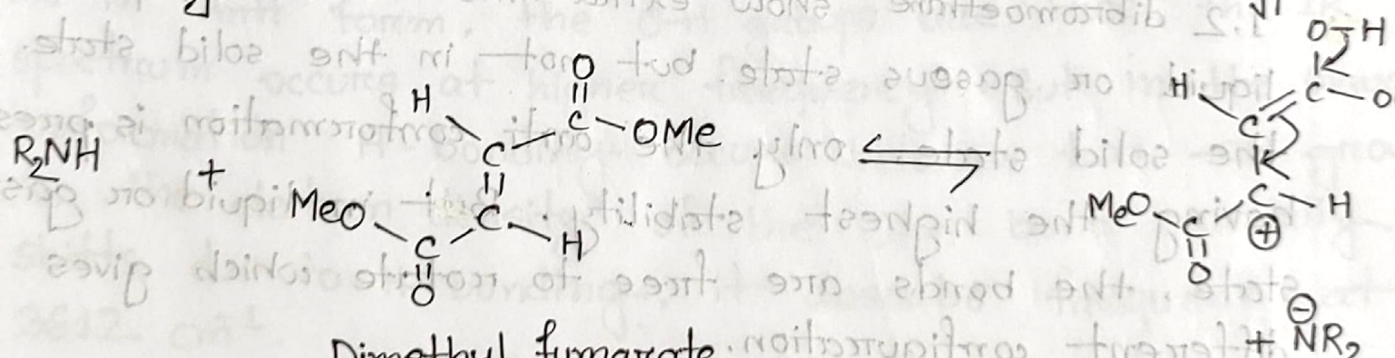
□ Dimethyl · maleate is isomerized to dimethyl fumarate by means of secondary amines. Explain.

Ans:

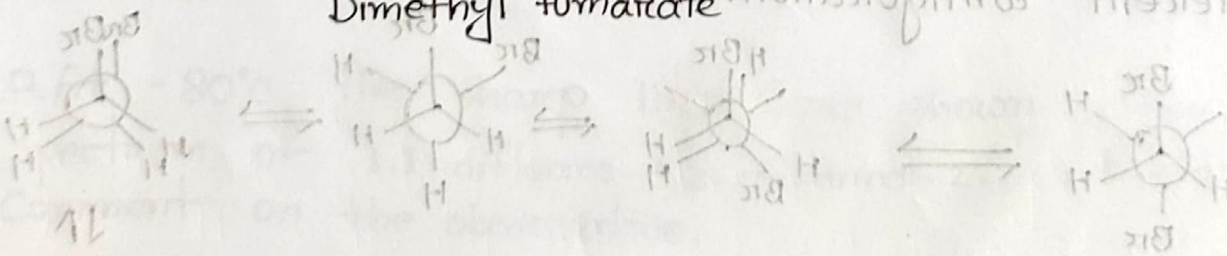
Interconversion is occured between cis-trans isomer by base catalyzed and secondary amine is also used.



Dimethyl maleate



Dimethyl fumarate



□ Name the methods used for interconversion of cis-trans isomers.

Ans 8

In this interconversion, at first the double bond is broken which causes free bond rotation to produce cis-trans isomer. Some methods are-

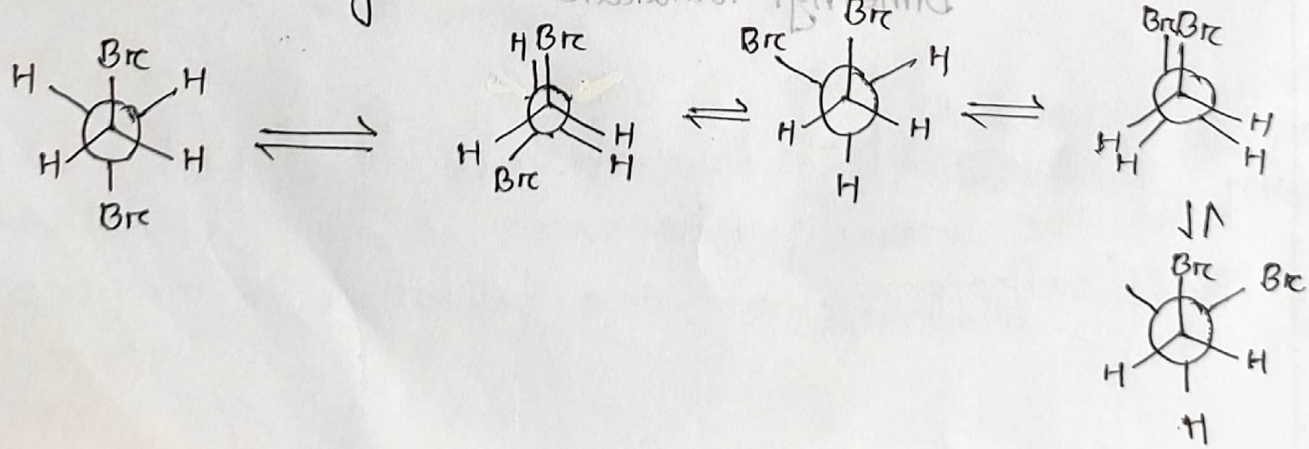
(i) By application of heat
(ii) Photochemical method
(iii) Acid - catalyzed isomerisation
(iv) Base catalyzed isomerisation
(v) Radical catalyzed

[2 এর চেয়ে বেশি Mark হলে Example দিতে হবে]

□ What conclusion you can draw from the fact that extra lines in the IR spectrum of liquid 1,2-dibromoethane are split?

Ans 8

1,2 dibromoethane show extra lines in the IR spectrum of liquid or gaseous state but not in the solid state. In the solid state, only the anti conformation is present having the highest stability. But in liquid or gasous state, the bonds are free to rotate which gives different configuration.

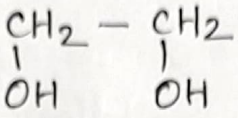


Thus for the rotation of the compound, the spectrum is splitted into different lines.

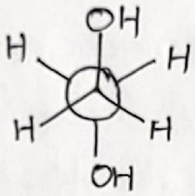
□ A dil. sol. in CCl_4 of ethylene glycol shows the following IR branch 3644, 3612 cm^{-1}. Explain.

Ans:

The structure of ethylene glycol-

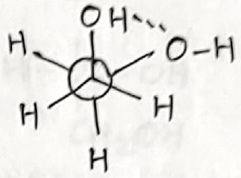


The conformation of ethylene glycol,



$\nu_{O-H} = 3644\ cm^{-1}$

Free O-H



$\nu_{OH} = 3612\ cm^{-1}$

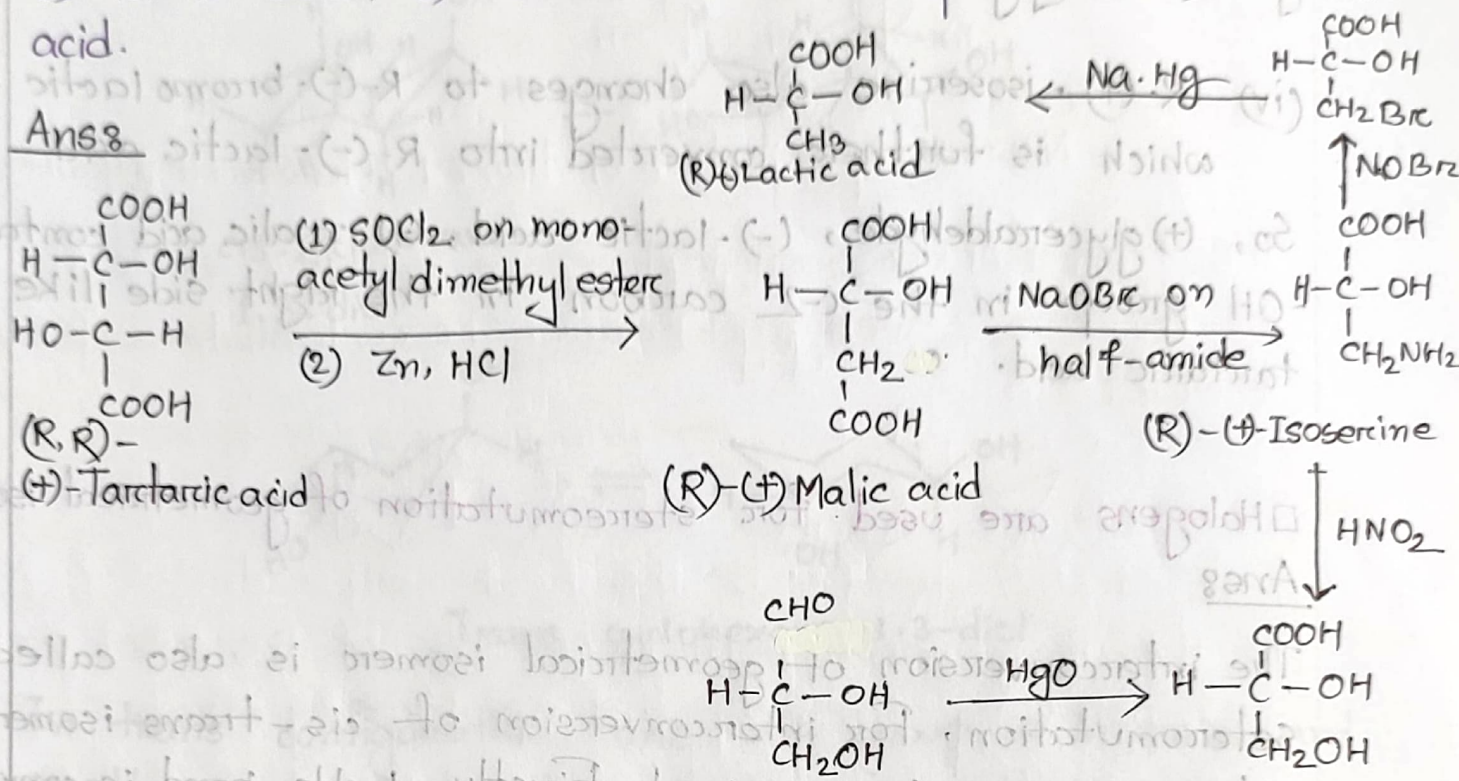
H-bonded OH.

As in Anti form, the O-H groups are free, the IR spectrum occurs at higher frequency. But in the Gauche conformation H-bonding occurs between two -OH group. As the electron density of the O-H bond partially shifts due to H-bonding, it absorbs frequency at 3612 cm^{-1}.

□ At -80°C five sharp lines are shown in the nmr spectrum of 1,1-difluoro-1,2-dibromo-2,2-dichloroethane. Comment on the observation.

□ Show the determination of configuration of (+) glyceraldehyde, (−)-lactic acid, (+) malic acid with respect to (+) tartartaric acid.

Ans:



Not necessary:-

Description:

(i) Here, the reaction is started from (R,R)-(+)-tartartaric acid where the C-3 -OH group in the left hand side and the C-2 OH group in the right hand side. Then the C-3 containing -OH group is converted into $-CH_2$ without breaking bond between C-2 and C-3 and gives R(+)malic acid with same configuration

(ii) R-(+) malic acid is converted into R(+) Isoserine with no change in the configuration as no bond of the chiral carbon is broken.

(iii) R-(+) Isoserine changes to glyceric acid. where

configuration is also same R(-) and R(+) glyceraldehyde gives R(-) glyceric acid.

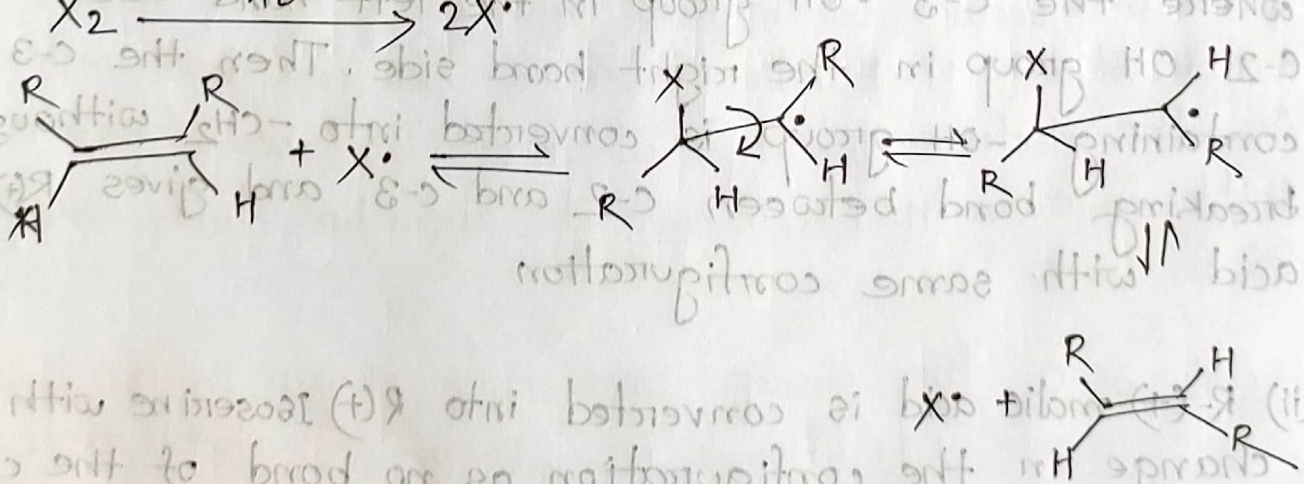
(iv) R-(+)-isoserine also changes to R-(-)-bromo lactic acid which is further converted into R-(-)-lactic acid.

So, (+) glyceraldehyde, (-)-lactic acid, (+)-malic acid contain -OH group in the C-2 carbon in the right side like (+) tartaric acid.

□ Halogens are used for stereomutation of geometrical isomers.

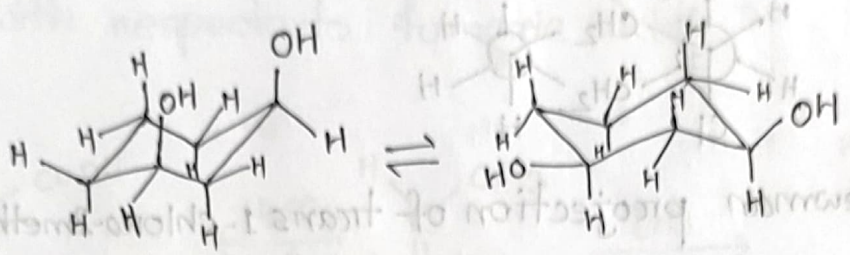
Ans:

The interconversion of geometrical isomer is also called setereomutation. For interconversion of cis-trans isomers extra energy is required. Firstly, double bond is converted into single bond then free bond rotation produces new isomer. Halogens are firstly converted into free radicals and then conversion occurs.

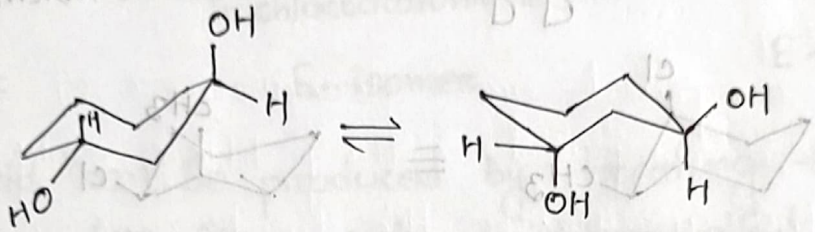


□ Cis -cyclohexane-1,3-diol is more stable than trans form.

Ans:



cis-cyclohexane-1,3-diol.



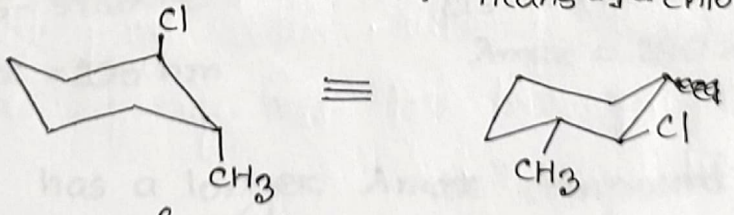
Trans-cyclohexane-1,3-diol

In trans configuration, one -OH group is in axial and another -OH is in equatorial position. But one of the cis conformation has two -OH groups in equatorial position which is the most stable one.

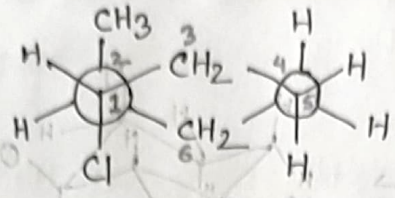
□ Draw the most stable conformation of trans-1-chloro-2-methylcyclohexane and its corresponding Newman projection. Also explain why is it more stable than cis-1-chloro-2-methylcyclohexane.

Ans:

The most stable conformation of trans-1-chloro-2-methylcyclohexane is:

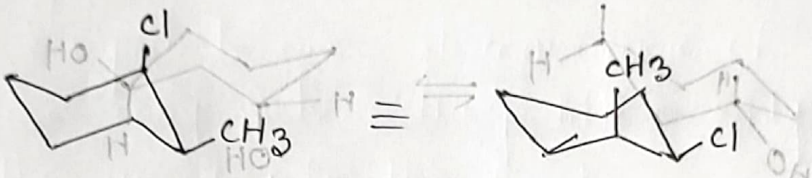


Chair conformation.



Newman projection of trans 1-chloro-2methylcyclohexane

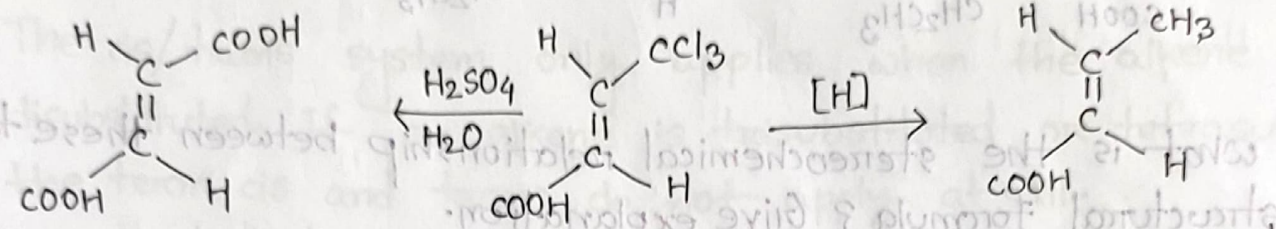
The trans 1-chloro-2methylcyclohexane is more stable than cis-1-chloro-2-methylcyclohexane. The conformations of cis isomer:



In the cis isomer, in both cases, one of the axial position is occupied by one group and equatorial by other.
Again, in the trans conformation, though once both group occupies axial position, by interconversion or ring flipping it becomes most stable having them at equatorial position.
Since diequatorial position refers to the least torsional strain than axial-equatorial position, trans conformation of the mentioned compound is more stable than cis.

□ Show how can you determine the configuration of crotonic acid with respect to fumaric acid.

Ans:



Fumaric acid — E-isomer

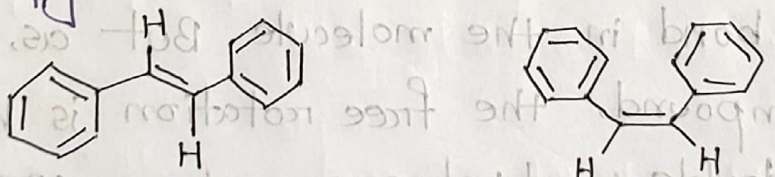
Trichlorocrotonic acid — E-isomer

Crotonic acid — E-isomer

Fumaric acid can be produced by treating trichlorocrotonic acid with water. Since only trichloromethyl group is converted into carboxyl group, the configuration remains same. Further, reduction of trichloromethyl group converts trichlorocrotonic acid into crotonic acid, it also remain as E isomer. As, both the fumaric acid and crotonic acid retain the configuration of trichlorocrotonic acid, their configuration is same and they are E isomers of each other.

□ Distinguish between cis and trans-stilbene by using UV-spectroscopy.

Ans:

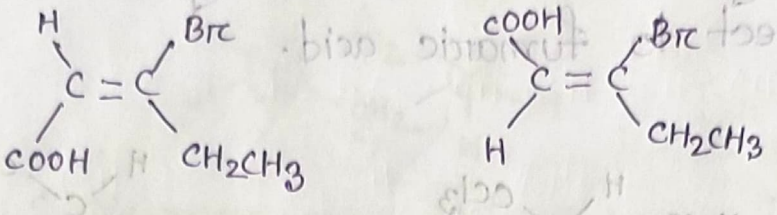


Trans-stilbene
$\lambda max = 295\ nm$

cis-stilbene
$\lambda max = 280\ nm$

Trans stilbene has a longer λmax compared to cis-stilbene. This is because the trans isomer is more planar, allowing for greater delocalization of pie electrons, which leads to absorption of longer wavelength.

□ A compound with molecular formula $C_5H_7BrO_2$:



(i) What is the stereochemical relationship between these two structural formula ? Give explanation.

Ans:

These two compounds are stereoisomers of each other. Between these compounds, they differ only by the arrangement of the groups. In the first compound, the higher priority groups in both the carbon are $-COOH$ and $-Br$ which are one the different side. In the second compound keeping other groups fixed, $-Br$ and $-COOH$ is at the same side. Thus they are (E)-(Z) isomers of each other respectively.

(ii) Are they conformational or configurational seteroisomers?

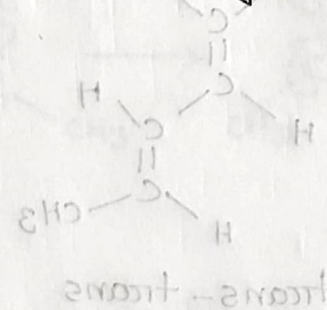
Ans:

Conformational isomers differ from each other by rotation of a single bond in the molecule. But as, in the mentioned compound, the free rotation is hindered because of double bond between carbons, conformational isomerism is not possible. Also, they differ only because of the arrangement of the groups in space, they are configurational isomers.

(iii) Either the cis/trans or the E/Z systems will you follow to give the names of these two compounds and why?

Ans:

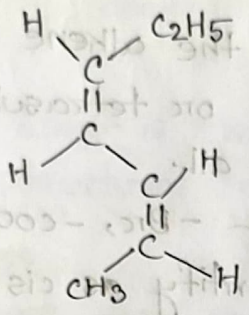
The cis/trans system only applies when the alkene is disubstituted. If an alkene is trisubstituted or tetrasubstituted the term cis and trans do not apply at all.
The mentioned alkenes are trisubstituted by -Br, -COOH and $-CH_2CH_3$. Thus it is not possible to identify as cis or trans. Rather by using chan Ingold prelog priority rules- naming can be done by (E)/(Z) system.



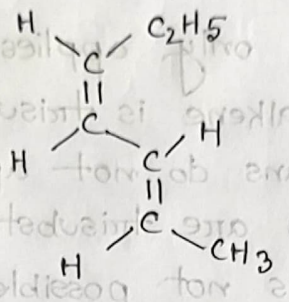
Class8

□ Draw isomers of $H_2C_5-CH=CH-CH=CH-CH_3$

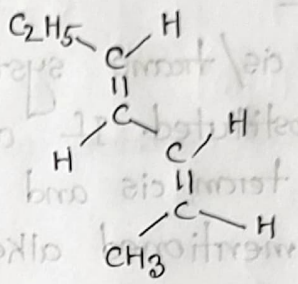
Ans:



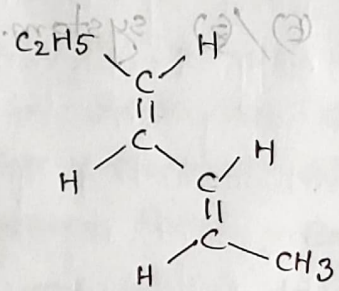
cis-cis



cis-trans



trans-cis

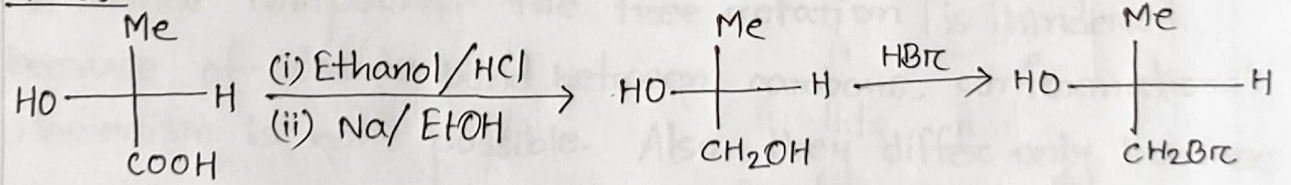


trans-trans

• Configuration: The arrangement of atoms that characterise a particular stereoisomer is called its configuration.

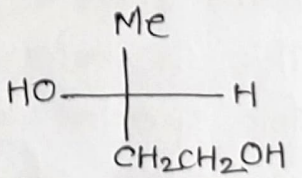
□ Determine the configuration of β-hydroxybutyric acid with respect to L-(-)-Lactic acid.

Ans:



L(-)-Lactic acid

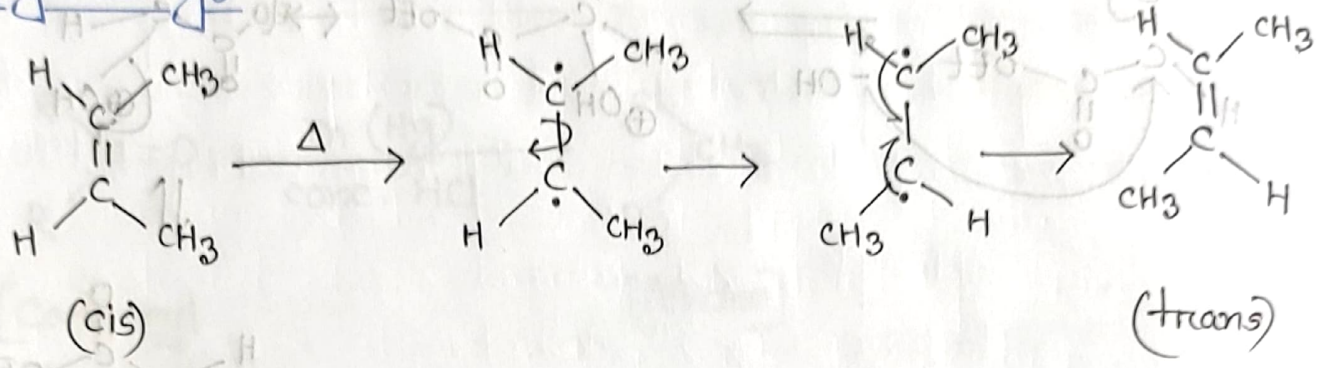
* Since no chiral bond is broken, configuration remains same.



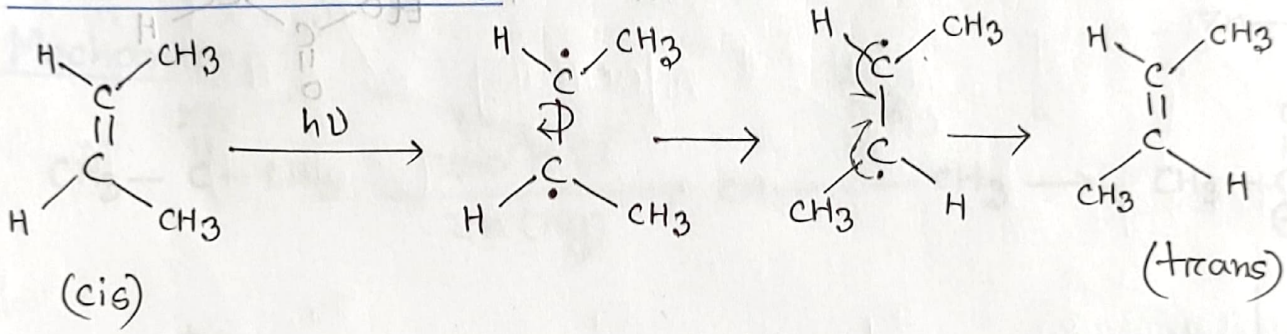
L-(-)-β-hydroxybutyric acid

□ Interconversion of geometrical isomers:

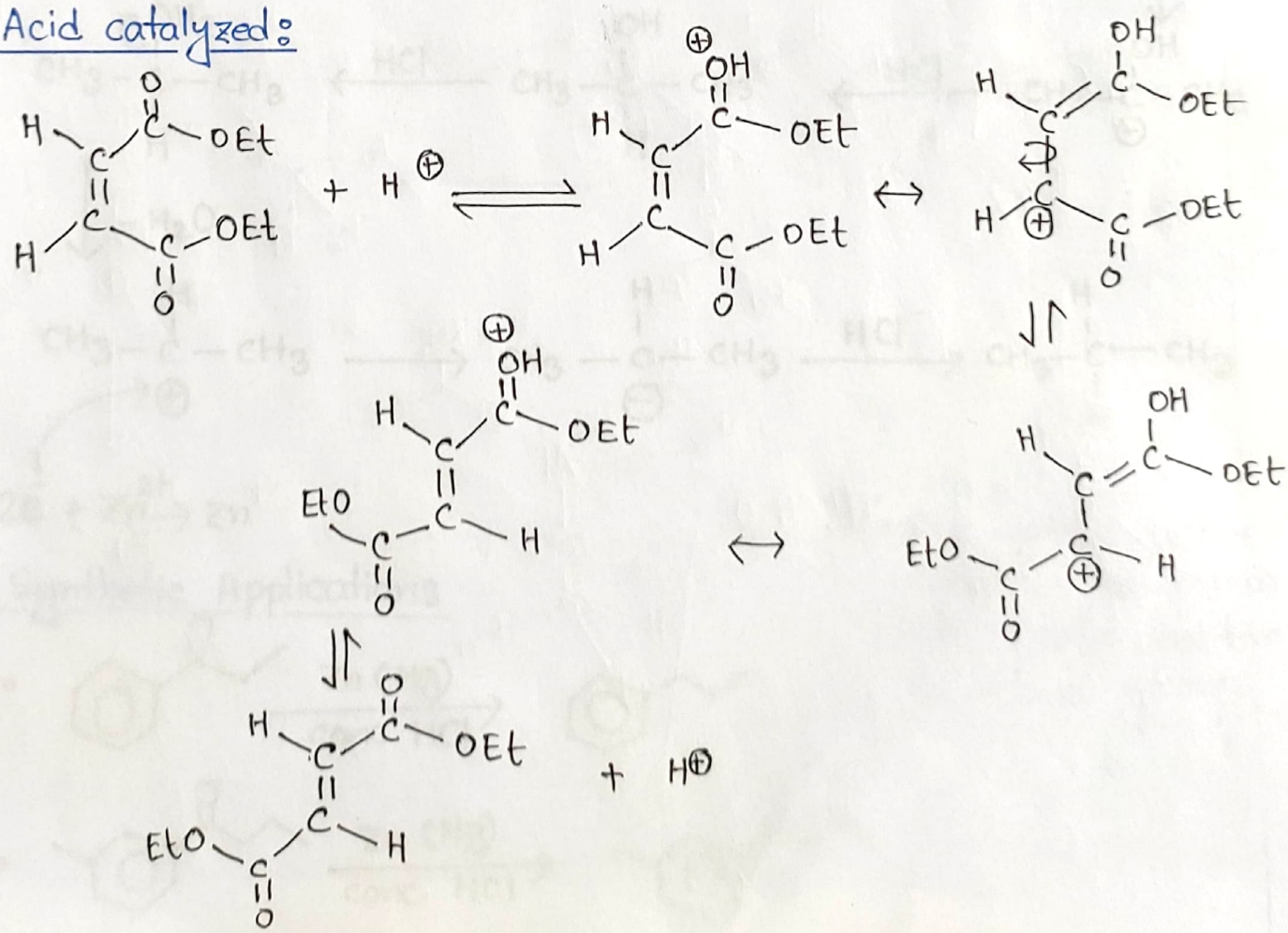
- **By heating:**



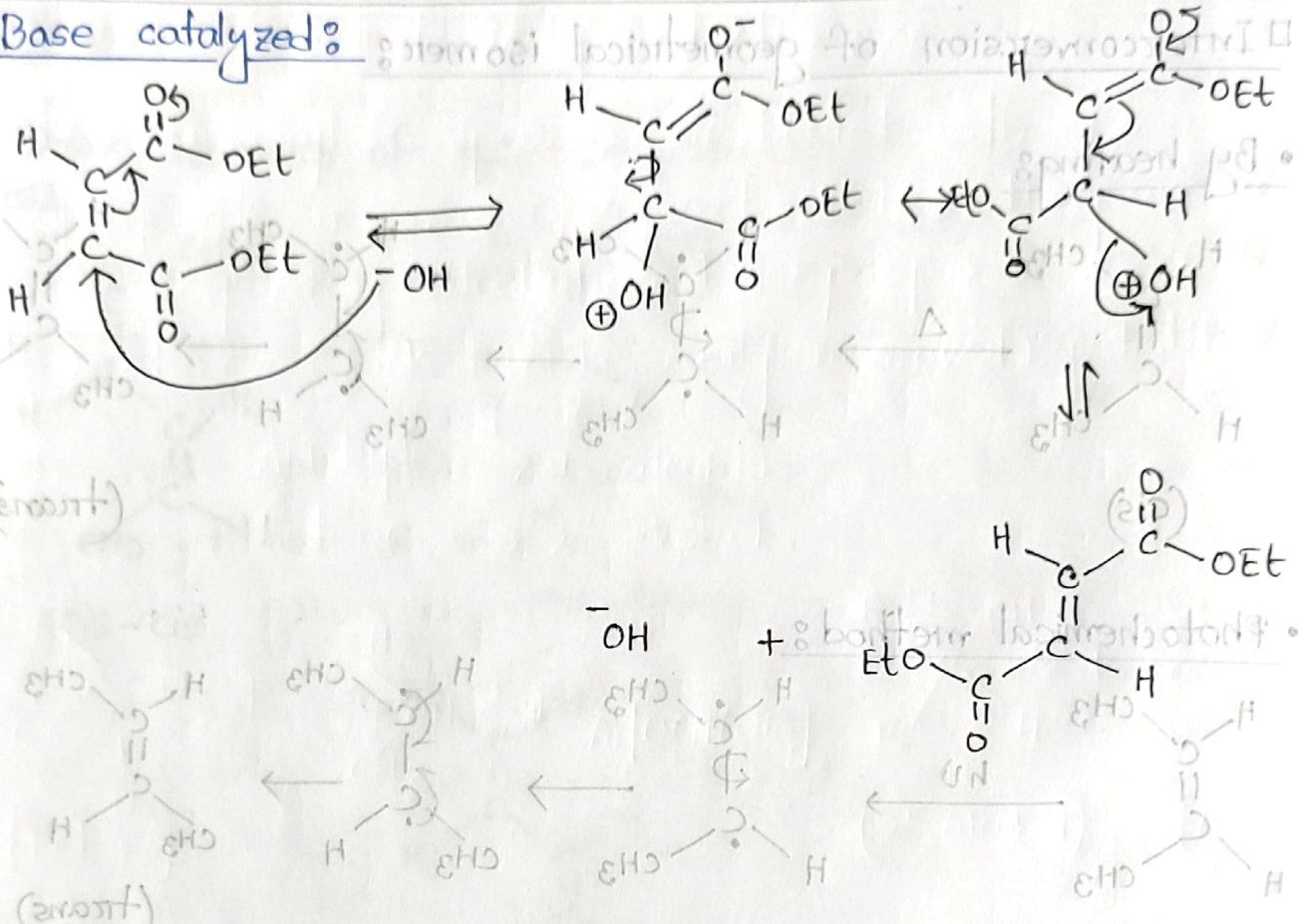
- **Photochemical method:**



- **Acid catalyzed:**

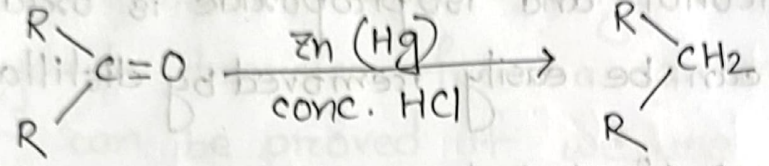


• Base catalyzed:



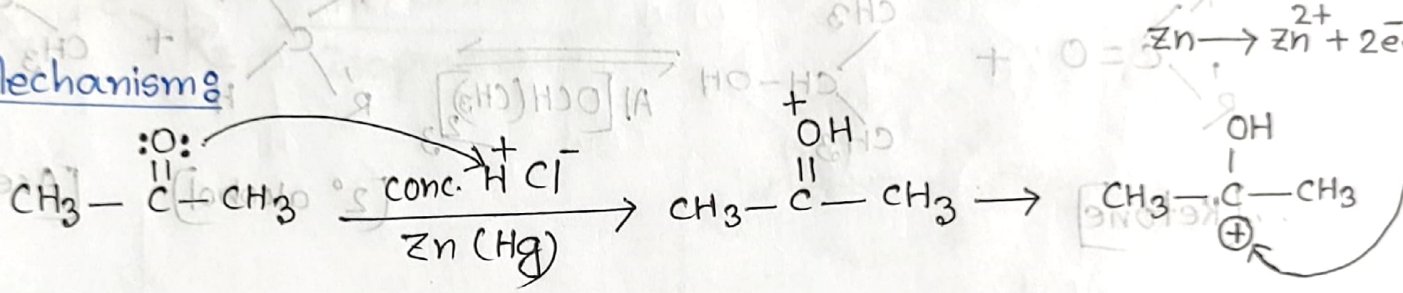
Name reactions

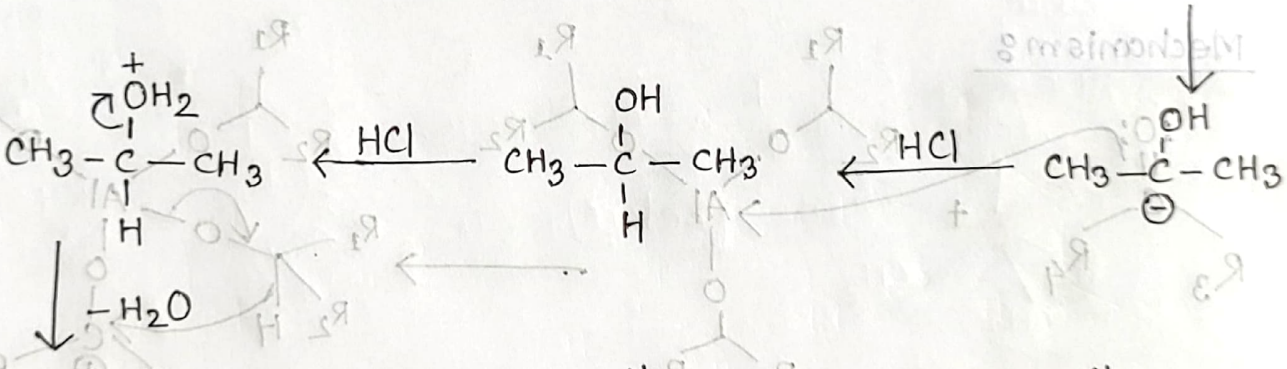
Clemmensen Reductions

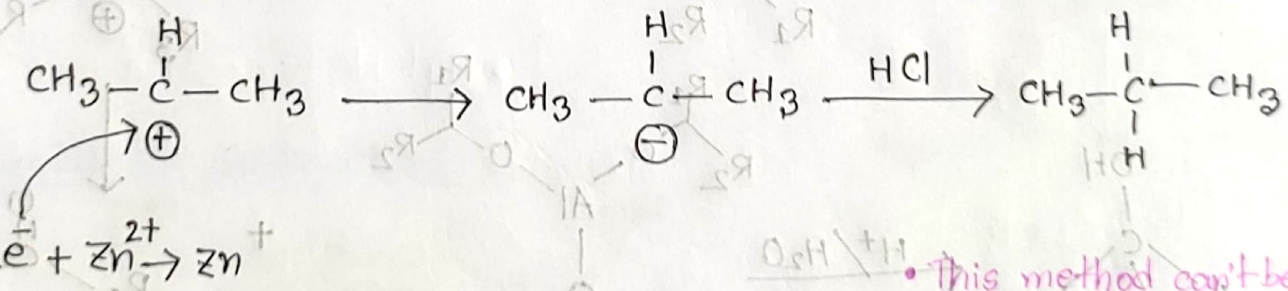


[Carbonyl Compound] [Alkane]

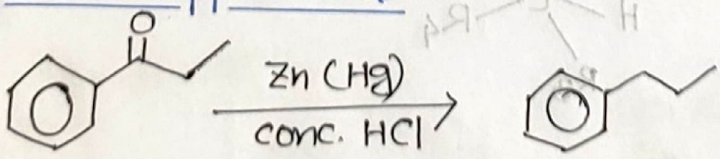
Mechanism:

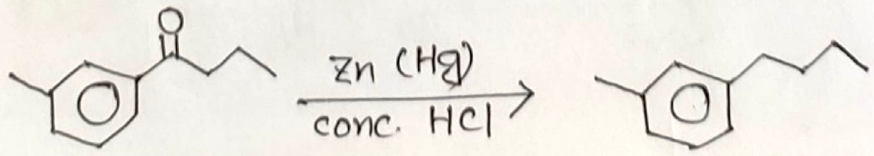






Synthetic Applications:

- 

- 

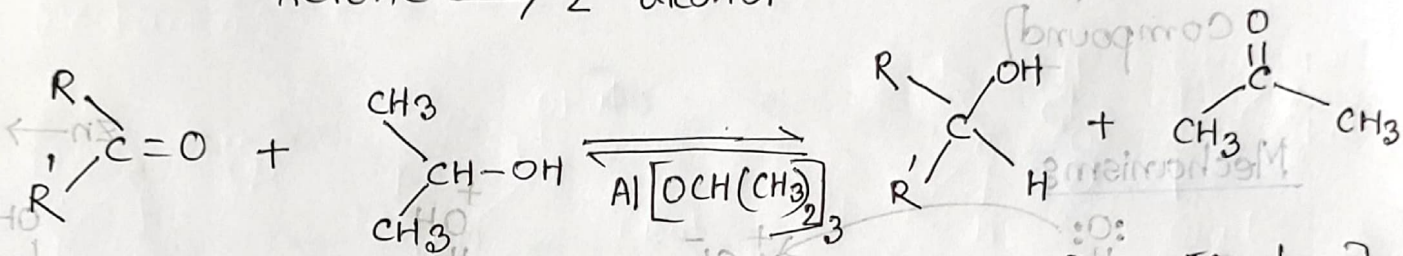
- This method can't be used for compounds having acid sensitive functional group.

Meerwein - Ponndorf - Verley reduction :

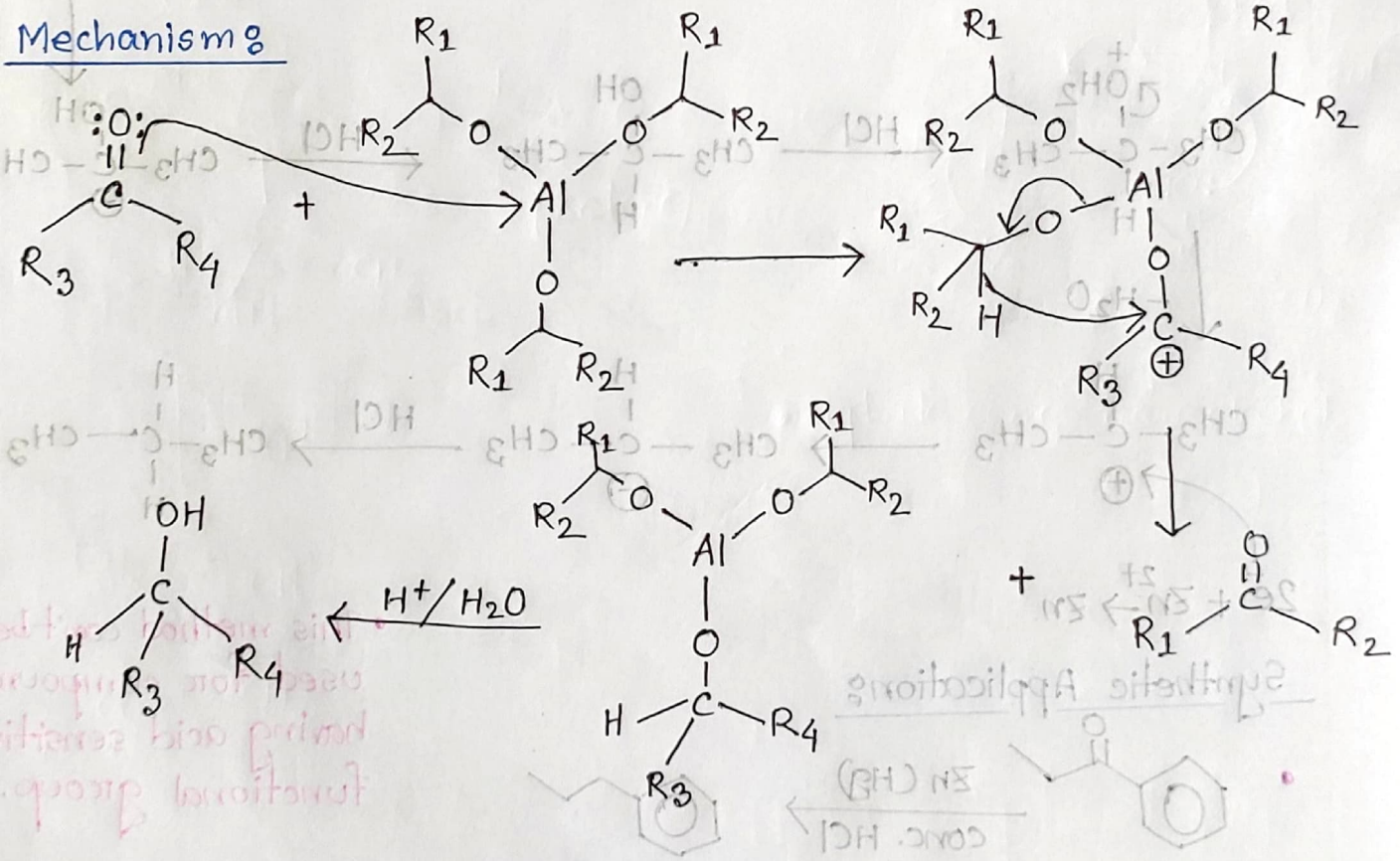
When carbonyl compounds are heated with aluminum isopropoxide in an isopropanol solution, they are reduced to corresponding alcohol and isopropoxide is oxidized to acetone, which can be easily removed by distillation.

Aldehyde $\longrightarrow$ 1° alcohol

Ketone $\longrightarrow$ 2° alcohol



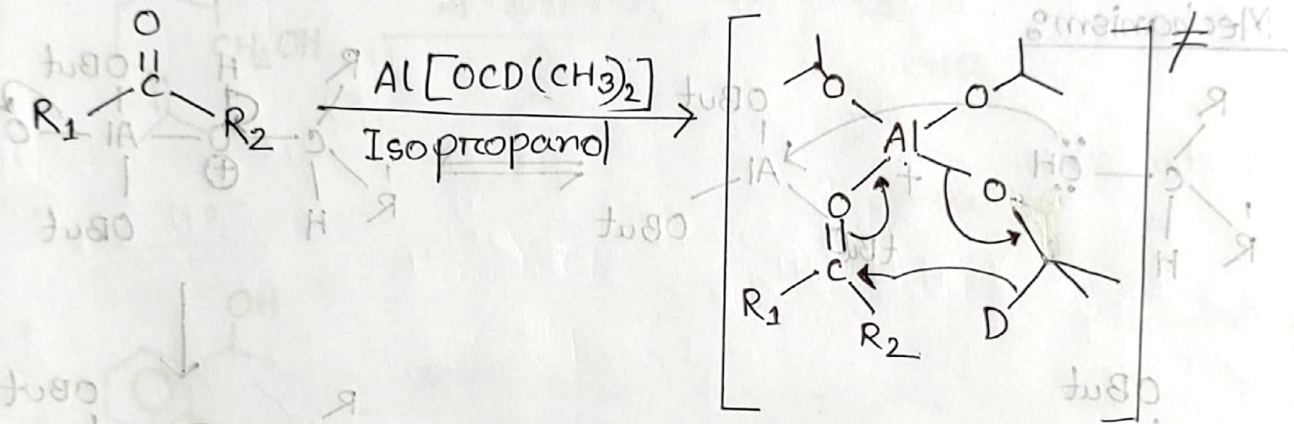
Mechanism :



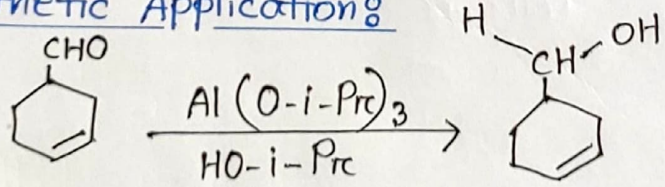
- How can you prove that hydride is transferred from iso-propoxide ?

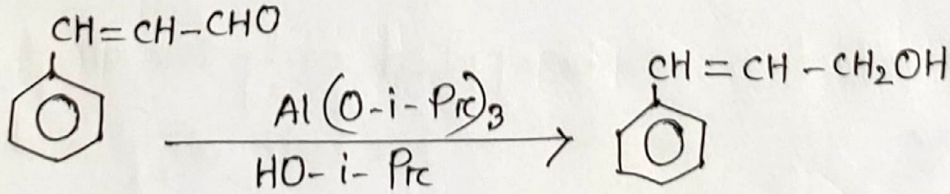
Ans 8

Meerwein-Ponndorf-Verly reduction occurs with hydride transfer from a carbon to carbonyl carbon. It can be proved if we use $Al[OCD(CH_3)_2]_3$ instead of $Al[OCH(CH_3)_2]_3$. Thus R_2CD-OH is formed.



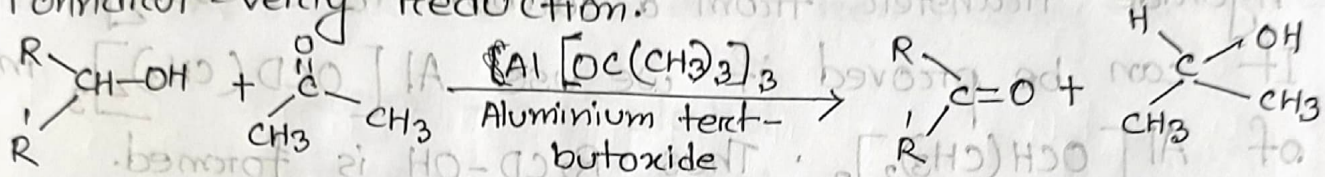
Synthetic Applications

- 

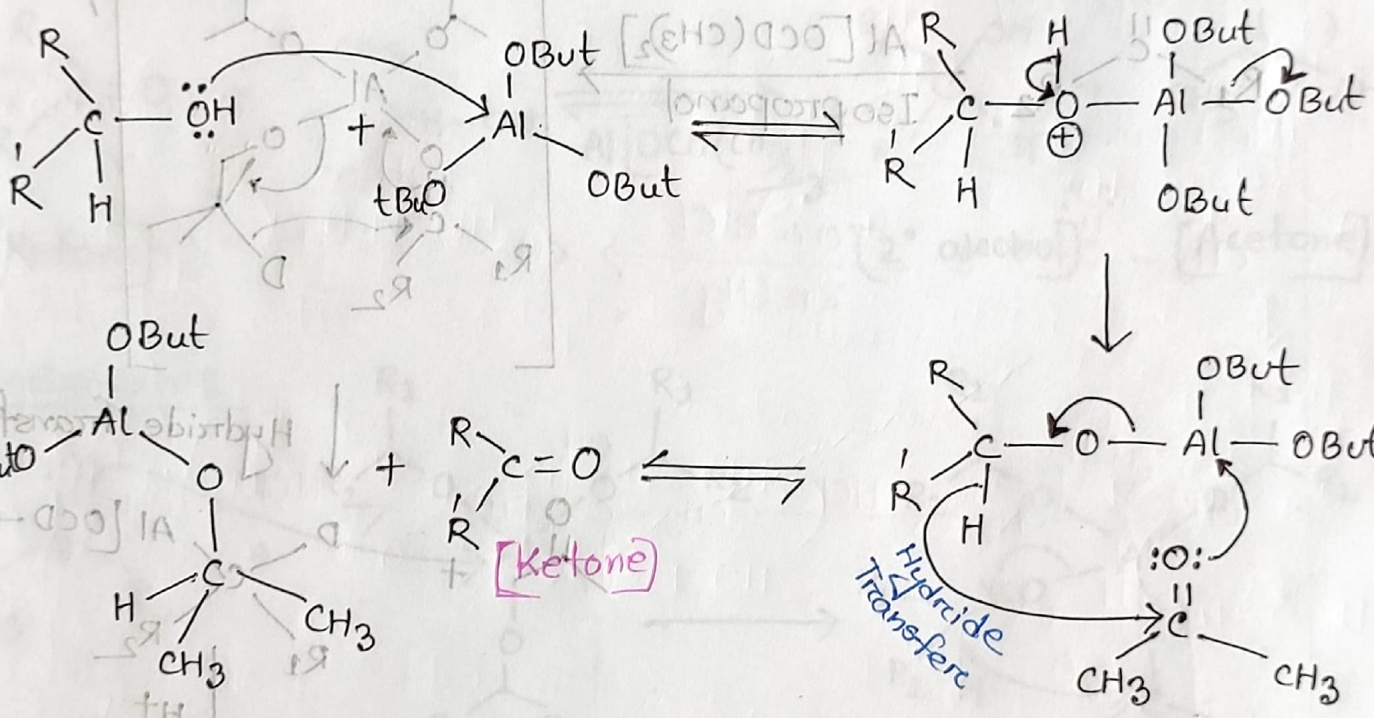
- 

Oppenauer Oxidation:

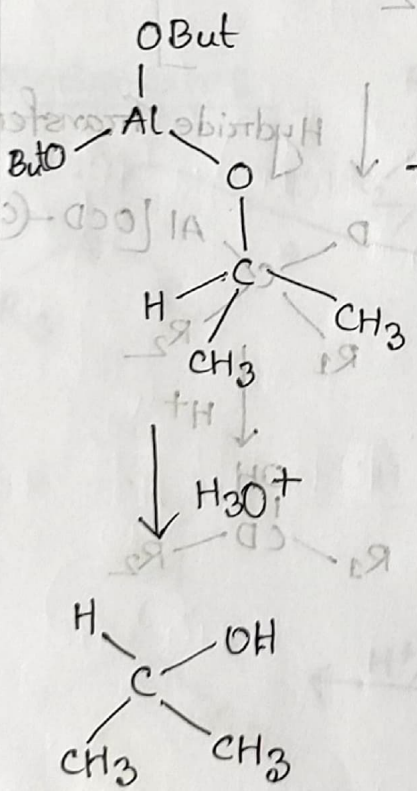
The oxidation of an alcohol with aluminium tert-butoxide as a catalyst in presence of acetone is known as Oppenauer oxidation. This oxidation is reverse to Meerwein Ponndrof-Verly reduction.



Mechanism:



[Ketone]

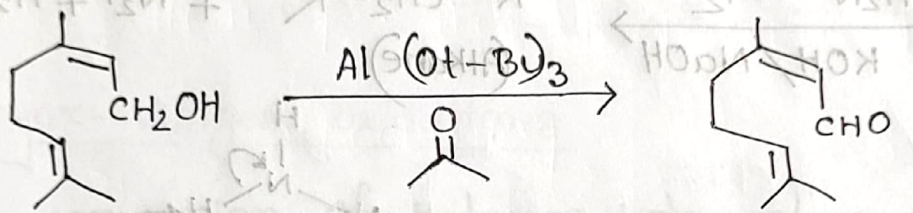


- Hydride comes from alcohol

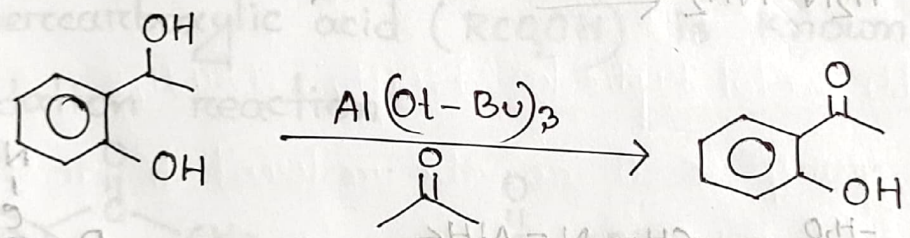
Disadvantages

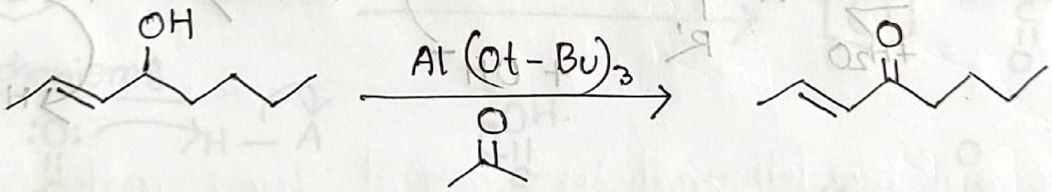
- Highly chemoselective
- Use of high temp.
- Large quantity ketone hydride acceptors
- Production of aldol condensation products

Synthetic Applications



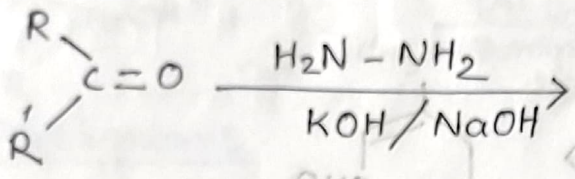
citral



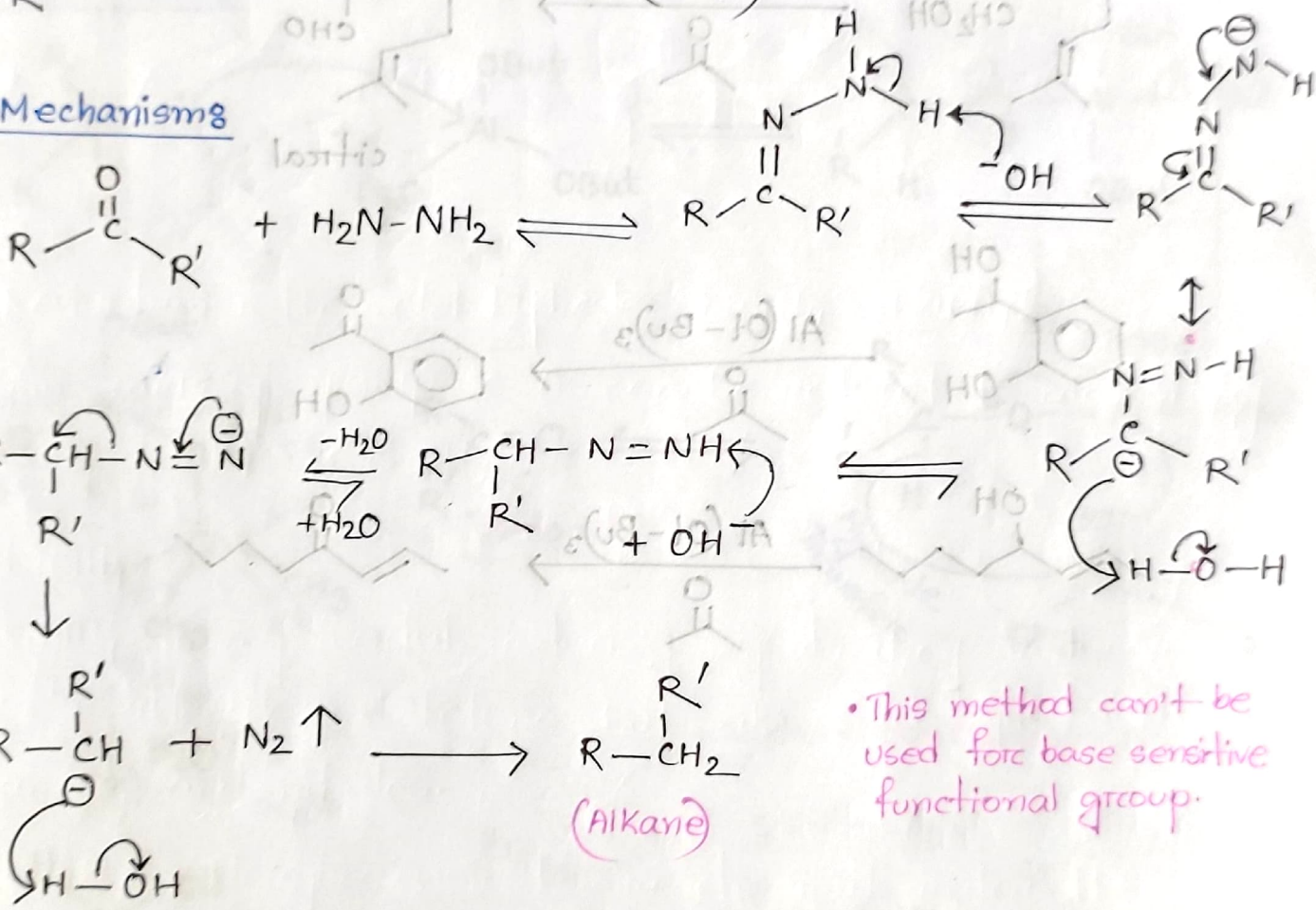


Wolf - Kishnere reduction :

The reduction of aldehyde ore ketone into $-CH_2$ group by heating with hydrazine in presence of a strong base usually KOH ore NaOH is known as Wolf - Kishnere reduction.



Mechanism :

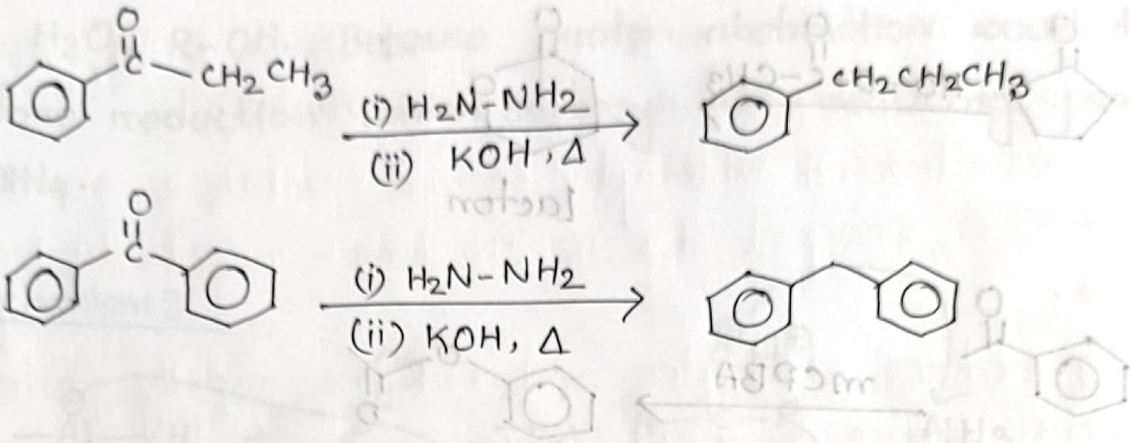


- This method can't be used fore base sensitive functional group.

NOTE:

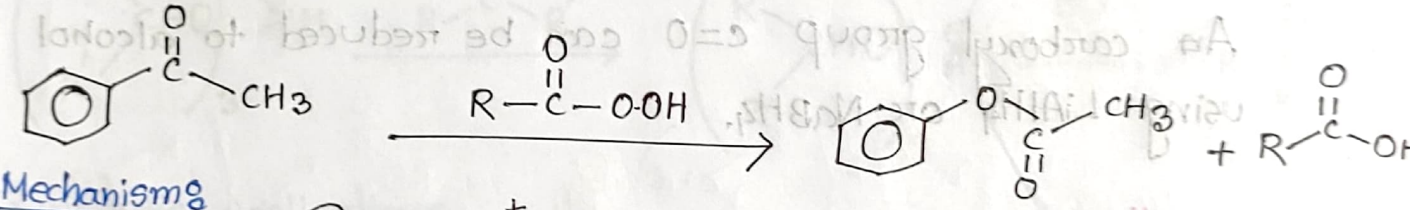
Clemmenson reduction can be applied only fore alkyl group but Wolf - Kishnere " " " " fore both alkyl and aryl groups.

Synthetic Applications

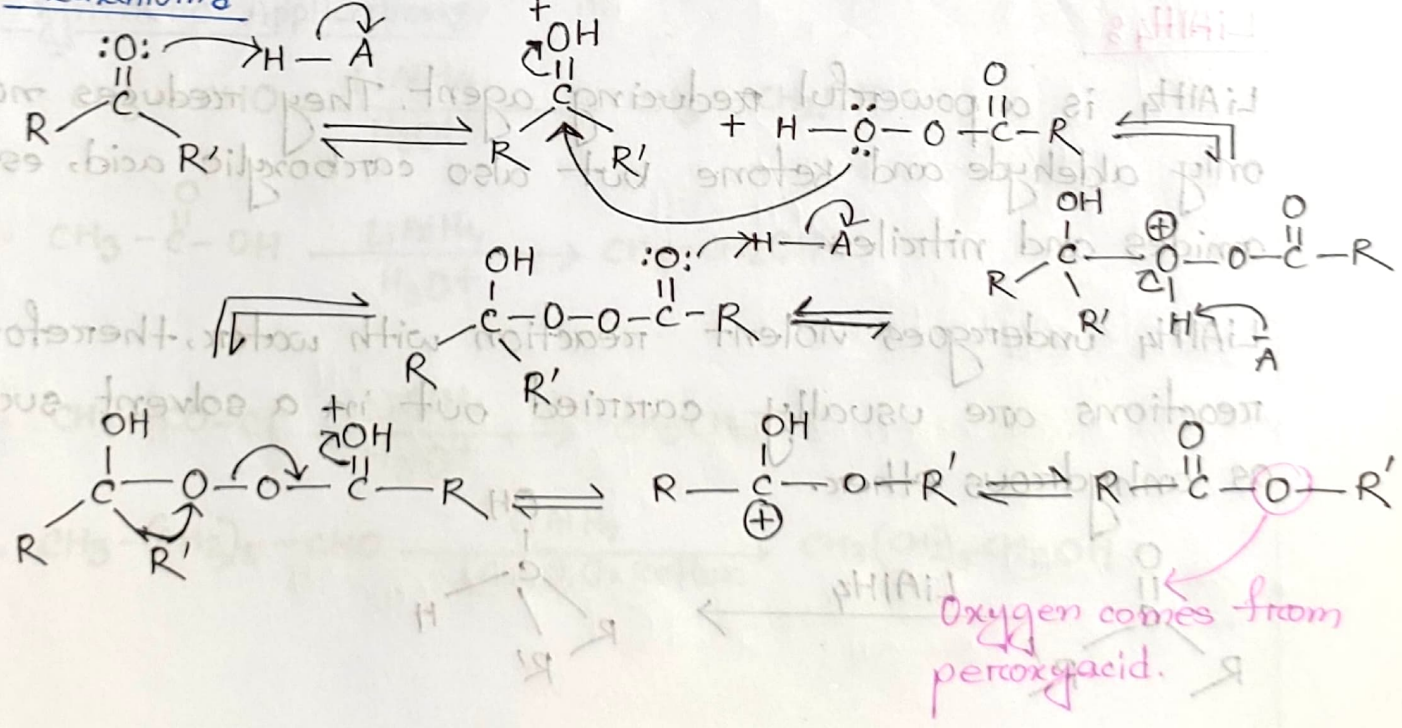


Baeyer-Villiger oxidations

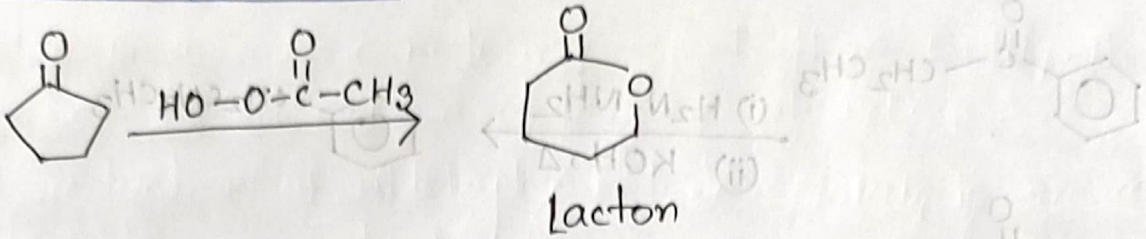
The conversion of ketones into esters and cyclic ketones into lactones by the insertion of an oxygen atom from a percarboxylic acid (RCO_2OH) is known as Baeyer-Villiger oxidation reaction.

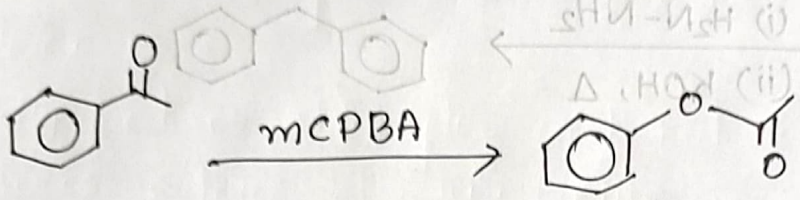


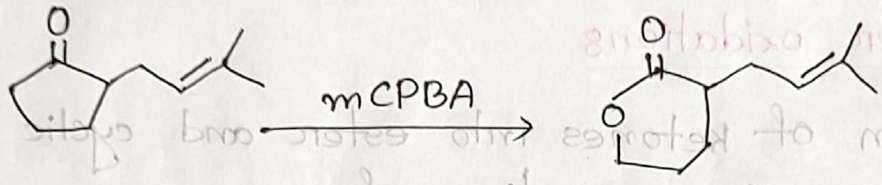
Mechanisms



Synthetic Application:

- 

- 

- 

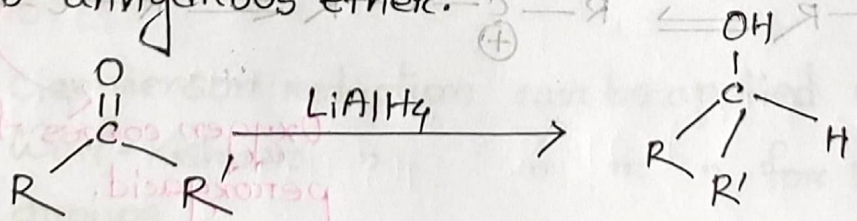
Reduction with metal hydrides $LiAlH_4$, $NaBH_4$:

An carboxyl group C=O can be reduced to alcohol by using $LiAlH_4$ or $NaBH_4$.

$LiAlH_4$:

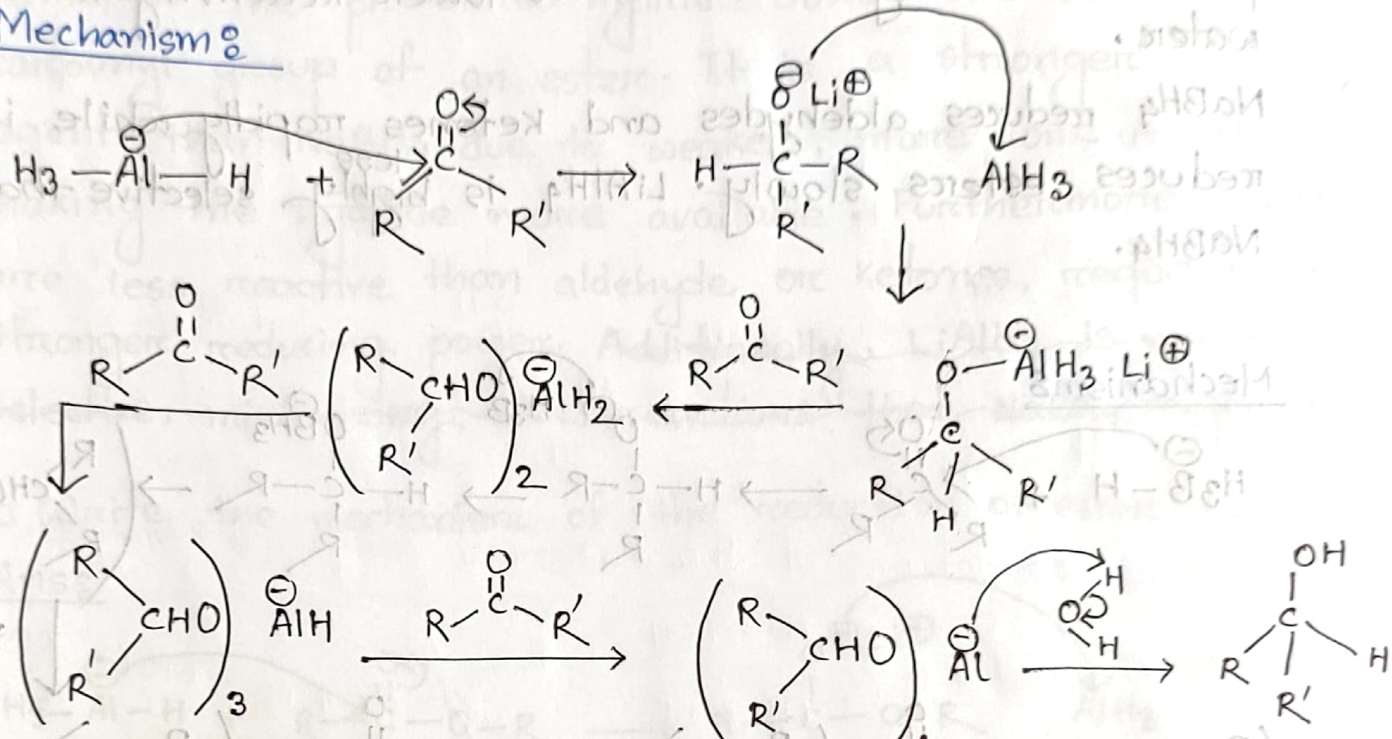
$LiAlH_4$ is a powerful reducing agent. They reduces not only aldehyde and ketone but also carboxylic acid, esters, amides and nitriles.

$LiAlH_4$ undergoes violent reaction with water, therefore reactions are usually carried out in a solvent such as anhydrous ether.



The reduction can not be carried out in protonic solvent e.g. H_2O, R-OH. Because proton abstraction would take place before reduction which destroy the reducing power of $LiAlH_4$.

Mechanism:



Synthetic Applications:

- $CH_3-CHO \xrightarrow[H_3O^+]{LiAlH_4} CH_3-CH_2OH$
- $CH_3-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[H_3O^+]{LiAlH_4} CH_3-CH_2OH$
- $CH_3-CO-Cl \xrightarrow[H_3O^+]{LiAlH_4} CH_3CH_2OH$
- $CH_3-(CH_2)_5-CHO \xrightarrow[(C_2H_5)_2O,\ Reflux]{LiAlH_4} CH_3(CH_2)_5-CH_2OH$

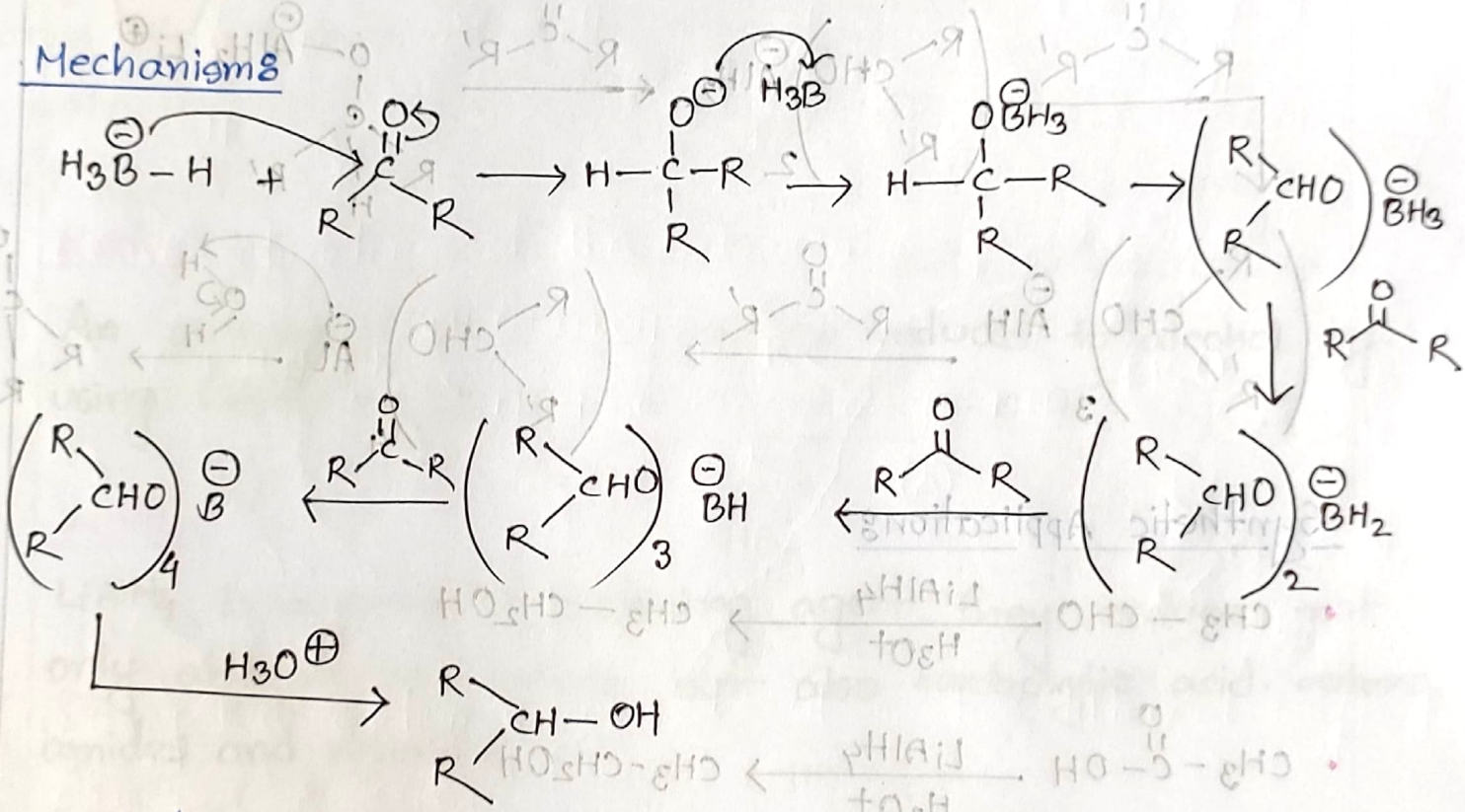
$NaBH_4$:

$NaBH_4$ is a milderc reducing agent than $LiAlH_4$. Its reaction can be carrcied out in waterc orc alcohol.

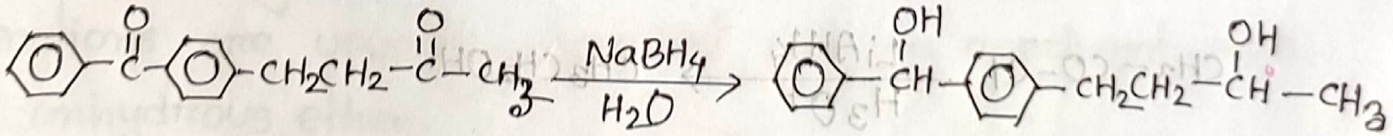
For reduction of aldehyde/ ketone $NaBH_4$ is morce prefercable reagent due to its lowerc reactivity towards waterc.

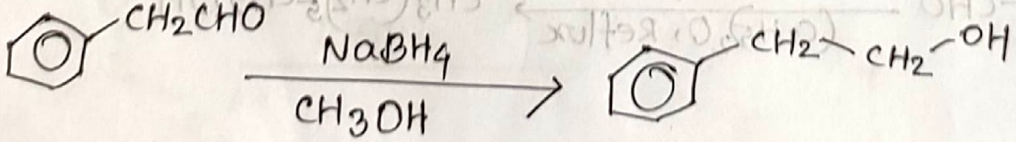
$NaBH_4$ reduces aldehydes and ketones rapidly while it reduces esterc slowly. $LiAlH_4$ is ~~highly~~ less selective than $NaBH_4$.

Mechanism:



Synthetic Applications:

- 

- 

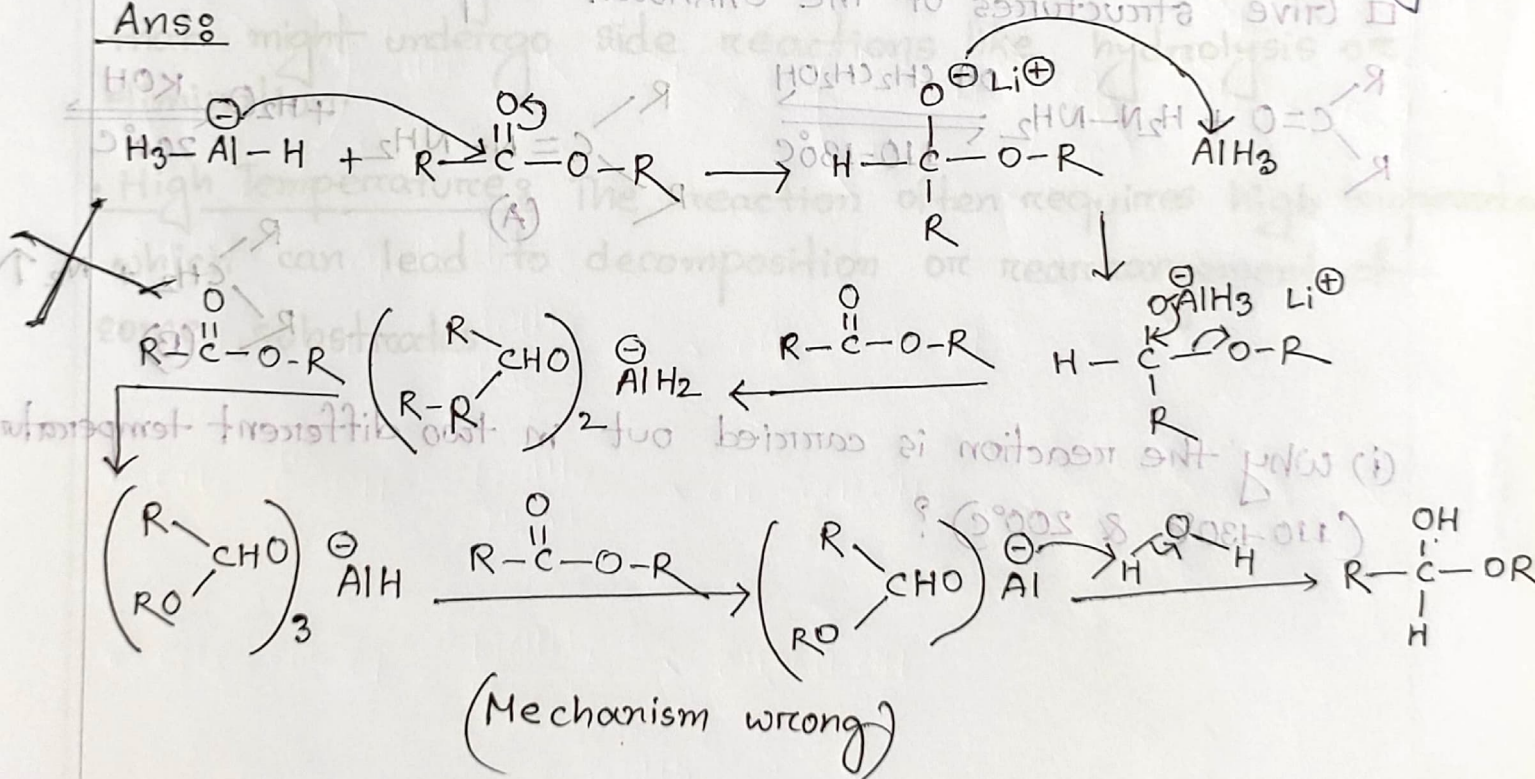
□ Both the $LiAlH_4$ and $NaBH_4$ are hydride donors. Which one of the hydride donors will you select to reduce the carbonyl group of an ester function? Give explanation.

Ans:

$LiAlH_4$ is the preferred hydride donor for reducing the carbonyl group of an ester. It is a stronger reducing agent than $NaBH_4$ due to weaker, more ionic Al–H bond, making the hydride more available. Furthermore, esters are less reactive than aldehyde or ketones, requiring the stronger reducing power. ~~Additionally, $LiAlH_4$ is more selective, minimizing side reactions that $NaBH_4$ might induce.~~

□ Write the mechanism of the reduction of ester using $LiAlH_4$

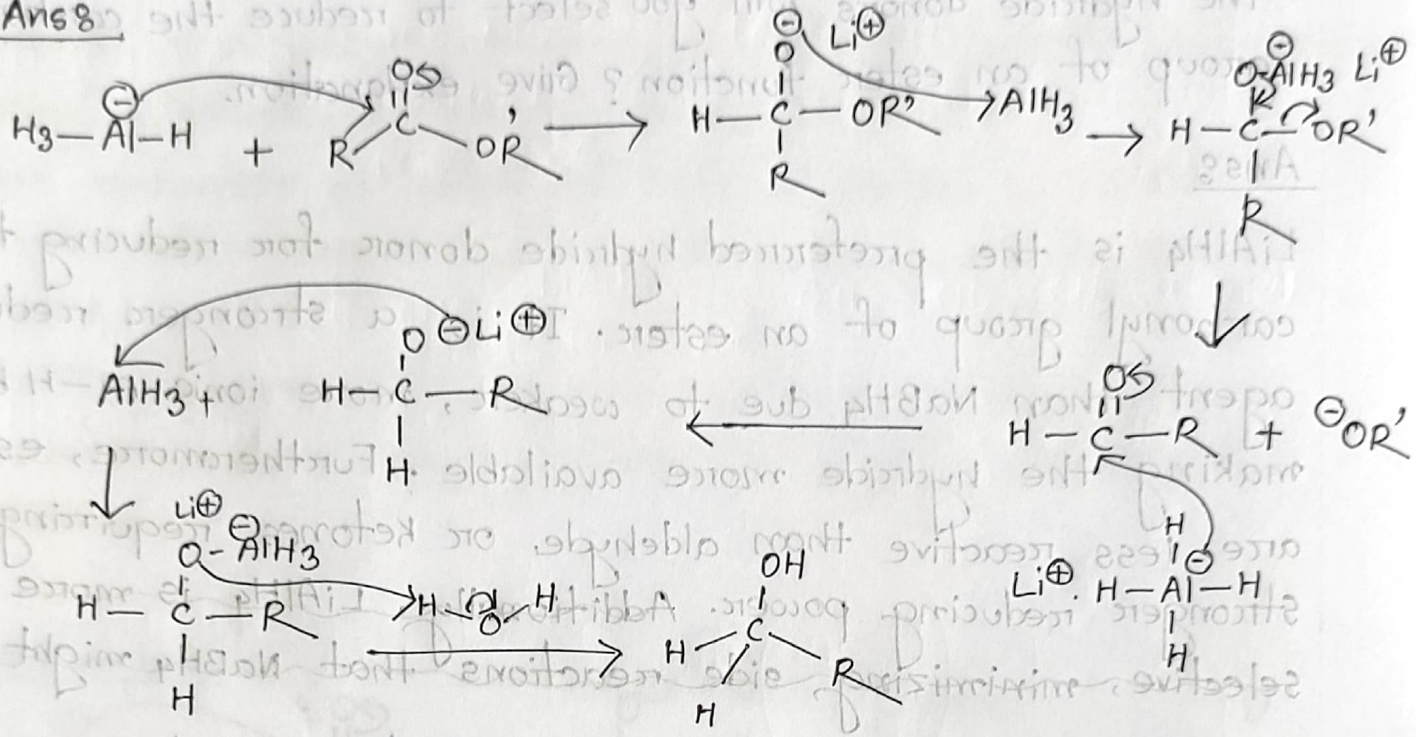
Ans:



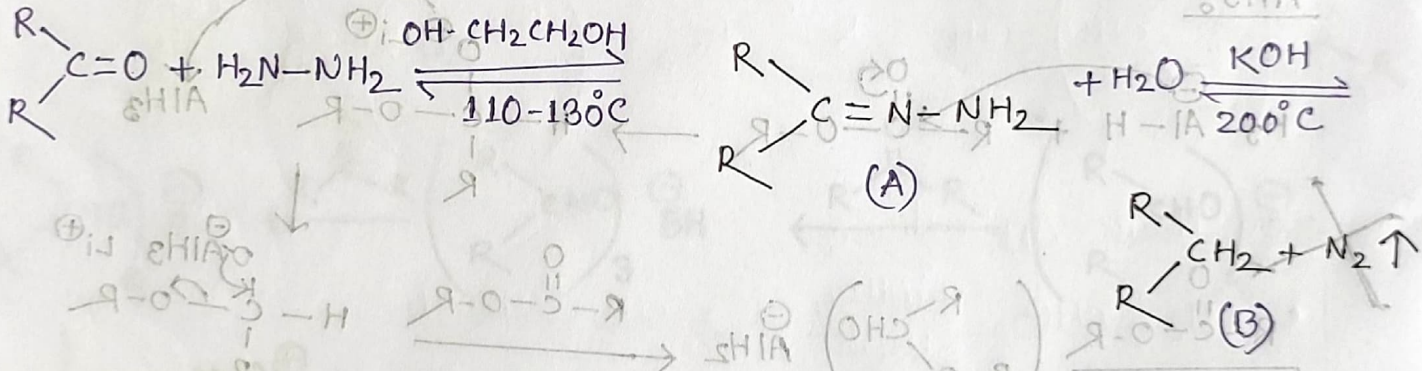
(Mechanism wrong)

□ Write the mechanism of reduction of an ester with $LiAlH_4$.

Ans 8



□ Give structures of the unknown compound A and B:



(i) Why the reaction is carried out in two different temperature (110-130°C & 200°C) ?

(ii) Can you use CH_3CH_2OH as a solvent instead of $OHCH_2CH_2OH$ to carry out above reaction?

Ans: No, ethanol is not suitable for wolf-Kishner reduction. This reduction requires very high temperature to proceed. But ethanol's boiling point is too low to achieve that. Ethylene glycol with much higher boiling point can only be suitable for the reaction.

(iii) Write the limitations of the reaction.

Ans:

Two limitations are:

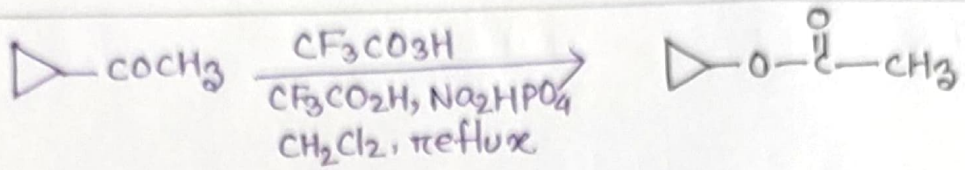
- Basic conditions: The reaction requires strongly basic conditions, making it unsuitable for base-sensitive reactant that might undergo side reactions like hydrolysis or elimination.
- High Temperature: The reaction often requires high temperatu which can lead to decomposition or rearrangement of some substrates.

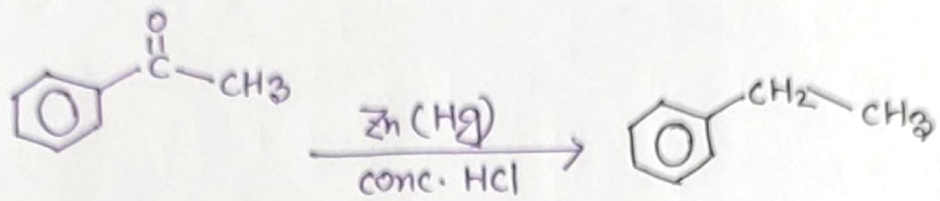
□ Complete the following reactions and give mechanism:

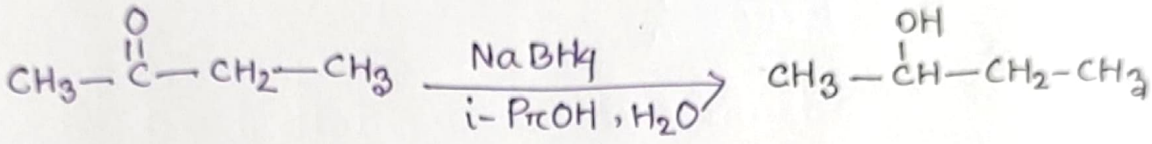
(i) $(C_6H_5)_2C=O \xrightarrow[t\text{-}C_4H_9O^-K^+,\ DMSO,\ 25^\circ C]{H_2N-NH_2} (C_6H_5)_2CH_2$

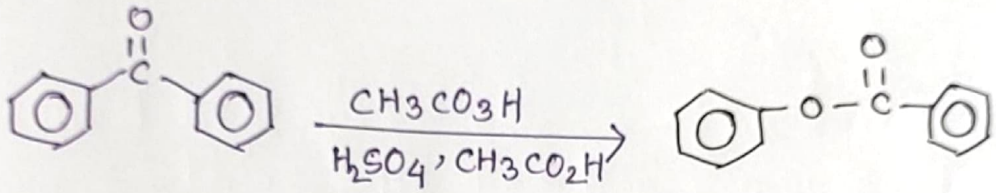
(ii) $C_6H_5-CH=CH-CHO \xrightarrow[(C_2H_5)_2O,\ -10^\circ C]{LiAlH_4} C_6H_5-CH_2-CH_2-CH_2OH$

Mechanism:

(iii) 
COCH3
CF3CO3H
CF3CO2H, Na2HPO4
CH2Cl2, reflux
O
C
CH3

(iv) 
O
C
CH3
Zn(Hg)
conc. HCl
CH2
CH3

(v) 
CH3
O
C
CH2
CH3
NaBH4
i-PrOH, H2O
OH
CH
CH3
CH2
CH3

(vi) 
O
C
CH3CO3H
H2SO4, CH3CO2H
O
O
C